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RAMSEY CHAINS IN GRAPHS

Ritabrato Chatterjee, Ph.D.

Western Michigan University, 2026

The research in this work deals with decomposition of graphs and Ramsey theory, two popular areas of research in graph theory due to numerous fascinating problems and the sheer mathematical beauty of the subjects. A decomposition $\{G_1, G_2, ..., G_k\}$ of a graph G into subgraphs of G is ascending if the subgraph G_i is isomorphic to a proper subgraph of G_{i+1} for $i = 1, 2, ..., k-1$. The well-known and long-standing of Ascending Subgraph Decomposition Conjecture states that every graph has an ascending subgraph decomposition, that is, the subgraphs (links) in the decomposition sequence are pairwise edge-disjoint in the given graph, monochromatic, and every link is a subgraph of all succeeding links in the sequence. One of the major topics in graph theory involving edge colorings takes place in Ramsey theory where typically for each red-blue edge coloring of a given graph, one of two prescribed monochromatic subgraph occurs. Ramsey theory is a combinatorial concept that deals with the structural analysis of a complex graphical system to assert the existence of prescribed unavoidable patterns or substructures within complete randomness. In graph theory, this translates to the $k-$edge coloring ($k \in \mathbb{N}$) of a complete graph of sufficiently large order such that any such coloring preserves the existence of a monochromatic copy of some graph of smaller order. The Ramsey number of a sequence of graphs $(G_1, G_2, ..., G_k)$, denoted by $R(G_1, G_2, ..., G_k)$, is the minimum positive integer n such that any $k-$ edge coloring of the complete graph K_n of order n produces a monochromatic copy of G_i for some $i \in \{1, 2, ..., k\}$.

In this research study, we investigate the concept of a Ramsey chain which involves the existence of pairwise edge-disjoint monochromatic subgraphs in a red-blue edge coloring of a given graph that satisfies

the Ascending Subgraph Decomposition (ASD) conditions. The color-specific Ramsey index of a graph G, denoted by $AR_c(G)$, is the maximum length of a Ramsey chain for a given red-blue edge coloring c of G. The Ramsey index of a graph, denoted by $AR(G)$, is the minimum taken over all $AR_c(G)$, for every red-blue edge coloring c of G (that is, $AR(G) = \min\{AR_c(G) : \text{for all red-blue edge coloring } c \text{ of } G\}$). The concepts of Ramsey chain and Ramsey index provide a different point of view on the Ascending Subgraph Decomposition Conjecture by means of edge-colorings of graphs. The primary goals of this work is to study the structures of Ramsey chains in several well-known classes of graphs, determine the Ramsey indices of these graphs, and investigate Ramsey chains in graphs with prescribed restrictions. Results and open questions are presented in this area of research.

RAMSEY CHAINS IN GRAPHS

by

Ritabrato Chatterjee

A dissertation submitted to the Graduate College
in partial fulfillment of the requirements
for the degree of Doctor of Philosophy
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Ritabrato Chatterjee

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Chapter 1

Introduction

In this chapter, we review some fundamental concepts and results on graph decomposition and Ramsey theory and provide background and motivation for research topics in these areas. We refer to the books [2, 4, 5] for graph theoretic notation and terminology not defined here. All graphs under consideration here are nonempty graphs in general.

1.1 Graph Decomposition

A *decomposition* $\mathcal{D}$ of a graph G with a partition $\{E_1, E_2, \ldots, E_t\}$ of nonempty subsets of $E(G)$ is a collection $\{H_1, H_2, \ldots, H_t\}$ of subgraphs of G such that $H_i = G[E_i]$ for some subset E_i of $E(G)$. Thus, no subgraph H_i in a decomposition of G contains isolated vertices. If $\mathcal{D}$ is a decomposition of G, then G is said to be *decomposed into* the subgraphs $H_1, H_2, \ldots, H_t$. If each subgraph H_i in $\mathcal{D}$ is a spanning subgraph of G, then $\{H_1, H_2, \ldots, H_t\}$ is a *factorization* of G. On the other hand, every factorization of a nonempty graph G also gives rise to a decomposition of G. We now describe two types of decompositions of graphs that have received much attention, namely, decompositions in which every two subgraphs are isomorphic and decompositions in which no two subgraphs are isomorphic.

If $\mathcal{D} = \{H_1, H_2, \ldots, H_t\}$ is a decomposition of a graph G such that H_i is isomorphic to some graph H for $1 \leq i \leq t$, then $\mathcal{D}$ is an *H-decomposition* of G. If there exists an H-decomposition of a graph G, then G is said to be *H-decomposable*. For example, the graph $G = K_{2,2,2}$ (the graph of the octahedron) is H-decomposable for the graph H shown in Figure 1.1, where an H-decomposition of G is also shown. Such a decomposition of G is referred to as an *isomorphic*

decomposition of G.

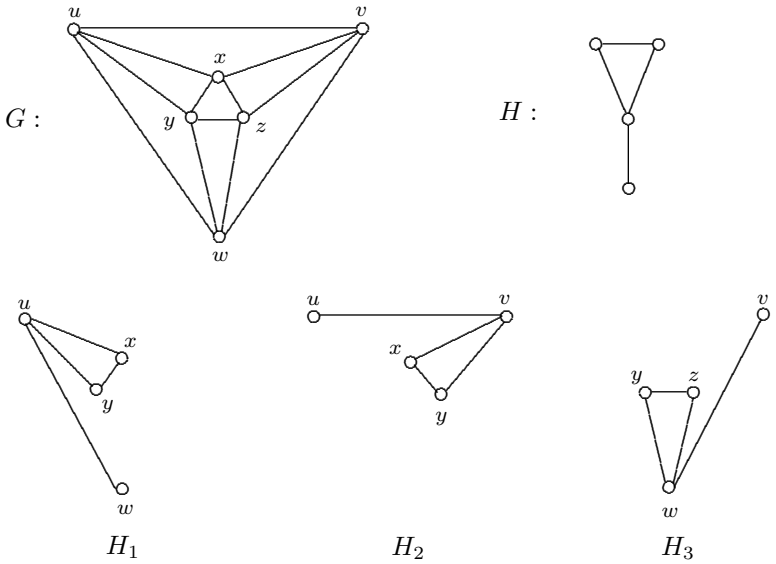


Figure 1.1: An H-decomposable graph

The following are some well-known results on isomorphic decompositions in graphs. For a positive integer n, K_n is the complete graph of order n and size $\binom{n}{2}$ and C_n is the cycle of order and size n. A matching of size m is denoted by mK_2 For positive integers r and k,

1. every r-regular bipartite graph can be decomposed into r matchings rK_2,
2. the complete graph K_{2k} can be decomposed into $2k-1$ matchings kK_2, and
3. the complete graph K_{2k+1} can be decomposed into k (Hamiltonian) cycles C_{2k+1}.

A *tree* is a connected graph with no cycles. It has been conjectured that for every tree T of size m, the complete graph K_{2m+1} is T-decomposable.

If $\mathcal{D} = \{H_1, H_2, \ldots, H_t\}$ is a decomposition of a graph G such that $H_i \ncong H_j$ for every pair i, j of integers with $1 \leq i < j \leq t$, then $\mathcal{D}$ is an *irregular decomposition* of G. Since every graph of size 3 or more can be decomposed into two non-isomorphic subgraphs, a primary question concerns decomposing graphs of a given size into a maximum number of pairwise non-isomorphic subgraphs (without isolated vertices). For example, the complete graph $H = K_6$ of size 15 in Figure 1.2 has an irregular decomposition into the five subgraphs

$$H_1 = K_3,\ H_2 = P_4,\ H_3 = 3K_2,\ H_4 = K_{1,3}, \text{ and } H_5 = P_2 + P_3,$$

all of size 3 shown in Figure 1.2. These five graphs are the five non-isomorphic graphs of size 3 containing no isolated vertices. Here, P_n denotes the path of order n, $K_{1,m}$ denotes the star of size m, and $H + G$ denotes the union of vertex-disjoint graphs G and H.

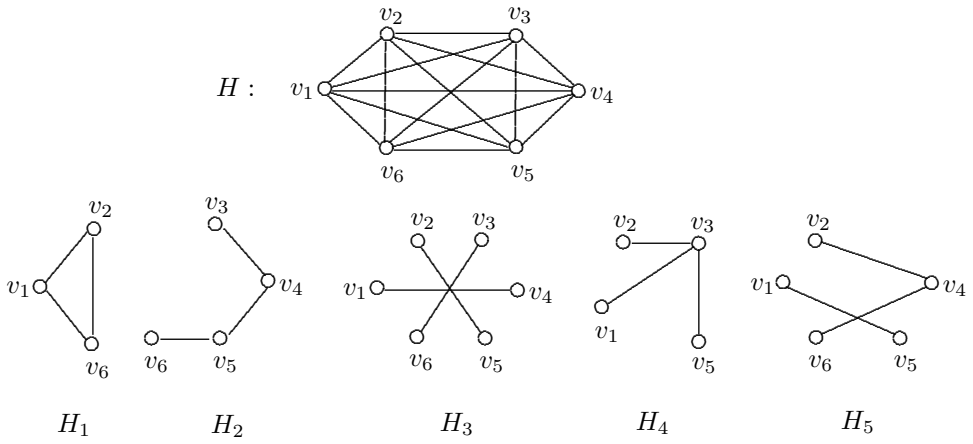


Figure 1.2: An irregular decomposition of a graph

A special type of an irregular decomposition of a graph of order n and a prescribed size is of particular interest. For an integer $n \geq 2$, a graph G of size $\binom{n+1}{2}$ is said to have an *ascending subgraph decomposition* (ASD) if G can be decomposed into n subgraphs $G_1, G_2, \ldots, G_n$ such that

(1) G_i has size i for $1 \leq i \leq n$ and

(2) G_i is isomorphic to a subgraph of G_{i+1} for $1 \leq i \leq n-1$.

For example, the graph G of order size $10 = \binom{4+1}{2}$ in Figure 1.3 has an irregular decomposition into four non-isomorphic subgraphs G_1, G_2, G_3, G_4, also shown in Figure 1.3, where

$$G_1 = P_2,\ G_2 = P_3,\ G_3 = P_4, \text{ and } G_4 = P_2 + P_4.$$

These subgraphs have the added property that G_i is isomorphic to a subgraph of G_{i+1} for $i = 1, 2, 3$, which is a characteristic of interest here. Thus, this decomposition of G is an ASD into the four graphs G_1, G_2, G_3, G_4.

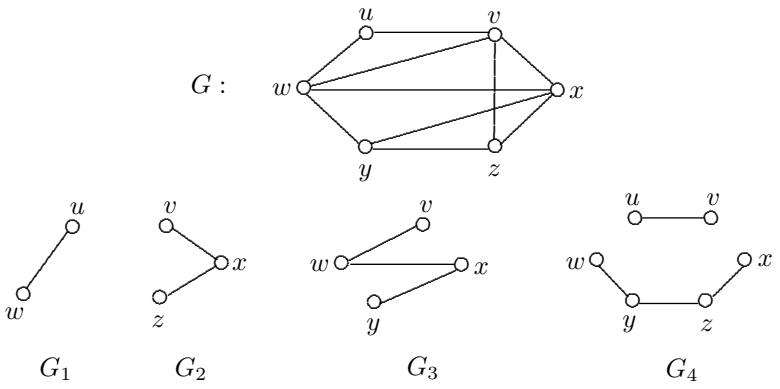


Figure 1.3: Another irregular decomposition of a different graph

In any ascending subgraph decomposition $\{G_1, G_2, G_3, \ldots, G_n\}$ of a graph G of size $\binom{n+1}{2}$ where $n \geq 3$, the graph $G_1 = K_2$, the graph $G_2 \in \{P_3, 2K_2\}$, and the graph $G_3 \in \{K_{1,3}, P_4, 3K_2, P_3 + K_2, K_3\}$. The most common example of a graph G of size $\binom{n+1}{2}$ is the complete graph K_{n+1}. In fact, K_{n+1} has an ASD into stars, necessarily $K_{1,1}, K_{1,2}, \ldots, K_{1,n}$. The following known result illustrates this.

Example 1.1.1 *Every complete graph K_{n+1} has an ASD into stars.*

Proof. Let $V(K_{n+1}) = \{v_0, v_1, \ldots, v_n\}$ and let G_n be the subgraph of size n in K_{n+1} induced by the n edges incident with v_n. Thus, G_n is the star $K_{1,n}$ of size n with central vertex v_n. Deleting v_n from K_{n+1} results in the subgraph K_n of order n and size $\binom{n}{2}$. Next, let $G_{n-1} = K_{1,n-1}$ be the subgraph of K_n induced by the $n-1$ edges incident with v_{n-1}. Thus, G_{n-1} is a star of size $n-1$ with central vertex v_{n-1}. Continuing in this manner, we arrive at an irregular decomposition $\{G_1, G_2, \ldots, G_n\}$ of K_{n+1} where $G_i \cong K_{1,i}$ is a star of size i with central vertex v_i for $i = 1, 2, \ldots, n$. This decomposition is an ASD of K_{n+1} into stars. For $n+1 = 6$, such a decomposition $\{G_1, G_2, G_3, G_4, G_5\}$ is shown in Figure 1.4, where an edge labeled i belongs to G_i for $1 \leq i \leq 5$. ∎

It has been shown that many graphs of size $\binom{n+1}{2}$ can be decomposed into n subgraphs $G_1, G_2, \ldots, G_n$ where not only G_i has size i for $1 \leq i \leq n$ but G_i is isomorphic to a subgraph of G_{i+1} for $1 \leq i \leq n-1$. That is, each of these graphs of size $\binom{n+1}{2}$ has an ASD. These ascending subgraph decompositions illustrate a well-known conjecture (see [1]).

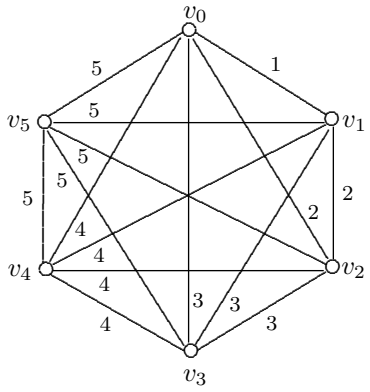


Figure 1.4: An ascending subgraph decomposition of K_6

The Ascending Subgraph Decomposition Conjecture *For each integer $n \geq 2$, every graph of size $\binom{n+1}{2}$ has an ascending subgraph decomposition.*

Upon learning the statement of this conjecture, the famous mathematician Paul Erdős doubted its truth and immediately offered \$5 for a counterexample. He then participated in a study of the conjecture (see [1]). This conjecture has drawn the attention of many researchers but has never been proved or disproved. There are many papers dealing with this conjecture (see [7, 8, 18, 19, 20, 22, 23, 30], for example). Among the several classes of graphs for which the conjecture has been verified are regular graphs (see [8, 24]). Two classes of graphs for which the conjecture can easily be verified are matchings mK_2 (consisting of m components K_2) and stars $K_{1,m}$ where $m = \binom{k+1}{2}$ for a positive integer k. Consequently, for every positive integer m, where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for a positive integer k, there is an ascending subgraph decomposition $\{G_1, G_2, \ldots, G_k\}$ of the graph G if either $G = mK_2$ or $G = K_{1,m}$ such that G_i is isomorphic to a proper subgraph of G_{i+1} for $1 \leq i \leq k-1$. If $G = mK_2$, the subgraphs G_i are matchings and if $G = K_{1,m}$, each subgraph G_i is a star. We refer to the books [2, 4, 5] for more information on decompositions and ascending subgraph decompositions in graphs.

1.2 Ramsey Theory

A well-known area within graph theory is Ramsey theory and a well-known concept in this theory is Ramsey numbers. The famous mathematician Ronald Graham (1935–2020) once stated that Ramsey theory is a branch of mathematics dedicated

to the proposition that *complete disorder is impossible* in the sense that within any sufficiently large system, some regularity must occur [26]. Ramsey theory has also been described as the study of unavoidable regularity in large structures, where the primary question is:

> *When is it the case that whenever the elements of some sufficiently large structure are partitioned into a finite number of classes, there is always at least one class within which a prescribed regular structure occurs?*

We are interested in the case where the structures in question are graphs whose edges are colored, where the classes are the subgraphs of graphs such that the edges in each subgraph are colored the same and where one of these subgraphs contains a prescribed graph. In this case, Ramsey theory being discussed is that in graph theory.

Let G be a graph without isolated vertices where each edge is assigned one of two given colors (a 2-*edge coloring* of G). In a *red-blue coloring* of G, every edge of G is colored red or blue. For two graphs F and H (without isolated vertices), the *Ramsey number* $R(F, H)$ of F and H is the minimum positive integer n for which every red-blue coloring of the complete graph K_n of order n results in either a subgraph of K_n isomorphic to F all of whose edges are colored red (a *red* F) or a subgraph of K_n isomorphic to H all of whose edges are colored blue (a *blue* H).

Ramsey numbers are named for Frank Ramsey (1903–1930), a British philosopher, economist and mathematician. The theorem for which Ramsey is best known was proved only as a lemma in a famous paper [32] by Ramsey. This lemma became the basis of the area of graph theory called Ramsey theory. It is a consequence of this lemma that the Ramsey number $R(F, H)$ exists for every two graphs F and H. If $F \cong H$, then $R(F, H) = R(F, F) = R(F)$ is the minimum positive integer n such that every red-blue coloring of K_n results in a *monochromatic* F. If F and H are both complete graphs, then $R(F, H)$ is called a *classical Ramsey number*. For example, $R(K_3) = 6$, $R(K_4) = 18$, and $R(K_5)$ is unknown [27, 29]. Erős contributed greatly to Ramsey theory [13, 14, 15, 16, 17] and often spoke of the notorious difficulty of many problems in Ramsey theory [9].

Many variations of Ramsey numbers have been studied, such as considering classes of graphs different from complete graphs and allowing the edges of the graphs in question to be colored with more than two colors. A book by Graham,

Rothschild and Spencer [26] is devoted to Ramsey theory and a chapter on Ramsey numbers by Faudree in the *Handbook of Graph Theory* [28, pp.1002-1025] is devoted as well to Ramsey numbers. As Roberts showed in [33], Ramsey theory has applications in such diverse fields as communication, information retrieval in computer science, and decision making.

The exact values of $R(F,H)$ have been determined only for pairs F, H of graphs belonging to relatively few well-known classes of graphs [31]. Some of these values are given below.

Theorem 1.2.1 [3] *For integers $m, n \geq 2$,*

$$R(K_{1,m}, K_{1,n}) = \begin{cases} m+n-1 & \text{if } m \text{ and } n \text{ are both even} \\ m+n & \text{otherwise.} \end{cases}$$

In particular, for $n \geq 3$, $R(K_{1,n}) = \begin{cases} 2n & \text{if } n \text{ is odd} \\ 2n-1 & \text{if } n \text{ is even.} \end{cases}$

Theorem 1.2.2 [25] *For integers n and m with $2 \leq m \leq n$,*

$$R(P_n, P_m) = n - 1 + \lfloor m/2 \rfloor .$$

In particular, for $n \geq 3$, $R(P_n) = \begin{cases} \frac{3(n-1)}{2} & \text{if } n \text{ is odd} \\ \frac{3n}{2} - 1 & \text{if } n \text{ is even.} \end{cases}$

Theorem 1.2.3 [21] *Let m and n be integers with $3 \leq m \leq n$.*

(1) *If m is odd, where $(m,n) \neq (3,3)$, then $R(C_m, C_n) = 2n-1$.*

(2) *If m and n are even, where $(m,n) \neq (4,4)$, then $R(C_m, C_n) = n + m/2 - 1$.*

(3) *If m is even and n is odd, then $R(C_m, C_n) = \max\{n + m/2 - 1,\ 2m-1\}$.*

(4) *$R(C_3) = R(C_4) = 6$.*

In particular, for $n \geq 5$, $R(C_n) = \begin{cases} n-1 & \text{if } n \text{ is odd} \\ \frac{3n}{2} - 1 & \text{if } n \text{ is even.} \end{cases}$

Theorem 1.2.4 [11, 12] *For integers m and n with $2 \leq m \leq n$,*

$$R(mK_2, nK_2) = m + 2n - 1.$$

In particular, $R(nK_2) = 3n - 1$ for $n \geq 2$.

Theorem 1.2.5 [10] *Let T be a tree of order $p \geq 2$. For every integer $n \geq 2$,*

$$R(T, K_n) = (p-1)(n-1) + 1.$$

We refer to the books [4, 5, 26] for more information on Ramsey numbers of graphs.

1.3 Ramsey Chains

In [2], a concept was introduced that involves both ascending subgraph decompositions and Ramsey-type colorings. Let G be a graph (without isolated vertices) of size m with a red-blue edge coloring c. A *Ramsey chain* of G *with respect to* c is a sequence $G_1, G_2, \ldots, G_\ell$ of pairwise edge-disjoint subgraphs of G such that each subgraph G_i $(1 \leq i \leq \ell)$ is monochromatic of size i and G_i is isomorphic to a subgraph of G_{i+1} for $1 \leq i \leq \ell - 1$. A Ramsey chain is sometime denoted by $(G_1, G_2, \ldots, G_\ell)$. Each subgraph G_i $(1 \leq i \leq \ell)$ in a Ramsey chain is called a *link* of the chain. The maximum length of a Ramsey chain of G with respect to c is the (ascending) *Ramsey index* $AR_c(G)$ of G. A Ramsey chain having maximum length $AR_c(G)$ is a maximum Ramsey chain of G with respect to c. The (ascending) *Ramsey index* $AR(G)$ of G is defined by

$$AR(G) = \min\{AR_c(G) : \ c \text{ is a red-blue edge coloring of } G\}.$$

A Ramsey chain of length $AR(G)$ is referred to as a *maximum* Ramsey chain. For example, for the red-blue coloring of the path P_7 of order 7 shown in Figure 1.5, two Ramsey chains $R_1 = (K_2, P_3)$ and $R_2 = (K_2, 2K_2, P_4)$ are also shown in Figure 1.5 where a bold line indicates a red edge and the thin line indicate a blue edge. Hence, the length of R_1 is 2 and the length of R_2 is 3. For this red-blue coloring of P_7, it follows that $AR_c(P_7) = 3$. In fact, it can be shown that $AR(P_7) = 3$ (since the three links in R_2 form a decomposition of P_7). Thus, R_2 is a maximum Ramsey chain of P_7 with respect to the red-blue coloring of P_7 shown in Figure 1.5.

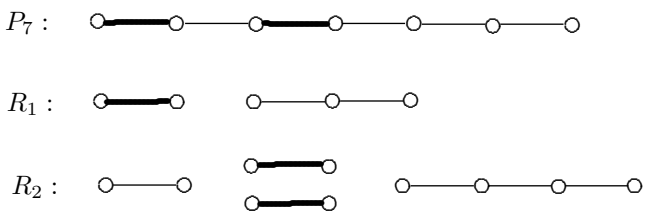


Figure 1.5: A red-blue coloring of P_7 and two Ramsey chains

These concepts were introduced and studied in [2, 6], using somewhat different terminology. Ramsey indexes have been determined for some well-known classes of graphs.

Theorem 1.3.1 [6] *If m and n are positive integer such that $\binom{n+1}{2} \leq m < \binom{n+2}{2}$, then $AR(mK_2) = AR(K_{1,m}) = n$. Consequently, if $G \in \{mK_2, K_{1,m}\}$ for some positive integer m, then*

$$AR(G) = \left\lfloor \frac{-1 + \sqrt{1+8m}}{2} \right\rfloor.$$

Theorem 1.3.2 [6] *For $4 \leq n \leq 6$, $AR(K_n) = n - 1$.*

Whether $AR(K_n) = n - 1$ for $n \geq 7$ is not known.

Ramsey indexes of the union of a matching and a star have been determined.

Theorem 1.3.3 [6] *For integers $a \geq 1$ and $b \geq 2$,*

$$AR(aK_2 + K_{1,b}) = \begin{cases} AR((a+1)K_2) & \text{if } b \leq a \\ AR(K_{1,b}) & \text{if } b \geq a+2 \\ AR(K_{1,b+1}) & \text{if } b = a+1. \end{cases}$$

For every graph G of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ that we have mentioned and for which $AR(G)$ has been determined, it has been shown that $AR(G) = k$. This brings up the question of finding graphs G of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for which $AR(G) \neq k$ and so $AR(G) \leq k - 1$. This led to an investigation of a special class of trees. The *distance* $d(u, v)$ between vertices u and v in a nontrivial connected graph G is the minimum length of a $u - v$ path in G. The *diameter*

diam(G) of G is the greatest distance between any two vertices of G. A *double star* is a tree of diameter 3. For integers a and b with $2 \leq a \leq b$, the double star $S_{a,b}$ has a vertex of degree a, a vertex of degree b, and all other vertices have degree 1. The double star $S_{a,b}$ has order $a+b$ and size $a+b-1$. Thus, if $\binom{k+1}{2} \leq a+b-1 < \binom{k+2}{2}$, then $AR(S_{a,b}) \leq k$. The following results have been obtained.

Theorem 1.3.4 [6] *Let $b \geq 2$ be an integer. Then*

$$AR(S_{2,b}) = k \text{ if and only if } \binom{k+1}{2} + 1 \leq b+1 < \binom{k+2}{2} + 1.$$

Theorem 1.3.5 [6] *Let $b \geq 3$ be an integer.*

If $b+2 = \binom{k+1}{2}$ or $b+2 = \binom{k+2}{2} + 1$, then $AR(S_{3,b}) = k-1$.

If $\binom{k+1}{2} + 2 \leq b+2 < \binom{k+2}{2}$, then $AR(S_{3,b}) = k$.

The following is a more general result dealing with double stars.

Theorem 1.3.6 [6] *Let $2 \leq a \leq b$ be integers. If*

$$a+b-1 \in \left\{\binom{k+1}{2}, \binom{k+1}{2}+1, \ldots, \binom{k+1}{2}+a-2\right\},$$

then $AR(S_{a,b}) = k-1$.

Consequently, for integers a and b with $2 \leq a \leq b$, if G is the double star $S_{a,b}$ of size $m = a+b-1$, where $\binom{k+1}{2} \leq m \leq \binom{k+2}{2} + a - 2$, then $AR(G) \neq k$.

The primary goal here is the investigation of Ramsey chains in graphs and the determination of Ramsey indexes of graphs in some well-known classes of graphs such as paths, cycles, linear forests (each of whose components is a path), and graphs related to stars and matchings. We also investigate Ramsey chains with prescribed properties.

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Chapter 2

Ramsey Chains in Paths and Cycles

For a graph G of size m and a red-blue coloring c of the edges of G, recall that a Ramsey chain in G with respect to c is a sequence $G_1, G_2, \ldots, G_\ell$ of pairwise edge-disjoint subgraphs of G such that each subgraph G_i $(1 \leq i \leq \ell)$ is monochromatic of size i and G_i is isomorphic to a subgraph of G_{i+1} $(1 \leq i \leq \ell - 1)$. The Ramsey index $AR_c(G)$ of G with respect to c is the maximum length of a Ramsey chain in G with respect to c. The Ramsey index $AR(G)$ of G is the minimum value of $AR_c(G)$ among all red-blue colorings c of G. Consequently, if G is a graph of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$, then $AR(G) \leq k$. It was proved in [6] that if $G = mK_2$ is a matching of size m or $G = K_{1,m}$ is a star of size m, then $AR(G) = k$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. One question was posed in [6] asking whether there are other classes S of graphs with the property that for every sufficiently large integer m, every graph G of size m in S has the property that $AR(G) = k$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. In this chapter, we show that all paths have this property and, as a consequence, all cycles have this property as well.

2.1 Preliminary Results

We begin by presenting some observations and preliminary results on Ramsey indexes of graphs. In general, all graphs under consideration contain no isolated vertices.

Observation 2.1.1 *If G is a graph of size m where $2 \leq m < \binom{k+2}{2}$ for a positive integer k, then $AR(G) \leq k$.*

Observation 2.1.2 *If H and G are graphs such that $H \subseteq G$, then $AR(H) \leq AR(G)$. Consequently, if $AR(H) \geq k$ for each graph H of size m, then $AR(G) \geq k$ for every graph G of size $m+1$.*

The following is a consequence of Observation 2.1.2.

Observation 2.1.3 *Let G be a graph without isolated vertices.*

(1) *If the maximum degree of G is Δ, then $AR(G) \geq AR(K_{1,\Delta})$.*

(2) *If kK_2 is a maximum matching in G, then $AR(G) \geq AR(kK_2)$.*

(3) *If P_k is a longest path in G, then $AR(G) \geq AR(P_k)$.*

(4) *If C_k is a largest cycle in G, then $AR(G) \geq AR(C_k)$.*

(5) *If K_ω is a largest complete subgraph in G, then $AR(G) \geq AR(K_\omega)$.*

For every graph G of size m with $2 \leq m \leq 9$, the Ramsey index $AR(G)$ is at most 3.

Observation 2.1.4 *Let G be a graph of size $m \geq 2$.*

(*a*) *If $m = 2$, then $AR(G) = 1$; while if $m > 2$, then $AR(G) \geq 2$.*

(*b*) *If $m = 3, 4, 5$, then $AR(G) = 2$.*

(*c*) *If $6 \leq m \leq 8$, then $AR(G) \in \{2, 3\}$. Furthermore, if $m \geq 6$ and c is a 2-edge coloring of G such that (i) there is a monochromatic subgraph F of G where $F \in \{P_4, P_3 + K_2\}$ and (ii) $G - E(F)$ has a monochromatic subgraph of size 2, then $AR_c(G) \geq 3$.*

By Observation 2.1.4, if G is a graph of size 8, then $AR(G) = 2$ or $AR(G) = 3$. The situation is clearer if G has size 9.

Proposition 2.1.5 *If G is a graph of size 9, then $AR(G) = 3$. Furthermore, if F is a graph of size 10 or more, then $AR(F) \geq 3$.*

Proof. Let G be a graph of size 9. By Observation 2.1.1, $AR(G) \leq 3$. It remains to show that $AR(G) \geq 3$. Let there be given a red-blue coloring c of G, where G_r is the red subgraph and G_b is the blue subgraph. Let m_r be the size of G_r and m_b the size of G_b. Thus, $m_r + m_b = 9$. We may assume that $m_r > m_b$ and so $m_r \geq 5$. If G_r is a star or a matching of size $m_r \geq 5$, then G has a Ramsey chain of length 3 and so $AR_c(G) \geq 3$. If G_r is neither a star nor a matching, then either $P_3 + K_2 \subseteq G_r$ or $P_4 \subseteq G_r$ and so $AR_c(G) \geq 3$ by Observation 2.1.4(c). Therefore, $AR(G) \geq 3$ and so $AR(G) = 3$.

Furthermore, if F is a graph of size 10 or more, then F contains a subgraph G of size 9. It then follows by Observations 2.1.2 that $AR(F) \geq AR(G) = 3$. ∎

2.2 Ramsey Chains in Paths

In order to present a result dealing with paths, it is useful to first present the following result dealing with positive integers.

Proposition 2.2.1 *For every integer $n \geq 5$, there is a collection $\{n_1, n_2, \ldots, n_t\}$ of t integers such that $1 \leq n_1 < n_2 < \cdots < n_t \leq \lceil n/2 \rceil$ and $\sum_{i=1}^{t} n_i = n$.*

Proof. We proceed by the strong form of induction.

For $n = 5$, $\lceil n/2 \rceil = 3$ and $2 + 3 = 5$.

For $n = 6$, $\lceil n/2 \rceil = 3$ and $1 + 2 + 3 = 6$.

For $n = 7$, $\lceil n/2 \rceil = 4$ and $3 + 4 = 1 + 2 + 4 = 7$.

For $n = 8$, $\lceil n/2 \rceil = 4$ and $1 + 3 + 4 = 8$.

For $n = 9$, $\lceil n/2 \rceil = 5$ and $2 + 3 + 4 = 4 + 5 = 1 + 3 + 5 = 9$.

For $n = 10$, $\lceil n/2 \rceil = 5$ and $1 + 4 + 5 = 2 + 3 + 5 = 1 + 2 + 3 + 4 = 10$.

Assume for an integer $k \geq 10$ that the statement is true for every integer i with $10 \leq i \leq k$. We show that this statement is true for $k+1$. Since $k \geq 10$, it follows that $k + 1 \geq 11$ and $\lceil (k+1)/2 \rceil \geq 6$. Therefore, $m = k + 1 - \lceil (k+1)/2 \rceil = \lfloor (k+1)/2 \rfloor \geq 5$. Thus, there are integers $n_1, n_2, \ldots, n_t$ with $1 \leq n_1 < n_2 < \cdots < n_t \leq \lceil m/2 \rceil$ such that $\sum_{i=1}^{t} n_i = m$. Since $n_t \leq \lceil m/2 \rceil < \lceil (k+1)/2 \rceil$, it follows

that $1 \leq n_1 < n_2 < \ldots < n_t < n_{t+1} = \lceil (k+1)/2 \rceil$ and $n_1 + n_2 + \cdots + n_t + n_{t+1} = k+1$. ■

The proof of Proposition 2.2.1 shows that the integer n_t in the statement can always be chosen to equal $\lceil n/2 \rceil$.

Corollary 2.2.2 *For every integer $n \geq 5$, there is a collection $\{n_1, n_2, \ldots, n_t\}$ of integers such that $1 \leq n_1 < n_2 < \cdots < n_t = \lceil n/2 \rceil$ and $\sum_{i=1}^{t} n_i = n$.*

While it is common to denote a path of order n (and size $n-1$) as P_n, since the emphasis here is on the size of a path, it is convenient to write Q_n to denote a path of size n, that is, $Q_n = P_{n+1}$. Thus, we often use Q_n to denote a path of *size* n.

Proposition 2.2.3 *For every integer $n \geq 5$ and every set $\{n_1, n_2, \ldots, n_t\}$ of t integers such that $1 \leq n_1 < n_2 < \cdots < n_t \leq \lceil n/2 \rceil$ and $\sum_{i=1}^{t} n_i = n$, the path Q_n can be decomposed into the matchings n_1K_2, n_2K_2, $\ldots$, n_tK_2.*

Proof. For $n \geq 5$, let $Q_n = (v_1, v_2, \ldots, v_{n+1})$ and let $e_i = (v_i, v_{i+1})$ for $1 \leq i \leq n$. Define a sequence $\mathcal{S}$ of the edges of Q_n as

$$\mathcal{S} = \begin{cases} (e_1, e_3, \ldots, e_{n-1}, e_2, e_4, \ldots, e_n) & \text{if } n \text{ is even} \\ (e_1, e_3, \ldots, e_n, e_2, e_4, \ldots, e_{n-1}) & \text{if } n \text{ is odd.} \end{cases}$$

Denote the sequence $\mathcal{S}$ by $(f_1, f_2, \ldots, f_n)$ and define a labeling ℓ of $\mathcal{S}$ by

$$\ell(f_i) = \begin{cases} t & \text{if } 1 \leq i \leq n_t \\ t-1 & \text{if } n_t + 1 \leq i \leq n_t + n_{t-1} \\ \vdots & \vdots \\ 1 & \text{if } n_t + n_{t-1} + \cdots + n_2 + 1 \leq i \leq n. \end{cases}$$

For every pair i, j of distinct integers with $1 \leq i, j \leq t$, if $\ell(f_i) = \ell(f_j)$, then f_i and f_j are not adjacent. Thus, for $1 \leq i \leq t$, the n_i edges labeled i form the matching n_iK_2 and so Q_n can be decomposed into the matchings n_1K_2, n_2K_2, $\ldots$, n_tK_2. ■

A *linear forest* is a forest every component of which is a path. Here, we are only concerned with linear forests with no isolated vertices. The following is a consequence of Proposition 2.2.3.

Corollary 2.2.4 *Let F_n be a linear forest of size $n \geq 5$. For every set $\{n_1, n_2, \ldots, n_t\}$ of t integers such that $1 \leq n_1 < n_2 < \cdots < n_t \leq \lceil n/2 \rceil$ and $\sum_{i=1}^{t} n_i = n$, the linear forest F_n can be decomposed into the matchings n_1K_2, n_2K_2, $\ldots$, n_tK_2.*

In the proof of Proposition 2.2.3, the decomposition $\{n_1K_2, n_2K_2, \ldots, n_tK_2\}$ of Q_n gives rise to an ascending *sequence* of Q_n which may not be an ascending *chain* since it is possible that $\{n_1, n_2, \ldots, n_t\} \neq \{1, 2, \ldots, t\}$. For example, Figure 2.1 shows such decompositions for Q_{12} and Q_{13}.

- ⋆ If $n = 12$, then $\lceil n/2 \rceil = 6$.

  For the set $\{1, 2, 3, 6\}$, the path Q_{12} can be decomposed into the matchings K_2, $2K_2$, $3K_2$, $6K_2$. For the set $\{3, 4, 5\}$, the path Q_{12} can be decomposed into the matchings $3K_2$, $4K_2$, $5K_2$.

- ⋆ If $n = 13$, then $\lceil n/2 \rceil = 7$.

  For the set $\{1, 2, 3, 7\}$, the path Q_{13} can be decomposed into the matchings K_2, $2K_2$, $3K_2$, $7K_2$. For the set $\{1, 3, 4, 5\}$, the path Q_{13} can be decomposed into the matchings K_2, $3K_2$, $4K_2$, $5K_2$.

Each of these decompositions gives rise to an ascending sequence (not an ascending chain) in Q_{12} or Q_{13}.

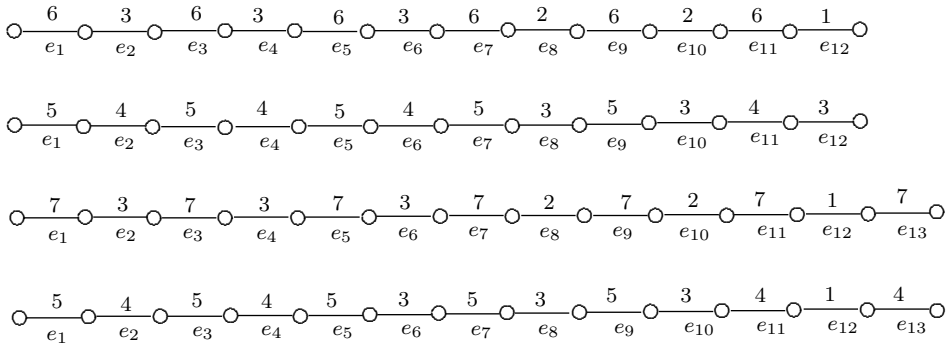


Figure 2.1: Illutrating the proof of Proposition 2.2.3 for $n = 12, 13$

The following corollary is a consequence of Propositions 2.2.1 and 2.2.3.

Corollary 2.2.5 *Every path of size 5 or more has a decomposition into distinct matchings.*

Let G be a graph without isolated vertices. For a proper subset X of $E(G)$, let $G - X$ be the subgraph of G induced by $E(G) - X$. Thus, $G - X$ has no isolated vertices. In particular, for each edge e in G, the graph $G - e$ is the subgraph of G induced by $E(G) - \{e\}$.

Observation 2.2.6 *Let H and G be two graphs such that H is isomorphic to a subgraph G' of G.*

(i) For each $e \in E(H)$, there is $f \in E(G') \subseteq E(G)$ such that

$$H - e \cong G' - f \subseteq G - f.$$

(ii) For each $f \in E(G') \subseteq E(G)$, there is $e \in E(H)$ such that

$$H - e \cong G' - f \subseteq G - f.$$

The following result gives a relationship between two paths whose sizes differ by 1.

Theorem 2.2.7 *For each integer $m \geq 3$, either*

$$AR(P_{m+1}) = AR(P_m) \text{ or } AR(P_{m+1}) = AR(P_m) + 1.$$

Proof. By Observation 2.1.2, $AR(P_m) \leq AR(P_{m+1})$ for each integer $m \geq 3$. Thus, it remains to show that $AR(P_m) \geq AR(P_{m+1}) - 1$. Let

$$P_{m+1} = (v_1, v_2, \ldots, v_{m+1})$$

and let $P_m = P_{m+1} - v_{m+1}$. Suppose that $AR(P_{m+1}) = t$. Let c be a red-blue coloring of P_m such that $AR_c(P_m) = AR(P_m)$. Then c can be extended to a red-blue coloring c' of P_{m+1} by assigning the color red, say, to the edge $e = v_m v_{m+1}$. Since $AR(P_{m+1}) = t$, there is a Ramsey chain $R = (G_1, G_2, \ldots, G_t)$ of length t in P_{m+1} with respect to c'. If $e \notin E(R)$, then R is a Ramsey chain of length t in P_m and so $AR(P_m) = AR_c(P_m) \geq t = AR(P_{m+1})$. This gives the desired result. Hence, we may assume that $e \in E(R)$ and so $e \in E(G_i)$ for some integer i where $1 \leq i \leq t$. If $e \in E(G_t)$, then $(G_1, G_2, \ldots, G_{t-1})$ is a Ramsey chain of length $t - 1$ in P_m and so $AR(P_m) = AR_c(P_m) \geq t - 1 = AR(P_{m+1}) - 1$. Thus, we may assume that $e \in E(G_i)$ where $1 \leq i \leq t - 1$.

⋆ First, suppose that $e \in E(G_1) \cup E(G_2)$. Let $f_2 \in E(G_2)$. If $e \in E(G_2)$, let $f_2 = e$. By Observation 2.2.6(i), there exists $f_3 \in E(G_3)$ such that $G_2 - f_2$ is isomorphic to a subgraph of $G_3 - f_3$. Next, for $f_3 \in E(G_3)$, there is $f_4 \in E(G_4)$ such that $G_3 - f_3$ is isomorphic to a subgraph of $G_4 - f_4$ by Observation 2.2.6(i) again. Continuing this procedure, we see that there is $f_i \in E(G_i)$ where $2 \leq i \leq t$ such that $G_i - f_i$ is isomorphic to a subgraph of $G_{i+1} - f_{i+1}$ for $2 \leq i \leq t-1$. Thus, $(G_2 - f_2, G_3 - f_3, \ldots, G_t - f_t)$ is a Ramsey chain of length $t-1$ in P_m. Hence, $AR(P_m) = AR_c(P_m) \geq t - 1 = AR(P_{m+1}) - 1$ if $e \in E(G_1) \cup E(G_2)$.

⋆ Next, suppose that $e \in E(G_p)$ where $3 \leq p \leq t-1$. Let $f_p = e$. By Observation 2.2.6(i), there is $f_i \in E(G_i)$, where $p+1 \leq i \leq t$, such that $G_i - f_i$ is isomorphic to a subgraph of $G_{i+1} - f_{i+1}$ for $p \leq i \leq t-1$. This produces a Ramsey chain

$$R_1 = (G_p - f_p, G_{p+1} - f_{p+1}, \ldots, G_t - f_t) \text{ in } P_m. \tag{2.1}$$

We now consider two possibilities, according to whether $G_{p-1} \cong G_p - f_p$ or $G_{p-1} \ncong G_p - f_p$.

- Assume first that $G_{p-1} \cong G_p - f_p$. If $p = 3$, then $G_2 - f_2 = K_2 \subseteq G_3 - f_3$. If $p \geq 4$, then, by Observation 2.2.6(i), there is $f_i \in E(G_{i-1})$ for $3 \leq i \leq p-1$ such that $G_i - f_i$ is isomorphic to a subgraph of $G_{i+1} - f_{i+1}$ for $2 \leq i \leq p-1$. This produces a Ramsey chain $R_2 = (G_2 - f_2, G_3 - f_3, \ldots, G_p - f_p)$ in P_m. Then R_2 and R_1 in (2.1) give rise to a Ramsey chain $(G_2 - f_2, G_3 - f_3, \ldots, G_t - f_t)$ of length $t-1$ in P_m.
- Assume next that $G_{p-1} \ncong G_p - f_p$. Thus, $G_{p-1} \cong G_p - f$ for some $f \in E(G_p - f_p)$. Since f_p is an edge of $G_p - f$, there is $f_{p-1} \in E(G_{p-1})$ such that $G_{p-1} - f_{p-1} \cong (G_p - f) - f_p$. By Observation 2.2.6(ii), there is $f_{p-2} \in G_{p-2}$ such that $G_{p-2} - f_{p-2}$ is isomorphic to a subgraph of $G_{p-1} - f_{p-1}$. Continuing this procedure, we see that for each integer i with $2 \leq i \leq p-1$, there is $f_i \in E(G_i)$ such that $G_i - f_i$ is isomorphic to a subgraph of $G_{i+1} - f_{i+1}$ for $2 \leq i \leq p-1$. This produces a Ramsey chain $R_2 = (G_2 - f_2, G_3 - f_3, \ldots, G_p - f_p)$ in P_m. Again, R_2 and R_1 in (2.1) give rise to a Ramsey chain $(G_2 - f_2, G_3 - f_3, \ldots, G_t - f_t)$ of length $t-1$ in P_m.

Hence, $AR(P_m) = AR_c(P_m) \geq t - 1 = AR(P_{m+1}) - 1$ if $e \in E(G_p)$ where $3 \leq p \leq t - 1$. ∎

The following is a consequence of Theorem 2.2.7.

Corollary 2.2.8 $\{AR(P_m)\}_{m=2}^{\infty} = \{1, 2, 3, \ldots\}$.

A logical question that now arises concerns the relationship of the Ramsey indexes of paths and cycles of the same size. In this connection, we have the following result.

Proposition 2.2.9 *For each integer $m \geq 3$, $AR(P_{m+1}) \leq AR(C_m)$.*

Proof. Let $AR(P_{m+1}) = k$. We claim that $AR(C_m) \geq k$. To verify this, we show that there is a Ramsey chain of length k in C_m for every red-blue coloring of C_m. Let c be a red-blue coloring of $C_m = (u_1, u_2, \ldots, u_m, u_{m+1} = u_1)$, where $e_i = u_i u_{i+1}$ for $1 \leq i \leq m$, so labeled such that $c(e_1)$ is red and $c(e_m)$ is blue. Let $P_{m+1} = (v_1, v_2, \ldots, v_m, v_{m+1})$ where $f_i = v_i v_{i+1}$ for $1 \leq i \leq m$. Define a red-blue coloring c' of P_{m+1} by $c'(f_i) = c(e_i)$ for $1 \leq i \leq m$. Since $AR(P_{m+1}) = k$, there is a Ramsey chain $R' = (G'_1, G'_2, \ldots, G'_k)$ of length k in P_{m+1}. Since $c'(f_1) = c(e_1) \neq c(e_m) = c'(f_m)$, it follows that f_1 and f_m do not belong to a same link in R' (if f_1 and f_m both belong to $E(R')$). Replacing f_i by e_i for $1 \leq i \leq m$, the monochromatic subgraph G'_j in P_{m+1} gives rise to a monochromatic subgraph G_j in C_m for $1 \leq j \leq k$. Then $R = (G_1, G_2, \ldots, G_k)$ is a Ramsey chain of length k in C_m. Therefore, $AR(C_m) \geq k = AR(P_{m+1})$. ∎

This gives rise to the following question:

Is $AR(P_{m+1}) = AR(C_m)$ for every integer $m \geq 3$?

We first answer this question when $3 \leq m \leq 20$. We write $k(G)$ for the number of components in a graph G. The following observation is a consequence of Observation 2.1.4 and Proposition 2.1.5.

Observation 2.2.10 *Let c be a 2-edge coloring of the path $H = Q_m$ of size $m \geq 3$. For $i = 1, 2$, let H_i be the subgraph of size m_i in H induced by the set of edges colored i. Then H_i is a linear forest with $k(H_i) \geq 1$ components for $i = 1, 2$ and $|k(H_1) - k(H_2)| \leq 1$.*

(a) *If* $m_i \geq 3$ *where* $i \in \{1,2\}$, *then* $2K_2 \subseteq H_i$. *Thus, if* $m \geq 6$ *such that* $2K_2 \subseteq H_i$ *and* $m_j \geq 3$, *where* $\{i,j\} = \{1,2\}$, *then* $AR_c(C_m) \geq 3$.

(b) *If* $m_1, m_2 \geq 3$, *then* $AR_c(Q_m) \geq 3$.

(c) *If* $m_i \geq 5$ *for some* $i \in \{1,2\}$, *then* $AR_c(Q_m) \geq 3$.

(d) *If* $m = 9$, *then* $AR(Q_9) = 3$.

The following lemma is useful for the next result.

Lemma 2.2.11 *Let* c *be a red-blue coloring of a path* $Q_m = (e_1, e_2, \ldots, e_m)$ *of size* $m \geq 3$. *If* $c(e_1) \neq c(e_m)$, *then there is a Ramsey chain of length* $AR(C_m)$ *in* Q_m.

Proof. Let $C_m = (u_1, u_2, \ldots, u_m, u_{m+1} = u_1)$, where $f_i = u_i u_{i+1}$ for $1 \leq i \leq m$ and $AR(C_m) = k$. Define a red-blue coloring c' of C_m by $c'(f_i) = c(e_i)$ for $1 \leq i \leq m$. Since $AR(C_m) = k$, there is a Ramsey chain $R' = (G'_1, G'_2, \ldots, G'_k)$ of length k in C_m. Since $c'(f_1) = c(e_1) \neq c(e_m) = c'(f_m)$, it follows that f_1 and f_m belong to different links in R'. Replacing f_i by e_i for $1 \leq i \leq m$, the monochromatic subgraph G'_j in C_m gives rise a monochromatic subgraph G_j in P_{m+1} for $1 \leq j \leq k$. Thus, $R = (G_1, G_2, \ldots, G_k)$ is a Ramsey chain of length k in Q_m. ∎

With the aid of Observations 2.1.4, 2.2.10, and Lemma 2.2.11, the following result can be proved with an extensive case-by-case analysis.

Proposition 2.2.12 $AR(P_{m+1}) = AR(C_m) = \begin{cases} 2 & \text{if } 3 \leq m \leq 5 \\ 3 & \text{if } 6 \leq m \leq 9 \\ 4 & \text{if } 10 \leq m \leq 14 \\ 5 & \text{if } 15 \leq m \leq 20. \end{cases}$

Proof. Let $Q_m = P_{m+1}$ be the path of size $m \geq 3$. By Observation 2.1.4(b), if $3 \leq m \leq 5$, then $AR(Q_m) = 2$. By Observation 2.2.10(d), $AR(Q_9) = 3$. We show that $AR(Q_m) = 3$ for $6 \leq m \leq 8$. Since $m \leq 8$, it follows that $AR(Q_m) \leq 3$ by Observation 2.1.1. It remains to show that $AR(Q_m) \geq 3$; that is, every 2-edge coloring of Q_m produces a Ramsey chain of length 3 in Q_m. By Theorem 2.2.7, it suffices to show that $AR(Q_6) \geq 3$. Let c be a 2-edge coloring of $H = Q_6$ using

the colors 1 and 2. For $i = 1, 2$, let H_i be the subgraph of size m_i induced by the edges colored i in H where $m_1 \leq m_2 \leq 5$. If $m_1 \geq 3$ or $m_2 = 5$, then $AR(Q_6) \geq 3$ by Observation 2.2.10. Thus, we may assume that $m_1 \leq 2$ and $m_2 \leq 4$. Since $m_1 + m_2 = 6$, it follows that $(m_1, m_2) = (2, 4)$. Since $AR(C_6) = 3$, we may assume (by Lemma 2.2.11) that the initial and terminal edges of H are colored the same. Hence, $|k(H_1) - k(H_2)| = 1$. If $H_1 = Q_2$, then $H_2 \in \{Q_3 + K_2, 2Q_2\}$; while if $H_1 = 2K_2$, then $H_2 \in \{Q_4, Q_2 + 2K_2\}$. In either situation, there is a Ramsey chain of length 3 in H. Thus, $AR(Q_6) = 3$. Therefore, $AR(Q_m) = 3$ for $6 \leq m \leq 9$.

Next, we show that $AR(Q_{10}) = 4$. Since $AR(Q_{10}) \leq 4$ by Observation 2.1.1, it remains to show that $AR(Q_{10}) \geq 4$. Let c be a 2-edge coloring of $H = Q_{10}$ using the colors 1 and 2. For $i = 1, 2$, let H_i be the subgraph of size m_i induced by the edges colored i in H where $m_1 \leq m_2$. Thus, $1 \leq m_1 \leq 5$ and $2 \leq m_2 \leq 9$ where $m_1 + m_2 = 10$. Since $AR(C_{10}) = 4$, we may assume (by Lemma 2.2.11) that the initial and terminal edges of H are colored the same. Thus, $|k(H_1) - k(H_2)| = 1$. We show that there is a Ramsey chain $R_c : G_1, G_2, G_3, G_4$ of length 4 in H with respect to c. Since the size of Q_{10} is 10, any such sequence results in a decomposition of Q_{10}. We now consider the possible ordered pairs (m_1, m_2).

- ⋆ If $(m_1, m_2) = (1, 9)$, then $G_1 = H_1 = K_2$ and H_2 is a linear forest of size 9. Since $2 + 3 + 4 = 9$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $2K_2, 3K_2, 4K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2$ is a Ramsey chain of length 4 in H

- ⋆ If $(m_1, m_2) = (2, 8)$, then $G_2 = H_1 \in \{Q_2, 2K_2\}$.

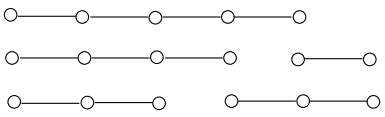
  - ◦ If $H_1 = 2K_2$, then H_2 is a linear forest of size 8. Since $1 + 3 + 4 = 8$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $K_2, 3K_2, 4K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2$ is a Ramsey chain of length 4 in H.
  - ◦ If $H_1 = Q_2$, then $H_2 \in \{Q_x + Q_y : 1 \leq x < y \leq 7, x + y = 8\}$. Hence, H_2 can decomposed into $K_2, Q_2 + K_2, Q_3 + K_2$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, Q_3 + K_2$ is a Ramsey chain of length 4 in H.

- ⋆ If $(m_1, m_2) = (3, 7)$, then H_1 can be decomposed into K_2 and $2K_2$. Since $3 + 4 = 7$ and H_2 is a linear forest of size 7, it follows by Corollary 2.2.4 that H_2 can be decomposed into $3K_2$ and $4K_4$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2$ is a Ramsey chain of length 4 in H.

⋆ If $(m_1, m_2) = (4, 6)$, then $H_1 \in \{Q_4, Q_3 + K_2, 2Q_2, Q_2 + 2K_2, 4K_2\}$.

- If $H_1 = Q_2 + 2K_2$ or $H_1 = 4K_2$, then H_2 is a linear forest of size 6. Since $1 + 2 + 3 = 6$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $K_2, 2K_2, 3K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, H_1$ is a Ramsey chain of length 4 in H.

  Hence, we may assume that $H_1 \in \{Q_4, Q_3 + K_2, 2Q_2\}$.



- If $H_1 = Q_4$, then $H_2 \in \{Q_x + Q_y : 1 \leq x < y \leq 5, x + y = 6\}$. Then H_1 can be decomposed into K_2 and $Q_2 + K_2$ and H_2 can be decomposed into Q_2 and $G_4 \in \{Q_3 + K_2, Q_4\}$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, G_4$ is a Ramsey chain of length 4 in H.
- If $H_1 \in \{Q_3 + K_2, 2Q_2\}$, then $k(H_2) \in \{1, 3\}$ and so

$$H_2 \in \{Q_6, Q_4 + 2K_2, Q_3 + Q_2 + K_2, 3Q_2\}.$$

  If $H_2 = Q_6$, then H_2 can be decomposed into $K_2, Q_2, Q_2 + K_2$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, H_1 = Q_3 + K_2$ is a Ramsey chain of length 4 in H.
  If $H_2 \in \{Q_4 + 2K_2, Q_3 + Q_2 + K_2, 3Q_2\}$, then H_2 can be decomposed into Q_2 and $G_4 \in \{Q_2 + 2K_2, 2Q_2\}$. Since $H_1 \in \{Q_3 + K_2, 2Q_2\}$ can be decomposed into K_2 and $Q_2 + K_2$, it follows that $R_c : K_2, Q_2, Q_2 + K_2, G_4$ is a Ramsey chain of length 4 in H.

⋆ If $(m_1, m_2) = (5, 5)$, then one of H_1 and H_2 is a graph in the set

$$\{Q_5, Q_4 + K_2, Q_3 + Q_2, Q_3 + 2K_2, 2Q_2 + K_2, Q_2 + 3K_2, 5K_2\}.$$

Then H_1 can be decomposed into K_2 and $G_4 \in \{Q_2 + 2K_2, 4K_2\}$. Since H_2 is a linear forest of size 5 and $2 + 3 = 5$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $2K_2$ and $3K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, G_4$ is a Ramsey chain of length 4 in H.

Therefore, $AR(Q_{10}) = 4$. First, $AR(Q_{14}) \geq AR(Q_{10}) = 4$ by Observation 2.1.2. Since $m = 14 < \binom{5+1}{2}$, it follows that $AR(Q_{14}) \leq 4$ and so $AR(Q_{14}) = 4$. Therefore, $AR(Q_m) = 4$ for $10 \leq m \leq 14$.

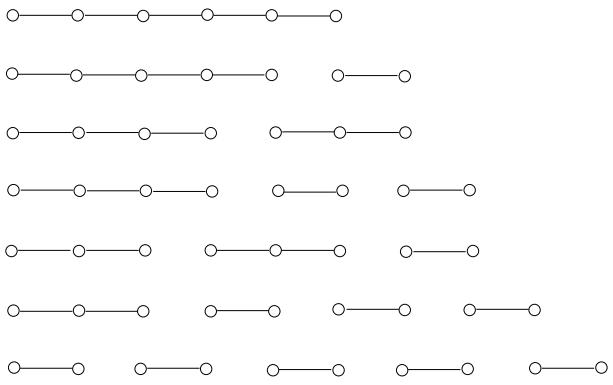


Figure 2.2: Linear forests of size 5

It remains to show that $AR(Q_m) = 5$ for $15 \leq m \leq 20$. Since $m < \binom{6+1}{2} = 21$ for $15 \leq m \leq 20$, it follows that $AR(Q_m) \leq 5$ for $15 \leq m \leq 20$. To show that $AR(Q_m) \geq 5$ for $15 \leq m \leq 20$, it suffices to show that $AR(Q_{15}) = 5$. We saw that the size of Q_{15} is $15 = \binom{5+1}{2}$ and so $AR(Q_{15}) \leq 5$. Next, we show that $AR(Q_{15}) \geq 5$. Let c be a 2-edge coloring of $H = Q_{15}$ using the colors 1 and 2, where m_i is the number of the edges in the subgraph H_i of H colored i for $i = 1, 2$ and $m_1 \leq m_2$. Thus, $1 \leq m_1 \leq 7$ and $2 \leq m_2 \leq 14$ where $m_1 + m_2 = 15$. We show that there is a Ramsey chain $R_c : G_1, G_2, G_3, G_4, G_5$ of length 5 in H with respect to c. Since the size of Q_{15} is 15, it follows that $\{G_1, G_2, G_3, G_4, G_5\}$ is a decomposition of Q_{15}. Since $AR(C_{15}) = 5$, we may assume (by Lemma 2.2.11) that the initial and terminal edges of H are colored the same. Thus, $|k(H_1) - k(H_2)| = 1$. We show that there is a Ramsey chain $R_c : G_1, G_2, G_3, G_4, G_5$ of length 5 in H with respect to c. Since the size of Q_{15} is 10, any such sequence results in a decomposition of Q_{15}. We now consider the possible ordered pairs (m_1, m_2).

- ⋆ If $(m_1, m_2) = (1, 14)$, then $G_1 = H_1 = K_2$ and H_2 is a linear forest of size 14. Since $2 + 3 + 4 + 5 = 14$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $2K_2, 3K_2, 4K_2, 5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, 5K_2$ is a Ramsey chain of length 5 in H.

- ⋆ If $(m_1, m_2) = (2, 13)$, then $G_2 = H_1 \in \{Q_2, 2K_2\}$.

  - ◦ First, suppose that $H_1 = 2K_2$. Since H_2 is a linear forest of size 13 and $1 + 3 + 4 + 5 = 13$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $K_2, 3K_2, 4K_2, 5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, 5K_2$

is a Ramsey chain of length 5 in H.

- If $H_1 = Q_2$, then $H_2 \in \{Q_x + Q_y : 1 \leq x < y \leq 7, x + y = 13\}$. Since $y \geq 7$, it follows that H_2 can decomposed into $G_1 = K_2$, $G_3 = Q_2 + K_2$, $G_4 = Q_3 + K_2$, and $G_5 = Q_4 + K_2$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, Q_3 + K_2, Q_4 + K_2$ is a Ramsey chain of length 5 in H. Such decompositions are illustrated in Figure 2.3 for $H_2 = Q_1 + Q_{12}$ and $H_2 = Q_6 + Q_7$ where an edge labeled i belongs to the subgraph G_i for $i \in \{1, 3, 4, 5\}$.

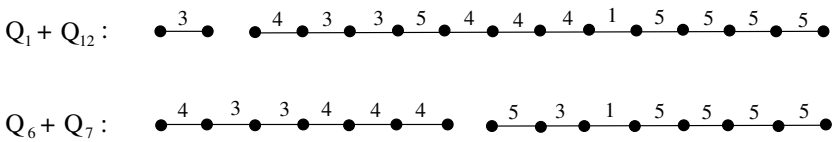


Figure 2.3: Decompositions of $H_2 = Q_1 + Q_{12}$ and $H_2 = Q_6 + Q_7$

⋆ If $(m_1, m_2) = (3, 12)$, then H_1 can be decomposed into K_2 and $2K_2$. Since $3 + 4 + 5 = 12$ and H_2 is a linear forest of size 12, it follows by Corollary 2.2.4 that H_2 can be decomposed into $3K_2, 4K_4, 5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, 5K_2$ is a Ramsey chain of length 5 in H.

⋆ If $(m_1, m_2) = (4, 11)$, then $H_1 \in \{Q_4, Q_3 + K_2, 2Q_2, Q_2 + 2K_2, 4K_2\}$.

- If $H_1 = 4K_2$, then H_2 is a linear forest of size 11. Since $1 + 2 + 3 + 5 = 11$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $K_2, 2K_2, 3K_2, 5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, H_1$ is a Ramsey chain of length 4 in H.
- If $H_1 \in \{Q_2 + 2K_2, Q_3 + K_2\}$, then H_1 can be decomposed into K_2 and $3K_2$. Since H_2 is a linear forest of size 11 and $2+4+5 = 11$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $2K_2, 4K_2, 5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, 5K_2$ is a Ramsey chain of length 5 in H.
- If $H_1 = Q_4$, then H_1 can be decomposed into $G_1 = K_2$ and $G_3 = Q_2 + K_2$ where $k(H_2) = 2$. Hence, $H_2 \in \{Q_x + Q_y : 1 \leq x < y \leq 10, x + y = 11\}$ and so H_2 can be decomposed into $G_2 = Q_2$, $G_4 = Q_2 + 2K_2$, and $G_5 \in \{Q_2 + 3K_2, 2Q_2 + K_2, Q_3 + 2K_2, Q_5\}$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, Q_2 + 2K_2, Q_4 + K_2, G_5$ is a Ramsey chain of length 5 in H. Such decompositions are illustrated in Figure 2.4 for $H_2 \in \{Q_1 + Q_{10}, Q_4 +$

$Q_7, Q_5 + Q_6\}$ where an edge labeled i belongs to the subgraph G_i for $i \in \{2, 4, 5\}$.

- If $H_1 = 2Q_2$, then H_1 can be decomposed into $G_1 = K_2$ and $G_3 = Q_2 + K_2$ and $k(H_2) \in \{1, 3\}$. Thus, either $H_2 = Q_{11}$ or $H_2 = Q_{\ell_1} + Q_{\ell_2} + Q_{\ell_2}$ where $\ell_1 \leq \ell_2 \leq \ell_3$. Since $\ell_1 + \ell_2 + \ell_3 = 11$, it follows that $\ell_3 \geq 4$ and $\ell_2 + \ell_3 \geq 8$. Hence, H_2 can be decomposed into $G_2 = Q_2$, $G_4 = Q_2 + 2K_2$, and $G_5 \in \{Q_2 + 3K_2, 2Q_2 + K_2, Q_3 + 2K_2, Q_4 + K_2, Q_5\}$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, Q_2 + 2K_2, G_5$ is a Ramsey chain of length 5 in H. Such decompositions are illustrated in Figure 2.4 for $H_2 \in \{Q_{11}, Q_1 + Q_5 + Q_5, Q_3 + 2Q_4\}$ where an edge labeled i belongs to the subgraph G_i for $i \in \{2, 4, 5\}$.

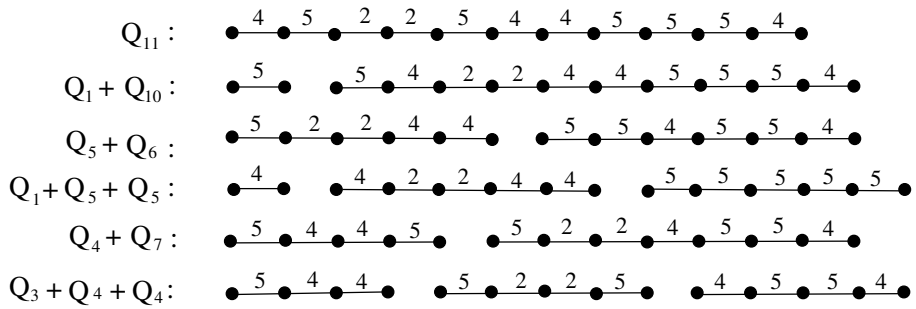


Figure 2.4: Decompositions of some linear forests of size 11

- If $(m_1, m_2) = (5, 10)$, then H_1 is a linear forest of size 5 and H_2 is a linear forest of size 10. Since $2+3 = 5$ and $1+4+5 = 10$, it follows by Corollary 2.2.4 that H_1 can be decomposed into $2K_2$ and $3K_2$ and H_2 can be decomposed into $K_2, 4K_2$ and $5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, 5K_2$ is a Ramsey chain of length 5 in H.

- If $(m_1, m_2) = (6, 9)$, then H_1 is a linear forest of size 6 and H_2 is a linear forest of size 9. Since $1 + 2 + 3 = 6$ and $4 + 5 = 9$, it follows by Corollary 2.2.4 that H_1 can be decomposed into $K_2, 2K_2$, $3K_2$ and that H_2 can be decomposed into $4K_2$, $5K_2$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, 5K_2$ is a Ramsey chain of length 5 in H.

- If $(m_1, m_2) = (7, 8)$, then H_1 is a linear forest of size 7 and H_2 is a linear forest of size 8. Since $1 + 2 + 4 = 7$, it follows by Corollary 2.2.4 that H_1 can

be decomposed into $K_2, 2K_2$ and $4K_2$ Furthermore, H_2 can be decomposed into $3K_2$ and $G_5 \in \{Q_2 + 3K_2, 5K_2\}$. Thus, $R_c : K_2, 2K_2, 3K_2, 4K_2, G_5$ is a Ramsey chain of length 5 in H.

Hence, $AR(Q_{15}) = 5$. Since $AR(Q_{20}) \geq AR(Q_{15}) = 5$ by Observation 2.1.2, it follows that $AR(Q_{20}) = 5$. Therefore, $AR(Q_m) = 5$ for $15 \leq m \leq 20$. ∎

This result suggests the following conjecture.

Conjecture 2.2.13 *If $m \geq 3$ is an integer, then $AR(P_{m+1}) = AR(C_m)$. Equivalently, if $k \geq 2$ is an integer, then $AR(P_{m+1}) = k$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$.*

2.3 Ramsey Indexes of Paths and Cycles

In this section, we verify Conjecture 2.2.13. We begin here with an observation.

Observation 2.3.1 *If $k \geq 7$ is an integer, then $k \leq \left\lceil \frac{\binom{k+1}{2}/2}{2} \right\rceil$ where strict inequality holds for $k \geq 8$.*

Proof. If $k = 7$, then $k = \left\lceil \frac{\binom{k+1}{2}/2}{2} \right\rceil$. If $k \geq 8$, then $k(k-7) > 0$. Thus, $8k < k^2 + k$ and so $4k < \binom{k+1}{2}$. Hence, $k < \frac{\binom{k+1}{2}/2}{2} \leq \left\lceil \frac{\binom{k+1}{2}/2}{2} \right\rceil$. ∎

Proposition 2.3.2 *Let $m = \binom{k+1}{2}$ for some integer $k \geq 4$. For every two positive integers m_1 and m_2 with $m = m_1 + m_2$ and $m_1, m_2 \notin \{2, 4\}$, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into two subsets*

$$A = \{a_1, a_2, \ldots, a_{k_1}\} \text{ and } B = \{b_1, b_2, \ldots, b_{k_2}\},$$

where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1}$, and $b_1 < b_2 < \cdots < b_{k_2}$ such that

$$a_{k_1} \leq \left\lceil \tfrac{m_1}{2} \right\rceil,\ b_{k_2} \leq \left\lceil \tfrac{m_2}{2} \right\rceil,\ \textstyle\sum_{i=1}^{k_1} a_i = m_1, \text{ and } \sum_{i=1}^{k_2} b_i = m_2.$$

Proof. We proceed by induction on $k \geq 4$. It is routine to show that the statement is true for $k = 4, 5$. If $k = 6$, then $m = \binom{k+1}{2} = 21$. Let m_1 and m_2 be positive integers such that $m_1 + m_2 = 21$ and $m_1, m_2 \notin \{2, 4\}$. We may assume that $m_1 \leq m_2$. Thus, $m_1 \leq 10$ and $m_1 \notin \{2, 4\}$. Partitions $\{A, B\}$ of $[6]$ with the desired properties are listed below for all possible such pairs (m_1, m_2):

For $m_1 = 1$, let $A = \{1\}$ and $B = \{2, 3, 4, 5, 6\}$.
For $m_1 = 3$, let $A = \{1, 2\}$ and $B = \{3, 4, 5, 6\}$.
For $m_1 = 5$, let $A = \{2, 3\}$ and $B = \{1, 4, 5, 6\}$.
For $m_1 = 6$, let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$.
For $m_1 = 7$, let $A = \{1, 2, 4\}$ and $B = \{3, 5, 6\}$.
For $m_1 = 8$, let $A = \{1, 3, 4\}$ and $B = \{2, 5, 6\}$.
For $m_1 = 9$, let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$.
For $m_1 = 10$, let $A = \{1, 4, 5\}$ and $B = \{2, 3, 6\}$.

Hence, the statement is true for $k = 6$. Assume that the statement is true for some integer $k \geq 6$. We show that the statement is true for $k + 1$. For an integer $m = \binom{k+2}{2}$ where $k \geq 6$, let m_1 and m_2 be two positive integers such that $m = m_1 + m_2$ and $m_1, m_2 \notin \{2, 4\}$. We may assume that $m_1 \leq m_2$. Thus, $m_2 \geq m/2$. Let $m_1' = m_1$ and $m_2' = m_2 - (k + 1)$. Thus, $m_1' = m_1 \notin \{2, 4\}$ and $m_1' + m_2' = \binom{k+1}{2}$. Since $k \geq 6$, it follows that $k(k - 1) \geq 30 > 22$ and so $(k^2 + 3k + 2) - 4k - 4 > 20$. Thus, $\frac{\binom{k+2}{2}}{2} - k - 1 \geq 5$. Since

$$m_2' = m_2 - (k + 1) \geq \tfrac{m}{2} - (k + 1) = \tfrac{\binom{k+2}{2}}{2} - (k + 1) \geq 5,$$

it follows that $m_2' \notin \{2, 4\}$. By the induction hypothesis, it follows that $[k]$ can be partitioned into two sets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$,

$$a_1 < a_2 < \cdots < a_{k_1} \leq \left\lceil \tfrac{m_1'}{2} \right\rceil = \left\lceil \tfrac{m_1}{2} \right\rceil,$$

and

$$b_1 < b_2 < \cdots < b_{k_2} \leq \left\lceil \tfrac{m_2'}{2} \right\rceil = \left\lceil \tfrac{m_2 - (k+1)}{2} \right\rceil$$

such that $\sum_{i=1}^{k_1} a_i = m_1' = m_1$ and $\sum_{i=1}^{k_2} b_i = m_2' = m_2 - (k + 1)$. Now, let

$$X = A = \{a_1, a_2, \ldots, a_{k_1}\} \text{ and } Y = B \cup \{k+1\} = \{b_1, b_2, \ldots, b_{k_2}, k+1\}.$$

Then $\{X, Y\}$ forms a partition of $[k+1]$ with

$$|X| + |Y| = k_1 + (k_2 + 1) = k + 1.$$

Since $m_2 \geq \frac{m}{2} = \frac{\binom{k+2}{2}}{2}$ and $k+1 \geq 7$, it follows by Observation 2.3.1 that $k+1 \leq \left\lceil \frac{\binom{k+2}{2}/2}{2} \right\rceil \leq \left\lceil \frac{m_2}{2} \right\rceil$. Furthermore,

$$\sum_{x \in X} x = m_1' = m_1 \text{ and } \sum_{y \in Y} y = m_2' + (k+1) = m_2.$$

Therefore, $\{X, Y\}$ is a partition of $[k+1]$ with the desired property. ∎

Theorem 2.3.3 *Let $m = \binom{k+1}{2}$ for some integer $k \geq 6$ and let c be a 2-edge coloring of $H = Q_m$ using the colors 1 and 2. For $i = 1, 2$, let H_i be the subgraph of H induced by the edges of H colored i where H_i has size m_i and $1 \leq m_1 \leq m_2$. If $m_1 \notin \{2, 4\}$, then $R_c : K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in Q_m.*

Proof. We consider three cases depending on whether $m_1 = 1$, $m_1 = 3$, or $m_1 \geq 5$.

Case 1. $m_1 = 1$. Let $G_1 = H_1 = K_2$. Since $k \leq \left\lfloor \frac{m-1}{2} \right\rfloor$ and $2 + 3 + \cdots + k = \binom{k+1}{2} - 1$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $2K_2, 3K_2, \ldots, kK_2$. Thus, $R_c : K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H.

Case 2. $m_1 = 3$. The subgraph H_1 can be decomposed into K_2 and $2K_2$. Since $k \leq \left\lfloor \frac{m-3}{2} \right\rfloor$ and $3 + \cdots + k = \binom{k+1}{2} - 3$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $3K_2, 4K_2, \ldots, kK_2$. Thus, $R_c : K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H.

Case 3. $m_1 \geq 5$. By Proposition 2.3.2, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into two sets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \left\lceil \frac{m_1}{2} \right\rceil$ and $b_1 < b_2 < \cdots < b_{k_2} \leq \left\lceil \frac{m_2}{2} \right\rceil$ such that $\sum_{i=1}^{k_1} a_i = m_1$ and $\sum_{i=1}^{k_2} b_i = m_2$. Since $m_2 \geq m_1 \geq 5$, it follows by Corollary 2.2.4 that H_1 can be decomposed into the matchings a_1K_2, a_2K_2, …, $a_{k_1}K_2$ and H_2

can be decomposed into the matchings b_1K_2, b_2K_2, ..., $b_{k_2}K_2$. Consequently, $R_c : K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H. ∎

Next, we consider the case when $m_1 \in \{2, 4\}$. First, we present two preliminary results. For an ordering $s = (e_1, e_2, \ldots, e_m)$ of the edges of a graph G of size m and an edge labeling ℓ of G, define $\ell(s) = (\ell(e_1), \ell(e_2), \ldots, \ell(e_m))$.

Proposition 2.3.4 *Let F_m be a linear forest of size m with at most two components where $m = \binom{k+1}{2} - 2$ for some integer $k \geq 6$. Then F_m can be decomposed into $k-1$ subgraphs $G_1, G_3, \ldots, G_k$ where $G_1 = K_2$ and $G_i = Q_2 + (i-2)K_2$ for $3 \leq i \leq k$, namely*

$$(G_1, G_3, \ldots, G_k) = (K_2, Q_2 + K_2, Q_2 + 2K_2, \ldots, Q_2 + (k-2)K_2).$$

Proof. Since F_m has at most two components, either $F_m = Q_m$ or $F_m = Q_x + Q_y$ for positive integers x and y where $x + y = m$. We consider these two cases. In each case, an edge labeling ℓ of F_m is defined such that for each integer i with $1 \leq i \leq k$ and $i \neq 2$, there are i edges of F_m are labeled i and the edges labeled i induce the subgraph G_i where $G_1 = K_2$ and $G_i = Q_2 + (i-2)K_2$ for $3 \leq i \leq k$. Once such a labeling has been established, it follows that F_m is decomposed into $G_1, G_3, G_4, \ldots, G_k$. There are $k-2$ edge-disjoint copies of Q_2 in this decomposition.

Case 1. $F_m = Q_m$. Let $s = (e_1, e_2, \ldots, e_m)$ where e_i and e_{i+1} are adjacent for $1 \leq i \leq m-1$. Since $m = \binom{k+1}{2} - 2 = (2k-3) + \binom{k-1}{2} = (2k-3) + \sum_{i=1}^{k-2} i$, the path Q_m can be decomposed and expressed as the sequence Q_{2k-3}, Q_{k-2}, Q_{k-3}, ..., Q_1. Let the edges of Q_{2k-3} be labeled $(k, k, k-1, k-1, \cdots, 3, 3, 1)$ and the edges of Q_j $(1 \leq j \leq k-2)$ labeled $(k, k-1, \ldots, k-i+1)$. Thus, the edge labeling ℓ of Q_m is

$$\begin{aligned} \ell(s) \;=\; & (k, k, k-1, k-1, \cdots, 3, 3, 1, \\ & k, k-1, \ldots, 3, k, k-1, \ldots, 4, \ldots, k, k-1, k). \end{aligned} \tag{2.2}$$

This results in the desired sequence $G_1, G_3, G_4, \ldots, G_k$. For example, if $F_{19} = Q_{19}$ where $k = 6$, then $\ell(s) = (6, 6, 5, 5, 4, 4, 3, 3, 1,\ 6, 5, 4, 3, 6, 5, 4, 6, 5, 6)$.

Case 2. $F_m = Q_x + Q_y$ *where* $x + y = m$. Let $Q_x = (e_1, e_2, \ldots, e_x)$ and $Q_y = (f_1, f_2, \ldots, f_y)$. Then $s = (e_1, e_2, \ldots, e_x, f_1, f_2, \ldots, f_y)$ is an ordering of the

edges of $Q_x + Q_y$. If x is even, then define an edge labeling ℓ of G as described in (2.2). If x is odd, then define a labeling ℓ of $Q_x + Q_y$ by

$$\begin{aligned} \ell(s) &= (1, k, k, k-1, k-1, \cdots, 3, 3, \\ &\quad k, k-1, \ldots, 3, k, k-1, \ldots, 4, \ldots, k, k-1, k). \end{aligned} \tag{2.3}$$

This results in the desired sequence $G_1, G_3, G_4, \ldots, G_k$. For example, if $F_{19} = Q_7 + Q_{12}$ where $k = 6$, then $\ell(s) = (1, 6, 6, 5, 5, 4, 4,\ 3, 3, 6, 5, 4, 3, 6, 5, 4, 6, 5, 6)$. ■

Proposition 2.3.5 *Let F_m be a linear forest of size m with at most three components where $m = \binom{k+1}{2} - 4$ for some integer $k \geq 6$. Then F_m can be decomposed into $k-2$ subgraphs $G_2, G_4, G_5, \ldots, G_k$ where $G_2 = Q_2$ and $G_i = Q_2 + (i-2)K_2$ for $4 \leq i \leq k$, namely*

$$(G_2, G_4, \ldots, G_k) = (Q_2, Q_2 + 2K_2, Q_2 + 3K_2, \ldots, Q_2 + (k-2)K_2).$$

Proof. Since F_m has at most three components, $F_m = Q_m$, $F_m = Q_x + Q_y$ where $x + y = m$, or $F_m = Q_x + Q_y + Q_z$ where $x + y + z = m$. We consider these three cases. In each case, we define an edge labeling ℓ of F_m such that for each integer i with $2 \leq i \leq k$ and $i \neq 3$, there are i edges of F_m labeled i and the edges labeled i induce the subgraph G_i where $G_2 = Q_2$ and $G_i = Q_2 + (i-2)K_2$ for $4 \leq i \leq k$. Once such a labeling has been established, it follows that F_m can be decomposed into $G_2, G_4, G_5, \ldots, G_k$. There are $k-2$ edge-disjoint copies of Q_2 in this decomposition.

Case 1. $F_m = Q_m$. Let $s = (e_1, e_2, \ldots, e_m)$ where e_i and e_{i+1} are adjacent for $1 \leq i \leq m-1$. Since $m = \binom{k+1}{2} - 4 = 2(k-2) + (k-3) + \binom{k-2}{2} = 2(k-2) + (k-3) + + \sum_{i=1}^{k-3} i$, the path Q_m can be decomposed and expressed as the sequence Q_{2k-4}, Q_{k-3}, Q_{k-3}, Q_{k-4}, ..., Q_1. Let the edges of Q_{2k-4} be labeled $(k, k, k-1, k-1, \cdots, 4, 4, 2, 2)$ and the edges of Q_j $(1 \leq j \leq k-3)$ be labeled $(k, k-1, \ldots, k-i+1)$. Thus, the edge labeling ℓ of Q_m is

$$\begin{aligned} \ell(s) &= (k, k, k-1, k-1, \cdots, 4, 4, 2, 2, k, k-1, \ldots, 4, \\ &\quad k, k-1, \ldots, 4, k, k-1, \ldots, 5, \ldots, k, k-1, k). \end{aligned} \tag{2.4}$$

This results in the desired sequence $G_2, G_4, G_5, \ldots, G_k$. For example, if $F_{17} = Q_{17}$ where $k = 6$, then $\ell(s) = (6, 6, 5, 5, 4, 4, 2, 2, 6, 5, 4, 6, 5, 4, 6, 5, 6)$.

Case 2. $F_m = Q_x + Q_y$ *where* $x + y = m$. Let $Q_x = (e_1, e_2, \ldots, e_x)$ and $Q_y = (f_1, f_2, \ldots, f_y)$. Then $s = (e_1, e_2, \ldots, e_x, f_1, f_2, \ldots, f_y)$ is an ordering of the edges of $Q_x + Q_y$. If x is even, then define the edge labeling ℓ of $Q_x + Q_y$ as described in (2.4). If x is odd, then define the edge labeling ℓ of $Q_x + Q_y$ such that

$$\begin{aligned}\ell(s) &= (4, k, k, k-1, k-1, \cdots, 4, 4, 2, 2, k, k-1, \ldots, 4,\\ &\quad k, k-1, \ldots, 5, k, k-1, \ldots, 5, k, k-1, \ldots, 6, \ldots, k, k-1, k). \quad (2.5)\end{aligned}$$

This results in the desired sequence $G_2, G_4, G_5, \ldots, G_k$. For example, if $F_{17} = Q_5 + Q_{12}$ where $k = 6$, then $\ell(s) = (4, 6, 6, 5, 5,\ 4, 4, 2, 2, 6, 5, 4, 6, 5, 6, 5, 6)$.

Case 3. $F_m = Q_x + Q_y + Q_z$ *where* $x+y+z = m$. We may assume that $x \geq y \geq z$. Let $Q_x = (e_1, e_2, \ldots, e_x)$, $Q_y = (f_1, f_2, \ldots, f_y)$, and $Q_z = (g_1, g_2, \ldots, g_z)$. Then $s = (e_1, e_2, \ldots, e_x, f_1, f_2, \ldots, f_y, g_1, g_2, \ldots, g_z)$ is an ordering of the edges of $Q_x + Q_y + Q_z$. There are $k-2$ edge-disjoint copies of Q_2 in the desired decomposition of F_m. Since $x \geq y \geq z$ and $k \geq 6$, it follows that $x + y \geq \frac{2}{3}m = \frac{2}{3}\left(\binom{k+1}{2} - 4\right) = \frac{k^2+k-8}{3} \geq 2(k-1)+1$. If x is even, then define an edge labeling ℓ of s as described in (2.4). If x is odd, then define an edge labeling ℓ of s as described in (2.5). ∎

Theorem 2.3.6 *Let* $m = \binom{k+1}{2}$ *for some integer* $k \geq 6$ *and let* c *be a* 2*-edge coloring of* $H = Q_m$ *using the colors* 1 *and* 2. *For* $i = 1, 2$, *let* H_i *be the subgraph of* H *induced by edges of* H *colored* i *where* H_i *has size* m_i *and* $1 \leq m_1 \leq m_2$. *If* $m_1 \in \{2, 4\}$, *then there is a Ramsey chain of length* k *in* Q_m.

Proof. Let $k(H_i)$, $i = 1, 2$, be the number of components of H_i. Thus, $|k(H_1) - k(H_2)| \leq 1$. First, suppose that $m_1 = 2$. Then $G_2 = H_1 \in \{Q_2, 2K_2\}$.

- First, suppose that $H_1 = 2K_2$. Since H_2 is a linear forest of size $m - 2 \geq 19$ and $1+3+4\cdots+k = \binom{k+1}{2} - 2 = m-2$ and $k \leq \left\lfloor \frac{m-2}{2} \right\rfloor$ when $k \geq 6$, it follows by Corollary 2.2.4 that H_2 can be decomposed into $K_2, 3K_2, 4K_2, \cdots, kK_2$. Thus, $R_c : K_2, 2K_2, \ldots, kK_2$ is a Ramsey chain of length k in H.

- If $H_1 = Q_2$, then $H_2 = Q_{m-2}$ or $H_2 = Q_x + Q_y$, where $1 \leq x < y$, $x+y = m-2$ and $y \geq \left\lceil \frac{m-2}{2} \right\rceil$. By Proposition 2.3.4, the subgraph H_2 can be decomposed into $K_2, Q_2 + K_2, Q_2 + 2K_2, \ldots, Q_2 + (k-2)K_2$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, Q_2 + 2K_2, \ldots, Q_2 + (k-2)K_2$ is a Ramsey chain of length k in H.

Next, suppose that $m_1 = 4$. Then $H_1 \in \{Q_4, Q_3 + K_2, 2Q_2, Q_2 + 2K_2, 4K_2\}$ and H_2 is a linear forest of size $m - 4$.

- If $H_1 \in \{Q_2 + 2K_2, Q_3 + K_2, 4K_2\}$, then H_1 can be decomposed into K_2 and $3K_2$. Since $2 + 4 + 5 + \cdots + k = \binom{k+1}{2} - 4 = m - 4 \geq 17$ and $k \leq \left\lfloor \frac{m-4}{2} \right\rfloor$, it follows that $k \geq 6$. By Corollary 2.2.4, H_2 can be decomposed into $2K_2, 4K_2, 5K_2, \cdots, kK_2$. Thus, $R_c : K_2, 2K_2, \ldots, kK_2$ is a Ramsey chain of length k in H.

- If $H_1 \in \{Q_4, 2Q_2\}$, then H_1 can be decomposed into $G_1 = K_2$ and $G_3 = Q_2 + K_2$. Furthermore, $H_2 = Q_{m-4}$ or $H_2 = Q_x + Q_y$ where $1 \leq x < y \leq m-1$ and $x + y = m - 4$, or $H_2 = Q_x + Q_y + Q_z$ where $1 \leq x \leq y \leq z \leq m - 2$ and $x + y + z = m - 4$. By Proposition 2.3.5, H_2 can be decomposed into $Q_2, Q_2 + K_2, Q_2 + 2K_2, \ldots, Q_2 + (k-2)K_2$. Thus, $R_c : K_2, Q_2, Q_2 + K_2, Q_2 + 2K_2, \ldots, Q_2 + (k-2)K_2$ is a Ramsey chain of length k in H. ∎

The following is a consequence of Observation 2.1.1, Proposition 2.2.12, and Theorems 2.3.3 and 2.3.6.

Corollary 2.3.7 *If $m = \binom{k+1}{2}$ for some integer $k \geq 2$, then $AR(Q_m) = k$.*

Proof. By Proposition 2.2.12, we may assume that $m \geq 21$ and so $k \geq 6$. Since $m = \binom{k+1}{2} < \binom{k+2}{2}$, it follows by Observation 2.1.1 that $AR(Q_m) \leq k$. By Propositions 2.3.3 and 2.3.6, every 2-edge coloring of Q_m results in a Ramsey chain of length k in Q_m and $AR(Q_m) \geq k$. Therefore, $AR(Q_m) = k$ if $m = \binom{k+1}{2}$ where $k \geq 6$. ∎

Theorem 2.3.8 *For every integer $m \geq 3$,*

$$AR(Q_m) = k \text{ if and only if } \binom{k+1}{2} \leq m < \binom{k+2}{2}.$$

Proof. By Proposition 2.2.12, we may assume that $m \geq 21$. First, suppose that $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ where $k \geq 6$. Since $m < \binom{k+2}{2}$, it follows that $AR(Q_m) \leq k$ by Observation 2.1.1. Since $AR(Q_m) = k$ when $m = \binom{k+1}{2}$ by Corollary 2.3.7, it follows by Theorem 2.2.7 that $AR(Q_m) \geq k$ for $m \geq \binom{k+1}{2}$. Therefore, $AR(Q_m) = k$ when $\binom{k+1}{2} \leq m < \binom{k+2}{2}$.

For the converse, suppose that $AR(Q_m) = k$ where $k \geq 6$. Since $AR(Q_m) = k$, every 2-edge coloring of $H = Q_m$ produces a Ramsey chain of length k and so $m \geq \binom{k+1}{2}$. Assume, to the contrary, that $m \geq \binom{k+2}{2}$. It then follows by Theorem 2.2.7

and Corollary 2.3.7 that $AR(Q_m) \geq k+1$, which is a contradiction. Therefore, $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. ∎

With the aid of Theorem 2.3.8, we are able to present the following result.

Theorem 2.3.9 *For every integer $m \geq 3$,*

$$AR(C_m) = k \text{ if and only if } \binom{k+1}{2} \leq m < \binom{k+2}{2}.$$

Proof. First, suppose that $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. Since $m < \binom{k+2}{2}$, it follows that $AR(C_m) \leq k$ by Observation 2.1.1. On the other hand, $AR(C_m) \geq AR(Q_m) = k$ by Proposition 2.2.9 and Theorem 2.3.8. Hence, $AR(C_m) = k$.

For the converse, suppose that $AR(C_m) = k$. Since $AR(C_m) = k$, every 2-edge coloring of C_m produces a Ramsey chain of length k and so $m \geq \binom{k+1}{2}$. Assume, to the contrary, that $m \geq \binom{k+2}{2}$. It then follows by Theorem 2.3.8 that $AR(Q_m) \geq k+1$. Hence, $AR(C_m) \geq AR(Q_m) \geq k+1$ by Proposition 2.2.9, which is a contradiction. Therefore, $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. ∎

The following is a consequence of Theorems 2.3.8 and 2.3.9.

Corollary 2.3.10 *If $G \in \{C_m, P_{m+1}\}$ where $m \geq 3$, then $AR(G) = \left\lfloor \frac{-1+\sqrt{1+8m}}{2} \right\rfloor$.*

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Chapter 3

Ramsey Chains in Linear Forests

A *linear forest* is a forest every component of which is a path. Here, we are only concerned with linear forests without isolated vertices. Both paths and matchings are linear forests, namely the linear forests with the minimum and maximum number of components. We saw in Chapters 1 and 2 that if G is a matching or a path of size $m \geq 3$, then

$$AR(G) = k \text{ if and only if } \tbinom{k+1}{2} \leq m < \tbinom{k+2}{2}. \tag{3.1}$$

Since a matching and a path are both linear forests, this brings up the question as to whether (3.1) holds for all linear forests of size $m \geq 3$. This question can be answered immediately for some forests. For example, the linear forest $F = Q_2 + (m-2)K_2$ consisting of $m-1$ components where $m = \binom{k+1}{2}$ with $k \geq 3$ has Ramsey index $k-1$, not k. The red-blue coloring of F that assigns red to both edges of Q_2 and blue to all other edges does not result in a Ramsey chain of length k since a Ramsey chain $F_1, F_2, \ldots, F_k$ of length k would require that $F_2 = Q_2$ and $F_3 = 3K_2$; however, $F_2 \not\subseteq F_3$. On the other hand, $\binom{k}{2}K_2 \subseteq F$ and $AR(\binom{k}{2}K_2) = k-1$; so, $AR(F) = k-1$. This is shown in Figure 3.1 where $F = Q_2 + 4K_2$ is a forest of size $m = 6 = \binom{3+1}{2}$, a bold edge is a red edge and a thin edge is a blue edge.

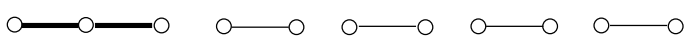


Figure 3.1: A linear forest of size 6 with Ramsey index 2

As another example, each of the linear forests $Q_2 + 8K_2$, $Q_3 + 7K_2$, $Q_4 + 6K_2$, $2Q_3 + 4K_2$, $Q_4 + Q_3 + 3K_2$, $Q_7 + 3K_2$ has size $10 = \binom{k+1}{2}$ where $k = 4$. In each

of these linear forests, a red-blue coloring is given in Figure 3.2 that shows that its Ramsey index is $3 = k - 1$. In Figure 3.2, a bold edge once again indicates a red edge and a thin edge indicates a blue edge. Each of the six linear forests in Figure 3.2 has t components for $t = 4, 5, \ldots, 9$. The examples in Figure 3.2 suggest that determining $AR(F)$ for a linear forest F may depend not only on its size but the number of components of F as well.

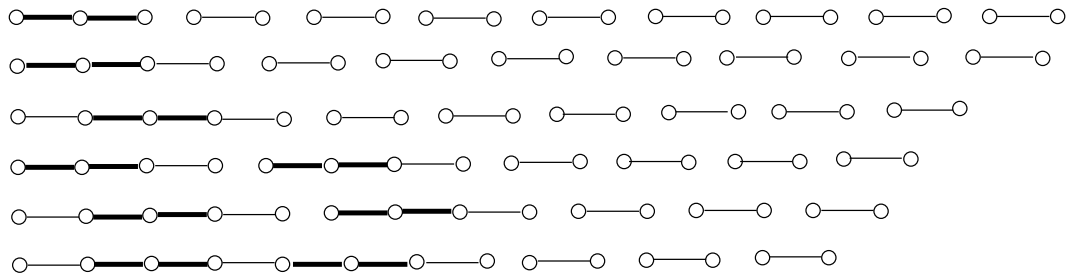


Figure 3.2: Linear forests of size 10 with Ramsey index 3

The goal here is to determine whether the result obtained for paths and matchings can be extended to include linear forests distinct from paths and matchings.

3.1 Ramsey Indexes of Linear Forests

In this section, we determine the Ramsey index of linear forests of size m where $m \neq \binom{k+1}{2}$ for any positive integer k. Prior to doing this, we recall some useful information from results presented in Chapters 1 and 2.

Theorem 3.1.1 *If G is a matching or a path of size $m \geq 3$, then*

$$AR(G) = k \text{ if and only if } \tbinom{k+1}{2} \leq m < \tbinom{k+2}{2}.$$

Observation 3.1.2 *If G is a graph of size m where $2 \leq m < \binom{k+2}{2}$ for a positive integer k, then $AR(G) \leq k$.*

Observation 3.1.3 *If H and G are graphs such that $H \subseteq G$, then $AR(H) \leq AR(G)$. Consequently, if $AR(H) \geq k$ for each graph H of size m, then $AR(G) \geq k$ for every graph G of size $m + 1$.*

Theorem 3.1.4 *Let $n \geq 5$ be an integer. For every set $\{n_1, n_2, \ldots, n_t\}$ of t integers such that $1 \leq n_1 < n_2 < \cdots < n_t \leq \lceil n/2 \rceil$ and $\sum_{i=1}^{t} n_i = n$, every linear forest of size n can be decomposed into the matchings n_1K_2, n_2K_2, ..., n_tK_2.*

Theorem 3.1.5 *Let $m = \binom{k+1}{2}$ for some integer $k \geq 4$. For every two positive integers m_1 and m_2 with $m = m_1 + m_2$ and $m_1, m_2 \notin \{2, 4\}$, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into two subsets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \left\lceil \frac{m_1}{2} \right\rceil$ and $b_1 < b_2 < \cdots < b_{k_2} \leq \left\lceil \frac{m_2}{2} \right\rceil$ such that $\sum_{i=1}^{k_1} a_i = m_1$ and $\sum_{i=1}^{k_2} b_i = m_2$.*

We now apply Observations 3.1.2 and 3.1.3, Theorems 3.1.4 and 3.1.5 to establish the following result.

Theorem 3.1.6 *If H is a linear forest of size m where $\binom{k+1}{2} < m < \binom{k+2}{2}$ for some integer $k \geq 3$, then $AR(H) = k$.*

Proof. Since $AR(H) \leq k$ by Observation 3.1.2, it remains only to verify that $AR(H) \geq k$, that is, to verify that H has a Ramsey chain of length k for every 2-edge coloring of H. It suffices to assume that $m = \binom{k+1}{2} + 1$ by Observation 3.1.3. Let c be a 2-edge coloring of H using the colors 1 and 2. We show that there is a Ramsey chain of length k in H. For $i = 1, 2$, let H_i be the linear forest in H induced by the edges of H colored i. Let H_i have size m_i where $1 \leq m_1 \leq m_2$. Therefore, $m_1 + m_2 = m = \binom{k+1}{2} + 1$. We now consider five cases, according to whether $m_1 \in \{1, 2, 3, 4\}$ or $m_1 \geq 5$.

Case 1. $m_1 = 1$. Thus, $m_2 = \binom{k+1}{2}$. Since $k \geq 3$, it follows that $k \leq \left\lceil \frac{\binom{k+1}{2}}{2} \right\rceil$.

Because $1 + 2 + \cdots + k = \binom{k+1}{2}$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $K_2, 2K_2, 3K_2, \cdots, kK_2$, which is a Ramsey chain of length k in H.

Case 2. $m_1 = 2$. Thus, $m_2 = \binom{k+1}{2} - 1$. Since $k \geq 3$, it follows that $k \leq \left\lceil \frac{\binom{k+1}{2}-1}{2} \right\rceil$. Because $2 + 3 + \cdots + k = \binom{k+1}{2} - 1$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $2K_2, 3K_2, \cdots, kK_2$. Since $K_2 \subseteq H_1$, there is a Ramsey chain of length k in H.

Case 3. $m_1 = 3$. Thus, $m_2 = \binom{k+1}{2} - 2$. Suppose first that $k = 3$. Then $m_2 = 4$. The linear forest H_1 can be decomposed into K_2 and $2K_2$, while the linear forest H_2 contains either $3K_2$ or $Q_2 + K_2$. In either case, H contains a Ramsey

chain of length $k = 3$. Hence, we may assume that $k \geq 4$. Let F be a linear forest of size $m' = \binom{k+1}{2} - 3$ in H_2. Since $k \geq 4$, it follows that $k \leq \left\lceil \frac{\binom{k+1}{2}-3}{2} \right\rceil$. Because $3 + 4 + \cdots + k = \binom{k+1}{2} - 3$, it follows by Theorem 3.1.4 that F can be decomposed into $3K_2, 4K_4, \ldots, kK_2$. Since H_1 can be decomposed into K_2 and $2K_2$, there is a Ramsey chain of length k in H.

Case 4. $m_1 = 4$. Thus, $m_2 = \binom{k+1}{2} - 3 \geq 7$ and so $k \geq 4$. Therefore, $k \leq \left\lceil \frac{\binom{k+1}{2}-3}{2} \right\rceil$. Since $3 + 4 + \cdots + k = \binom{k+1}{2} - 3$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $3K_2, 4K_4, \ldots, kK_2$. Let F be a linear forest of size 3 in H_1. Since F can be decomposed into K_2 and $2K_2$, there is a Ramsey chain of length k in H.

Case 5. $m_1 \geq 5$. Thus, $m_2 = \binom{k+1}{2} + 1 - m_1$ and $k \geq 5$. Let F be a linear forest of size $m_2' = m_2 - 1 \geq 5$ in H_2. Consequently, $m_1 \notin \{2, 4\}$, $m_2' \notin \{2, 4\}$, and $m_1 + m_2' = \binom{k+1}{2}$. By Proposition 3.1.5, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into two subsets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \lceil \frac{m_1}{2} \rceil$ and $b_1 < b_2 < \cdots < b_{k_2} \leq \left\lceil \frac{m_2'}{2} \right\rceil$ such that $\sum_{i=1}^{k_1} a_i = m_1$ and $\sum_{i=1}^{k_2} b_i = m_2'$. Hence, H_1 can be decomposed into $a_1K_2, a_2K_2, \ldots a_{k_1}K_2$ and F can be decomposed into $b_1K_2, b_2K_2, \ldots b_{k_2}K_2$, resulting in a Ramsey chain $K_2, 2K_2, \ldots, kK_2$ of length k in H. ∎

We now illustrate the proof of Theorem 3.1.6 for the linear forest $Q_5 + Q_6$ of size $m = 11$ consisting of paths of sizes 5 and 6. Then $k = 4$ and $\binom{4+1}{2} < 11 < 15 = \binom{4+2}{2}$. Five red-blue colorings of $Q_5 + Q_6$ are given in Figure 3.3 to reflect the five cases for $m_1 = 1, 2, 3, 4, 5$ in the proof, where again a bold edge indicates a red edge and a thin edge indicates a blue edge. For $i = 1, 2, 3, 4$, an edge labeled i belongs to the link G_i in a Ramsey chain $R : G_1, G_2, G_3, G_4$ of length 4 in $Q_5 + Q_6$ and the unlabeled edge is not used in R.

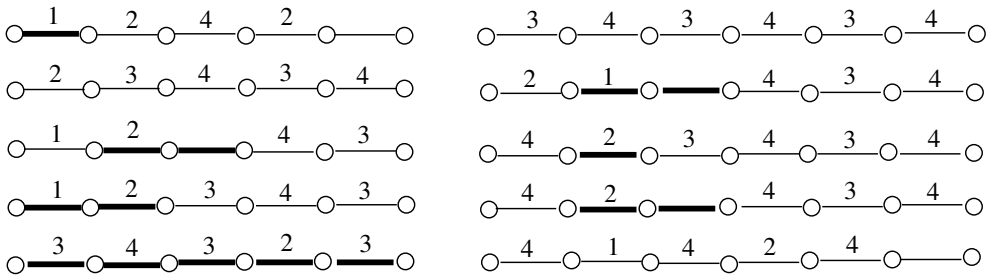


Figure 3.3: Red-blue colorings of $Q_5 + Q_6$ of size 11

3.2 Binomial Linear Forests

From the results obtained in Section 3.1, it follows that investigating $AR(F)$ for a linear forest F of size m, we need only be concerned with $m = \binom{k+1}{2}$ for some integer $k \geq 3$. Thus, it is convenient to introduce terminology for this class of linear forests. A linear forest is *k-binomial* (or simply *binomial*) if its size is $\binom{k+1}{2}$ for some positive integer k. For example, a 3-binomial linear forest has size 6, a 4-binomial linear forest has size 10, and a 5-binomial linear forest has size 15. We begin with an observation that is a consequence of Theorem 3.1.6 and Observation 3.1.3.

Corollary 3.2.1 *If H is a k-binomial linear forest where $k \geq 3$, then*

$$AR(H) \in \{k-1, k\}.$$

Furthermore, both values $k-1$ and k are attainable.

Proof. First, $AR(H) \leq k$ by Observation 3.1.2. By Theorem 3.1.6, every linear forest of size $\binom{k+1}{2} - 1$ has Ramsey index $k-1$. Hence, $AR(H) \geq k-1$ by Observation 3.1.3. Therefore, $AR(H) \in \{k-1, k\}$. Since

$$AR(Q_2 + (\tbinom{k+1}{2} - 2)K_2) = k-1 \text{ and } AR(Q_{\binom{k+1}{2}}) = k,$$

both values $k-1$ and k are attainable. ∎

By Theorem 3.1.6 and Corollary 3.2.1, if H is a 3-binomial linear forest, then $AR(H) \in \{2, 3\}$. We now determine the exact value of the Ramsey index of every 3-binomial linear forest. All 3-binomial linear forests are listed below:

$$Q_6,\ Q_5 + K_2,\ Q_4 + Q_2,\ Q_4 + 2K_2,\ 2Q_3,\ Q_3 + Q_2 + K_2,$$
$$Q_3 + 3K_2,\ 3Q_2,\ 2Q_2 + 2K_2,\ Q_2 + 4K_2,\ 6K_2.$$

Proposition 3.2.2 *Let H be a 3-binomial linear forest. Then*

$$AR(H) = 2 \text{ if and only if } H \in \{Q_4 + 2K_2, Q_3 + 3K_2, Q_2 + 4K_2\}.$$

Proof. Let $X = \{Q_4 + 2K_2, Q_3 + 3K_2, Q_2 + 4K_2\}$. First, observe that the linear forests H in X are the only 3-binomial linear forests containing a subgraph $F = Q_2$ such that $H - E(F) = 4K_2$ and consequently contains no copies of Q_2. Let c be a 2-edge coloring of H that assigns the color 1 to the two edges of F and the color 2 to all other edges of H. We claim that there is no Ramsey chain of length 3 in H. Assume, to the contrary, that there is a Ramsey chain G_1, G_2, G_3 of length 3 in H. Since the size of H is 6, it follows that $\{G_1, G_2, G_3\}$ is a decomposition of H. Necessarily, $G_2 = F = Q_2$ and so $Q_2 \subseteq G_3$. Since $H - E(F) = 4K_2$, this is impossible. Therefore, $AR(H) = 2$ by Corollary 3.2.1.

For the converse, suppose that $H \notin X$. We show that $AR(H) = 3$. Since $AR(H) \leq 3$ by Observation 3.1.2, it suffices to show that $AR(H) \geq 3$. Let c be a 2-edge coloring of H using the colors 1 and 2 where m_i is the number of the edges in the subgraph H_i of H colored i for $i = 1, 2$ and $m_1 \leq m_2$. Thus, $1 \leq m_1 \leq 3$ and $m_1 + m_2 = 6$. We consider all possible pairs (m_1, m_2).

- ⋆ If $(m_1, m_2) = (1, 5)$, then $G_1 = H_1 = K_2$ and H_2 is a linear forest of size 5. Since $2+3 = 5$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $2K_2, 3K_2$. Thus, $K_2, 2K_2, 3K_2$ is a Ramsey chain of length 3 in H

- ⋆ If $(m_1, m_2) = (2, 4)$, then $G_2 = H_1 \in \{Q_2, 2K_2\}$. First, suppose that $H_1 = 2K_2$. Then H_2 is a linear forest of size 4 and so H_2 can be decomposed into K_2 and a subgraph G_3 of size 3. Since every graph of size 3 contains $2K_2$, it follows that $K_2, H_1 = 2K_2, G_3$ is a Ramsey chain of length 3 in H. Next, suppose that $H_1 = Q_2$. Since $H \notin X$, it follows that $H_2 \neq 4K_2$. Thus, $1 \leq k(H_2) \leq 3$. If $k(H_2) = 1$, then $H_2 = Q_4$ can be decomposed into K_2 and Q_3. Thus K_2, Q_2, Q_3 is a Ramsey chain of length 3 in H. If $k(H_2) = 2$, then $H_2 = 2Q_2$ or $H_2 = Q_3 + K_2$. Since H_2 can be decomposed into K_2 and $Q_2 + K_2$, it follows that $K_2, Q_2, Q_2 + K_2$ is a Ramsey chain of length 3 in H. If $k(H_2) = 3$, then $H_2 = Q_2 + 2K_2$. Since H_2 can be decomposed into K_2 and $Q_2 + K_2$, it follows that $K_2, Q_2, Q_2 + K_2$ is a Ramsey chain of length 3 in H.

⋆ If $(m_1, m_2) = (3, 3)$, then H_1 can be decomposed into K_2 and $2K_2$; furthermore, H_2 contains $2K_2$. Thus, $K_2, 2K_2, H_2$ is a Ramsey chain of length 3 in H. ■

We now consider k-binomial linear forests for integers $k \geq 4$. The following two results provide useful information on 2-edge colorings of k-binomial linear forests for $k \geq 4$.

Proposition 3.2.3 *For a k-binomial linear forest H where $k \geq 4$, let c be a 2-edge coloring of H using the colors 1 and 2. For $i = 1, 2$, let H_i be the subgraph of H induced by the edges of H colored i where H_i has size m_i and $1 \leq m_1 \leq m_2$. If $H_1 \notin \{Q_2, 2Q_2, Q_4\}$, then $K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H and $AR_c(H) = k$.*

Proof. We consider five cases depending on whether $m_1 \in \{1, 2, 3, 4\}$ or $m_1 \geq 5$.

Case 1. $m_1 = 1$. Let $H_1 = K_2$. Since $k \leq \left\lfloor \frac{\binom{k+1}{2}-1}{2} \right\rfloor$ for $k \geq 4$ and $2 + 3 + \cdots + k = \binom{k+1}{2} - 1$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $2K_2, 3K_2, \ldots, kK_2$. Thus, $K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H and $AR_c(H) = k$.

Case 2. $m_1 = 2$. Thus, $H_1 \in \{2K_2, Q_2\}$ and $m_2 = \binom{k+1}{2} - 2$. Since $H_1 \neq Q_2$, it follows that $H_1 = 2K_2$. Because H_2 is a linear forest of size $\binom{k+1}{2} - 2$ and $1 + 3 + 4 \cdots + k = \binom{k+1}{2} - 2$ and $k \leq \left\lfloor \frac{\binom{k+1}{2}-2}{2} \right\rfloor$ when $k \geq 4$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $K_2, 3K_2, 4K_2, \cdots, kK_2$. Thus, $K_2, 2K_2, 3K_2, \ldots, kK_2$ is a Ramsey chain of length k in H and $AR_c(H) = k$.

Case 3. $m_1 = 3$. The subgraph H_1 can be decomposed into K_2 and $2K_2$. Since $k \leq \left\lfloor \frac{\binom{k+1}{2}-3}{2} \right\rfloor$ for $k \geq 4$ and $3 + \cdots + k = \binom{k+1}{2} - 3$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $3K_2, 4K_2, \ldots, kK_2$. Thus, $K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H and $AR_c(H) = k$.

Case 4. $m_1 = 4$. Then $H_1 \in \{Q_4, Q_3 + K_2, 2Q_2, Q_2 + 2K_2, 4K_2\}$ and $m_2 = \binom{k+1}{2} - 4$. Since $H_1 \notin \{2Q_2, Q_4\}$, it follows that $H_1 \in \{Q_2 + 2K_2, Q_3 + K_2, 4K_2\}$.

⋆ Let $k = 4$. Then H_2 is a linear forest of size 6. Since $1 + 2 + 3 = 6$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $K_2, 2K_2, 3K_2$. Thus, $K_2, 2K_2, 3K_2, H_1$ is a Ramsey chain of length 4 in H.

⋆ Let $k \geq 5$. Then H_1 can be decomposed into K_2 and $3K_2$. Since $2 + 4 + 5 + \cdots + k = \binom{k+1}{2} - 4$ and $k \leq \left\lfloor \frac{\binom{k+1}{2}-4}{2} \right\rfloor$ when $k \geq 5$, it follows by Theorem 3.1.4 that H_2 can be decomposed into $2K_2, 4K_2, 5K_2, \cdots, kK_2$. Thus, $K_2, 2K_2, \ldots, kK_2$ is a Ramsey chain of length k in H and $AR_c(H) = k$.

Case 5. $m_1 \geq 5$. By Proposition 3.1.5, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into two sets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \lceil \frac{m_1}{2} \rceil$, and $b_1 < b_2 < \cdots < b_{k_2} \leq \lceil \frac{m_2}{2} \rceil$ such that $\sum_{i=1}^{k_1} a_i = m_1$ and $\sum_{i=1}^{k_2} b_i = m_2$. Since $m_2 \geq m_1 \geq 5$, it follows by Theorem 3.1.4 that H_1 can be decomposed into the matchings a_1K_2, a_2K_2, ..., $a_{k_1}K_2$ and H_2 can be decomposed into the matchings b_1K_2, b_2K_2, ..., $b_{k_2}K_2$. Consequently, $K_2, 2K_2, \ldots kK_2$ is a Ramsey chain of length k in H and $AR_c(H) = k$. ∎

Proposition 3.2.4 *For a k-binomial linear forest H where $k \geq 4$, let c be a 2-edge coloring of H using the colors 1 and 2. For $i = 1, 2$, let H_i be the subgraph of H induced by the edges of H colored i where H_i has size m_i and $1 \leq m_1 \leq m_2$. If*

(a) $H_1 = Q_2$ and H_2 has at least $k - 2$ pairwise edge-disjoint copies of Q_2 or

(b) $H_1 \in \{Q_4, 2Q_2\}$ and H_2 has at least $k-3$ pairwise edge-disjoint copies of Q_2,

then

$$Q_1,\ G_2 \in \{Q_2, 2Q_1\},\ Q_2 + Q_1,\ Q_2 + 2Q_1,\ \ldots,\ Q_2 + (k-2)Q_1$$

is a Ramsey chain of length k in H.

Proof. First, suppose that $H_1 = Q_2$ and H_2 has at least $k - 2$ pairwise edge-disjoint copies of Q_2, which are denoted by A_3, A_4, ..., A_k. Then the number of edges of H_2 not belonging to any A_i $(3 \leq i \leq k)$ is

$$\binom{k+1}{2} - 2 - 2(k-2) = \binom{k+1}{2} - 2k + 2 = \binom{k-1}{2} + 1$$

and so these $\binom{k-1}{2}+1$ edges of H_2 can be decomposed into

$$B_1 = Q_1, B_3 = Q_1, B_4 = 2Q_1, B_5 = 3Q_1, \ldots, B_k = (k-2)Q_1$$

in such a way that $A_i + B_i = Q_2 + (i-2)Q_1$ for $3 \leq i \leq k$. Consequently, Q_1, Q_2, $Q_2 + Q_1$, $Q_2 + 2Q_1$, ..., $Q_2 + (k-2)Q_1$ is a Ramsey chain of length k in H.

Next, suppose that $H_1 \in \{Q_4, 2Q_2\}$. Then H_1 can be decomposed into $G_1 = Q_1$ and $G_3 = Q_2 + Q_1$. The subgraph H_2 has at least $k-3$ pairwise edge-disjoint copies of Q_2, which are denoted by A_4, A_5, ..., A_k. Then the number of edges of H_2 not belonging to any A_i $(4 \leq i \leq k)$ is

$$\binom{k+1}{2} - 4 - 2(k-3) = \binom{k+1}{2} - 2k + 2 = \binom{k-1}{2} + 1$$

and so these $\binom{k-1}{2}+1$ edges of H_2 can be decomposed into

$$B_2 = 2Q_1, B_4 = 2Q_1, B_5 = 3Q_1, \ldots, B_k = (k-2)Q_1$$

in such a way that $A_i + B_i = Q_2 + (i-2)Q_1$ for $4 \leq i \leq k$. Consequently,

$$Q_1,\ 2Q_1,\ Q_2 + Q_1,\ Q_2 + 2Q_1,\ \ldots,\ Q_2 + (k-2)Q_1$$

is a Ramsey chain of length k in H. ∎

With the aid of Observation 3.1.2 and Propositions 3.2.3 and 3.2.4, we are able to establish the following result.

Theorem 3.2.5 *If H is a k-binomial linear forest where $k \geq 4$ having at most $\binom{k-1}{2}$ components, then $AR(H) = k$.*

Proof. Since $AR(H) \leq k$ by Observation 3.1.2, it remains to show that $AR(H) \geq k$. Thus, we show that every 2-edge coloring of H produces a Ramsey chain of length k in H. Let c be a 2-edge coloring of H using the colors 1 and 2. For $i = 1, 2$, let H_i be the linear forest in H induced by the edges of H colored i, where H_i has size m_i with $1 \leq m_1 \leq m_2$ and $k(H_i)$ components. By Proposition 3.2.3, if $H_1 \notin \{Q_2, 2Q_2, Q_4\}$, then there is a Ramsey chain of length k in H. Thus, we may assume that $H_1 \in \{Q_2, 2Q_2, Q_4\}$. We consider two cases, depending on whether $H_1 = Q_2$ or $H_1 \in \{Q_4, 2Q_2\}$.

Case 1. $H_1 = Q_2$. Then H_2 is a linear forest of size $\binom{k+1}{2} - 2$. Let $k(H_2) = \ell$. Then $\ell \leq k(H) + 1 \leq \binom{k-1}{2} + 1$. Let $J_1, J_2, \ldots, J_\ell$ be the components of H_2

where J_i has size q_i for $1 \le i \le \ell$. Thus, $\sum_{i=1}^{\ell} q_i = m_2 = \binom{k+1}{2} - 2$. Let p be the maximum number of pairwise edge-disjoint copies of Q_2 in H_2. In each component J_i $(1 \le i \le \ell)$, the maximum number of pairwise edge-disjoint copies of Q_2 is $\lfloor q_i/2 \rfloor$ and so at most one edge of J_i does not belong to these $\lfloor q_i/2 \rfloor$ pairwise edge-disjoint copies of Q_2 in J_i. Hence, at most one edge in each J_i $(1 \le i \le \ell)$ does not belong to any of these p pairwise edge-disjoint copies of Q_2 in H_2. Since $\ell \le \binom{k-1}{2} + 1$, it follows that

$$p \ge \tfrac{1}{2}\left[\binom{k+1}{2} - 2 - \ell\right] \ge \tfrac{1}{2}\left[\binom{k+1}{2} - 2 - \binom{k-1}{2} - 1\right] = k - 2,$$

Therefore, there are at least $k-2$ pairwise edge-disjoint copies of Q_2 in H_2. By Proposition 3.2.4, there is a Ramsey chain of length k in H.

Case 2. $H_1 \in \{Q_4, 2Q_2\}$. Then H_1 can be decomposed into $G_1 = Q_1$ and $G_3 = Q_2 + Q_1$. Here, H_2 is a linear forest of size $\binom{k+1}{2} - 4$. Let $k(H_2) = \ell$. Then $\ell \le k(H) + 2 \le \binom{k-1}{2} + 2$. Let $J_1, J_2, \ldots, J_\ell$ be the components of H_2 where J_i has size q_i for $1 \le i \le \ell$. Thus, $\sum_{i=1}^{\ell} q_i = m_2 = \binom{k+1}{2} - 4$. Let ℓ' be the number of these components having odd size. If J_i has even size, then let $J_i' = J_i$. If J_i has odd size, then let J_i' be the subgraph of J_i obtained by removing a pendant edge from J_i, where J_i' is empty if $q_i = 1$. Hence, every subgraph J_i' has even size q_i' for $1 \le i \le \ell$ and every nonempty linear forest J_i' can be decomposed into $\frac{q_i'}{2}$ copies of Q_2. The size of the linear forest H_2' consisting of $J_1', J_2', \cdots, J_\ell'$ is therefore $\sum_{i=1}^{\ell} q_i' = \binom{k+1}{2} - 4 - \ell'$, which is an even number. Since $\ell \le k(H) + 2 \le \binom{k-1}{2} + 2$, it follows that

$$\binom{k+1}{2} - 4 - \ell' \ge \binom{k+1}{2} - 4 - \ell \ge \binom{k+1}{2} - 4 - \binom{k-1}{2} - 2 = 2k - 7.$$

Since $\binom{k+1}{2} - 4 - \ell'$ is even, it follows that $\binom{k+1}{2} - 4 - \ell' \ge 2k - 6$. Therefore, the number of pairwise edge-disjoint copies of Q_2 in H_2' (and in H_2 as well) is at least $\frac{1}{2}(2k-6) = k-3$. By Proposition 3.2.4, there is a Ramsey chain of length k in H. ∎

Next, we illustrate Theorem 3.2.5 for some 4-binomial forests of size 10 for which the number of components is 1, 2, or $3 = \binom{4-1}{2}$. Figure 3.4 shows red-blue colorings of the linear forests Q_{10}, $Q_3 + Q_7$, and $Q_2 + Q_3 + Q_5$ of size 10, where four bold edges are red edges and six thin edges are blue edges. For $i = 1, 2, 3, 4$, an edge labeled i belongs to the link G_i in a Ramsey chain G_1, G_2, G_3, G_4 of length 4 in the linear forest.

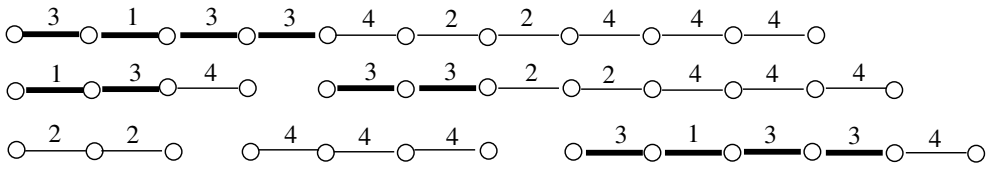


Figure 3.4: Red-blue colorings of three 4-binomial linear forests of size 10

3.3 Intermediate Value Results

By Theorems 3.1.1 and 3.2.5, every k-binomial linear forest where $k \geq 4$ with t components such that either $t = \binom{k+1}{2}$ or $1 \leq t \leq \binom{k-1}{2}$ has Ramsey index k. A natural question concerns whether these bounds on t can be improved. As we will see, no such improvement is possible. First, we provide a necessary and sufficient condition for a k-binomial linear forest to have Ramsey index $k-1$.

Theorem 3.3.1 *A k-binomial linear forest H where $k \geq 4$ has Ramsey index $k-1$ if and only if*

(a) *H contains a subgraph $F = Q_2$ such that $H - E(F)$ has at most $k-3$ pairwise edge-disjoint copies of Q_2 or*

(b) *H contains a subgraph $F \in \{Q_4, 2Q_2\}$ such that $H - E(F)$ has at most $k-4$ pairwise edge-disjoint copies of Q_2.*

Proof. First, we show that if H is a k-binomial linear forest where $k \geq 4$ such that neither (a) nor (b) holds, then $AR(H) = k$. Since $AR(H) \leq k$ by Observation 3.1.2, it remains to show that $AR(H) \geq k$. Let c be a 2-edge coloring of H using the colors 1 and 2. For $i = 1, 2$, let H_i be the linear forest in H induced by the edges of H colored i. Let H_i have size m_i where $1 \leq m_1 \leq m_2$. By Proposition 3.2.3, there is a Ramsey chain of length k in H if $H_1 \notin \{Q_2, 2Q_2, Q_4\}$. Thus, we may assume that $H_1 \in \{Q_2, 2Q_2, Q_4\}$.

- ⋆ If $H_1 = Q_2$, let $F = H_1 = Q_2$. Since (a) does not occur, it follows that $H - E(F)$ has at least $k-2$ pairwise edge-disjoint copies of Q_2. Hence, H has a Ramsey chain of length k by Proposition 3.2.4.

- ⋆ If $H_1 \in \{2Q_2, Q_4\}$, let $F = H_1$. Since (b) does not occur, it follows that $H - E(F)$ has at least $k-3$ pairwise edge-disjoint copies of Q_2. Hence, H has a Ramsey chain of length k by Proposition 3.2.4.

For the converse, suppose that H is a k-binomial linear forest where $k \geq 4$ such that either (a) or (b) occurs. We show that $AR(H) = k-1$. Since $AR(H) \geq k-1$ by Corollary 3.1.6, it remains to show that $AR(H) \leq k-1$. We consider two cases, according to whether (a) or (b) occurs.

Case 1. (*a*) *occurs.* Let $F = Q_2$ be a subgraph of H such that $H - E(F)$ has at most $k-3$ pairwise edge-disjoint copies of Q_2. Let c be a 2-edge coloring of H that assigns the color 1 to the two edges of F and the color 2 to all other edges of H. Then $H_1 = F = Q_2$ and $H_2 = H - E(F)$. We claim that there is no Ramsey chain of length k with respect to c. Assume, to the contrary, that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ of length k in H. Since the size of H is $\binom{k+1}{2}$, it follows that $\{G_1, G_2, \ldots, G_k\}$ is a decomposition of H. Necessarily, $G_2 = H_1 = F = Q_2$ and so $Q_2 \subseteq G_i$ for $3 \leq i \leq k$, which implies that $H - E(F)$ contains at least $k-2$ pairwise edge-disjoint copies of Q_2, which contradicts the fact that $H - E(F)$ has at most $k-3$ pairwise edge-disjoint copies of Q_2. Thus, $AR(H) \leq k-1$ and so $AR(H) = k-1$.

Case 2. (*b*) *occurs.* Let $F \in \{Q_4, 2Q_2\}$ be a subgraph of H such that $H - E(F)$ has at most $k-4$ pairwise edge-disjoint copies of Q_2 and let c be a 2-edge coloring of H that assigns the color 1 to the four edges of F and the color 2 to all other edges of H. Then $H_1 = F \in \{Q_4, 2Q_2\}$ and $H_2 = H - E(F)$. We claim that there is no Ramsey chain of length k with respect to c. Assume, to the contrary, that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ of length k in H. Since $\{G_1, G_2, \ldots, G_k\}$ is a decomposition of H, it follows that either (i) H_1 is decomposed into $G_1 = Q_1$ and $G_3 \in \{Q_3, Q_2 + Q_1\}$ or (ii) $G_4 = H_2 = F$. If (i) occurs, then $Q_2 \subseteq G_i$ for $4 \leq i \leq k$, which implies that $H - E(F)$ contains at least $k-3$ pairwise edge-disjoint copies of Q_2, which contradicts the fact that $H - E(F)$ has at most $k-4$ pairwise edge-disjoint copies of Q_2. If (ii) occurs, then G_3 contains a copy of Q_2 and G_i contains two edge-disjoint copies of Q_2 for $5 \leq i \leq k$, which implies that $H - E(F)$ contains at least $1 + 2(k-4) = 2k-7 > k-4$ pairwise edge-disjoint copies of Q_2. This contradicts the fact that $H - E(F)$ has at most $k-4$ pairwise edge-disjoint copies of Q_2. Thus, $AR(H) \leq k-1$ and so $AR(H) = k-1$. ∎

We have now seen for a linear forest H of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ and $k \geq 4$ that by Theorems 3.1.1, 3.1.6, and 3.2.5 if (i) $\binom{k+1}{2} < m < \binom{k+2}{2}$ or (ii) $m = \binom{k+1}{2}$ and H has t components where $t = \binom{k+1}{2}$ or $1 \leq t \leq \binom{k-1}{2}$, then $AR(H) = k$.

Consequently, it remains only to consider those k-binomial linear forests with an intermediate number t of components, namely $\binom{k-1}{2} < t < \binom{k+1}{2}$. Let c be any 2-edge coloring of H using the colors 1 and 2 resulting in the monochromatic subgraphs H_1 and H_2 of sizes m_1 and m_2, respectively, where $m_1 \leq m_2$. By Proposition 3.2.3, if $H_1 \notin \{Q_2, 2Q_2, Q_4\}$, then $AR_c(H) = k$. By Proposition 3.2.4 and Theorem 3.3.1, if $H_1 \in \{Q_2, 2Q_2, Q_4\}$, then $AR_c(H) \in \{k, k-1\}$. Furthermore, by Theorem 3.3.1, if $H_1 = Q_2$, then $AR_c(H) = k$ only when H_2 contains at least $k-2$ pairwise edge-disjoint copies of Q_2; otherwise $AR_c(H) = k-1$. Moreover, if $H_1 \in \{2Q_2, Q_4\}$, then $AR_c(H) = k$ only when H_2 has at least $k-3$ pairwise edge-disjoint copies of Q_2; otherwise $AR_c(H) = k-1$. All of these suggest that we need only consider those 2-edge colorings of H for which $H_1 \in \{Q_2, 2Q_2, Q_4\}$ and determine whether there exists

(a) a 2-edge coloring of H where $H_1 = Q_2$ such that H_2 has fewer than $k-2$ pairwise edge-disjoint copies of Q_2 or

(b) a 2-edge coloring of H where $H_1 \in \{2Q_2, Q_4\}$ such that H_2 has fewer than $k-3$ pairwise edge-disjoint copies of Q_2.

If there is a 2-edge coloring of H resulting in (a) or (b), then $AR(H) = k-1$; otherwise, $AR(H) = k$. Therefore, to determine the Ramsey index of a k-binomial linear forest H where $k \geq 4$ with an intermediate number t of components with $\binom{k-1}{2} < t < \binom{k+1}{2}$, it suffices to study the 2-edge colorings of H such that $H_1 \in \{Q_2, 2Q_2, Q_4\}$ such that $H_2 = H - E(H_1)$ possesses the minimum number of pairwise edge-disjoint copies of Q_2.

Each k-binomial linear forest H with t of components can be expressed as

$$Q_{q_1} + Q_{q_2} + \cdots + Q_{q_t}$$

where $q_1 \geq q_2 \geq \cdots \geq q_t \geq 1$ and $\sum_{i=1}^{t} q_i = \binom{k+1}{2}$. The maximum number of pairwise edge-disjoint copies of Q_2 in H is $s = \sum_{i=1}^{t} \left\lfloor \frac{q_i}{2} \right\rfloor$. Now, let c be any 2-edge coloring of H using the colors 1 and 2 resulting in the monochromatic subgraphs H_1 and H_2 of sizes m_1 and m_2, respectively, where $m_1 \leq m_2$ and $H_1 \in \{Q_2, 2Q_2, Q_4\}$. First, we make some observations. If $H_1 = Q_2$, then the maximum number of pairwise edge-disjoint copies of Q_2 in H_2 is either $s-1$ or $s-2$; while if $H_1 \in \{Q_4, 2Q_2\}$, then the maximum number of pairwise edge-disjoint

copies of Q_2 in H_2 is $s-2$, $s-3$, or $s-4$. We are now prepared to present the following result.

Theorem 3.3.2 *Let $H = Q_{q_1} + Q_{q_2} + \cdots + Q_{q_t}$ be a k-binomial linear forest of size $\sum_{i=1}^{t} q_i = \binom{k+1}{2}$ for some integer $k \geq 4$ with t components where $\binom{k-1}{2} < t < \binom{k+1}{2}$ and $s = \sum_{i=1}^{t} \left\lfloor \frac{q_i}{2} \right\rfloor$ is the maximum number of pairwise edge-disjoint copies of Q_2 in H.*

(1) *If $s > k$, then $AR(H) = k$.*

(2) *If $s = k$ and H contains two components of even size 4 or more, then $AR(H) = k-1$; otherwise, $AR(H) = k$.*

(3) *If $s = k-1$ and H contains at least one component of even size 4 or more, then $AR(H) = k-1$; otherwise, $AR(H) = k$.*

(4) *If $s \leq k-2$, then $AR(H) = k-1$.*

Proof. We first verify (1). Suppose that $s > k$. By Observation 3.1.2, it suffices to show that $AR_c(H) = k$ for every 2-edge coloring c of H. Let c be a 2-edge coloring of H using the colors 1 and 2. For $i = 1, 2$, let H_i be the linear forest in H induced by the edges of H colored i where H_i has size m_i and $1 \leq m_1 \leq m_2$. By Proposition 3.2.3, we may assume that $H_1 \in \{Q_2, 2Q_2, Q_4\}$. If $H_1 = Q_2$, then the number of pairwise edge-disjoint copies of Q_2 in H_2 is at least $s-2$. Since $s-2 > k-3$, it follows that $AR_c(H) = k$ by Theorem 3.3.1. If $H_1 \in \{Q_4, 2Q_2\}$, then the number of pairwise edge-disjoint copies of Q_2 in H_2 is at least $s-4$. Since $s-4 > k-4$, it follows that $AR_c(H) = k$ by Theorem 3.3.1. Therefore, $AR(H) = k$.

Next, we verify (2). Suppose that $s = k$.

⋆ First, assume that H contains two components Q_x and Q_y where $x, y \geq 4$ ans x and y are both even. Define a 2-edge coloring c of H using the colors 1 and 2 such that $H_1 = 2Q_2$ and H_1 is placed in H in such a way that each of Q_x and Q_y contains a copy of Q_2 of H_1 and the number of pairwise edge-disjoint copies of Q_2 in H_2 is $s-4 = k-4$. Thus, $AR_c(H) = k-1$ by Theorem 3.3.1 and so $AR(H) = k-1$.

⋆ Next, assume that H contains at most one component of even order 4 or more. Let c be any 2-edge coloring of H using the colors 1 and 2. For $i = 1, 2$, let H_i be the linear forest in H induced by the edges of H colored i where H_i has size m_i and $1 \leq m_1 \leq m_2$. By Proposition 3.2.3, we may assume that $H_1 \in \{Q_2, 2Q_2, Q_4\}$. If $H_1 = Q_2$, then the number of pairwise edge-disjoint copies of Q_2 in H_2 is at least $s - 2 = k - 2$ and so $AR_c(H) = k$ by Theorem 3.3.1. If $H_1 \in \{Q_4, 2Q_2\}$, then, since H contains at most one component of even order 4 or more, any placement of H_1 in H produces at least $s - 3 = k - 3$ pairwise edge-disjoint copies of Q_2 in H_2. It follows by Theorem 3.3.1 that $AR_c(H) = k$. Therefore, $AR(H) = k$.

We now verify (3). Suppose that $s = k - 1$.

⋆ First, assume that H contains at least one component Q_x where $x \geq 4$ and x is even. Define a 2-edge coloring c of H using the colors 1 and 2 such that $H_1 = Q_2$ and H_1 is placed in Q_x in such a way that the number of pairwise edge-disjoint copies of Q_2 in H_2 is $s - 2 = k - 3$. It follows by Theorem 3.3.1 that $AR_c(H) = k - 1$ and so $AR(H) = k - 1$.

⋆ Next, assume that H contains no component of even order 4 or more. Let c be any 2-edge coloring of H using the colors 1 and 2 where H_i is the linear forest of size m_i in H induced by the edges of H colored i and $1 \leq m_1 \leq m_2$. By Proposition 3.2.3, we may assume that $H_1 \in \{Q_2, 2Q_2, Q_4\}$. If $H_1 = Q_2$, then the number of pairwise edge-disjoint copies of Q_2 in H_2 is at least $s-1 = k-2$ and so $AR_c(H) = k$ by Theorem 3.3.1. If $H_1 \in \{Q_4, 2Q_2\}$, then, since H contains no component of even order 4 or more, any placement of H_1 in H produces at least $s - 2 = k - 3$ pairwise edge-disjoint copies of Q_2 in H_2. It follows by Theorem 3.3.1 that $AR_c(H) = k$. Therefore, $AR(H) = k$.

Finally, we verify (4). Suppose that $s \leq k - 2$. Define a 2-edge coloring c of H using the colors 1 and 2 such that $H_1 \in \{Q_2, Q_4, 2Q_2\}$. Then the number of pairwise edge-disjoint copies of Q_2 in H_2 is at least $s - 1$. Since $s - 1 \leq k - 3$, it follows by Theorem 3.3.1 that $AR_c(H) = k - 1$ and so $AR(H) = k - 1$. ∎

We have seen (by Theorems 3.1.1 and 3.2.5) that if H is a k-binomial linear forest where $k \geq 4$ with t components where $t = \binom{k+1}{2}$ or $1 \leq t \leq \binom{k-1}{2}$, then

$AR(H) = k$. With the aid of Theorem 3.3.2, we are now able to show that these bounds on t cannot be improved.

Theorem 3.3.3 *For every two integers t and k where $\binom{k-1}{2} < t < \binom{k+1}{2}$ and $k \geq 4$, there is a k-binomial linear forest H with t components such that*

$$AR(H) = k - 1.$$

Proof. For each integer i with $1 \leq i \leq 2k-2$ where $k \geq 4$, we construct a k-binomial linear forest F_i with $t = \binom{k-1}{2} + i$ components such that $AR(F_i) = k-1$. In particular, F_1 has $t = \binom{k-1}{2}+1$ components and F_{2k-2} has $t = \binom{k-1}{2}+(2k-2) = \binom{k+1}{2} - 1$ components.

For an integer $k \geq 4$, let $F_1 = Q_x + Q_y + \left[\binom{k-1}{2} - 1\right] Q_1$, where $x = y = k$ if k is even and $x = k+1$ and $y = k-1$ if $k \geq 5$ is odd. Thus, $x, y \geq 4$ are both even. Since the size of F_1 is $\binom{k+1}{2}$, it follows that F_1 is a k-binomial linear forest with $t = \binom{k-1}{2} + 1$ components. The maximum number of pairwise edge-disjoint copies of Q_2 in F_1 is $s = \left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{y}{2} \right\rfloor = k$. Since $s = k$ and F_1 has two components of even size 4 or more, it follows by Theorem 3.3.2 that $AR(F_1) = k-1$.

Next, let $F_2 = Q_{x-1} + Q_y + \binom{k-1}{2} Q_1$. Then F_2 is a k-binomial linear forest with $t = \binom{k-1}{2} + 2$ components. The maximum number of pairwise edge-disjoint copies of Q_2 in F_2 is $s = \left\lfloor \frac{x-1}{2} \right\rfloor + \left\lfloor \frac{y}{2} \right\rfloor = k-1$ and F_2 has the component Q_y of even size 4 or more. By Theorem 3.3.2, $AR(F_2) = k-1$. Next, let $F_3 = Q_{x-2} + Q_y + \left[\binom{k-1}{2} + 1\right] Q_1$. Then F_3 is a k-binomial linear forest with $t = \binom{k-1}{2} + 3$ components. The maximum number of pairwise edge-disjoint copies of Q_2 in F_3 is $s = \left\lfloor \frac{x-2}{2} \right\rfloor + \left\lfloor \frac{y}{2} \right\rfloor = k-1$ and F_3 has the component Q_y of even size 4 or more. By Theorem 3.3.2, $AR(F_3) = k-1$.

Next, let $F_4 = Q_{x-2} + Q_{y-1} + \left[\binom{k-1}{2} + 2\right] Q_1$. Then F_4 is a k-binomial linear forest with $t = \binom{k-1}{2} + 4$ components. The maximum number of pairwise edge-disjoint copies of Q_2 in F_4 is $s = \left\lfloor \frac{x-2}{2} \right\rfloor + \left\lfloor \frac{y-1}{2} \right\rfloor = k-2$. We continue this procedure, reducing the size of components greater than 1 until we arrive at

$$\begin{aligned} F_{2k-2} &= Q_2 + Q_1 + \left[\binom{k-1}{2} + 2k - 4\right] Q_1 \\ &= Q_2 + \left[\binom{k-1}{2} + 2k - 3\right] Q_1 = Q_2 + \left[\binom{k+1}{2} - 2\right] Q_1. \end{aligned}$$

Here, F_{2k-2} is a k-binomial linear forest with $t = \binom{k+1}{2} - 1$ components. For each integer i with $4 \leq i \leq 2k-2$, the maximum number of pairwise edge-disjoint copies of Q_2 in F_i is $s \leq k-2$ and so $AR(F_i) = k-1$ by Theorem 3.3.2. ■

To illustrate Theorem 3.3.2, we consider $k = 5$ and $\binom{4}{2} < 7 \leq t \leq 14 < \binom{6}{2}$. For $i = 1, 2, \ldots, 8$, a 5-binomial linear forest F_i with $t = \binom{5-1}{2} + i = 6 + i$ components is constructed such that $AR(F_i) = 4$.

- ⋆ Let $F_1 = Q_6 + Q_4 + 5Q_1$ where $t = 7$ and $s = 5$. Since F_1 has two components Q_6 and Q_4 of even size 4 or more, $AR(F_1) = 4$ by Theorem 3.3.2.

- ⋆ Let $F_2 = Q_5 + Q_4 + 6Q_1$ where $t = 8$ and $s = 4$. Since F_2 has the component Q_4 of size 4, it follows by Theorem 3.3.2 that $AR(F_2) = 4$.

- ⋆ Let $F_3 = Q_4 + Q_4 + 7Q_1$ where $t = 9$ and $s = 4$. Since F_3 has the component Q_4 of size 4, it follows by Theorem 3.3.2 that $AR(F_3) = 4$.

- ⋆ For $4 \leq i \leq 8$, let

  $F_4 = Q_4 + Q_3 + 8Q_1$ where $t = 10$ and $s = 3$,

  $F_5 = Q_3 + Q_3 + 9Q_1$ where $t = 11$ and $s = 2$,

  $F_6 = Q_3 + Q_2 + 10Q_1$ where $t = 12$ and $s = 2$,

  $F_7 = Q_2 + Q_2 + 11Q_1$ where $t = 13$ and $s = 2$, and

  $F_8 = Q_2 + Q_1 + 12Q_1$ where $t = 14$ and $s = 1$.

  Since $s \leq 5 - 2 = 3$, it follows by Theorem 3.3.2 that $AR(F_i) = 4$ for $4 \leq i \leq 8$.

We saw that the 5-binomial linear forest $F_1 = Q_6 + Q_4 + 5Q_1$ has 7 components and $AR(F_1) = 4$. This however does not imply that every 5-binomial linear forest with 7 components has Ramsey index 4. For example, $F = 4Q_3 + 3Q_1$ is also a 5-binomial linear forest with 7 components and the maximum number of pairwise edge-disjoint copies of Q_2 in F is $s = 4$. Since F has no component of even size 4 or more, it follows by Theorem 3.3.2 that $AR(F) = 5$.

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Chapter 4

Ramsey Chains in Galaxies

Matchings and stars are all subclasses of the well-known class of graphs called galaxies. A *galaxy* is a graph each of whose components is a star. For a positive integer s, an *s-galaxy* consists of s stars. Thus, for an s-galaxy of size m, it follows that $1 \leq s \leq m$. In this chapter, we denote the star $K_{1,m}$ of size m by S_m for convenience. A 1-galaxy of size m is the star S_m and an m-galaxy of size m is the matching $mK_2 = mS_1$. Hence, stars and matchings are galaxies with the minimum or maximum number of components. The Ramsey indexes of 1-galaxies are stated in Chapter 1. Hence, the next natural question is:

> *What are the Ramsey indexes of 2-galaxies $S_x + S_{m-x}$ of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some positive integer k?*

In this chapter, we determine the Ramsey index $AR(S_x + S_{m-x})$ for integers x and m with $1 \leq x \leq 4$ and $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some positive integer k. Additional results and conjectures are also presented dealing with Ramsey indexes of 2-galaxies.

4.1 Preliminary Results

We begin by presenting some preliminary results on Ramsey indexes of 2-galaxies. First, recall the following observations stated in Chapter 2.

Observation 4.1.1 *If G is a graph of size m where $2 \leq m < \binom{k+2}{2}$ for a positive integer k, then $AR(G) \leq k$.*

Observation 4.1.2 *If H and G are graphs such that $H \subseteq G$, then $AR(H) \leq AR(G)$. Consequently, if $AR(H) \geq k$ for each graph H of size m, then $AR(G) \geq k$ for every graph G of size $m+1$.*

The Ramsey index of any galaxy of size $m \geq 3$ is bounded above by the Ramsey index of the star S_m, as we show next. Recall that $AR(S_m) = k$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. In particular, $AR\left(S_{\binom{k+1}{2}}\right) = k$.

Proposition 4.1.3 *If H is a galaxy of size $m \geq 3$, then $AR(H) \leq AR(S_m)$.*

Proof. Let $AR(H) = k$. Then every 2-edge coloring of H produces a Ramsey chain of length k in H and so $m \geq \binom{k+1}{2}$. Thus, $S_{\binom{k+1}{2}} \subseteq S_m$. It then follows by Observation 4.1.2 that $AR(S_m) \geq AR\left(S_{\binom{k+1}{2}}\right) = k = AR(H)$. ∎

Strict inequality in Proposition 4.1.3 can occur. For example, consider the 3-galaxies $2S_1 + S_4$ of size 6 and $2S_1 + S_5$ of size 7. Figure 4.1 shows a red-blue coloring for each of these two graphs, where a bold edge indicates a red edge and a thin edge indicates a blue edge. Since there is no Ramsey chain of length 3 with respect to each of these colorings, it follows that $AR(2S_1+S_4) = AR(2S_1+S_5) = 2$. On the other hand, $AR(S_6) = AR(S_7) = 3$. Thus,

$$AR(2S_1 + S_4) < AR(S_6) \text{ and } AR(2S_1 + S_5) < AR(S_7).$$

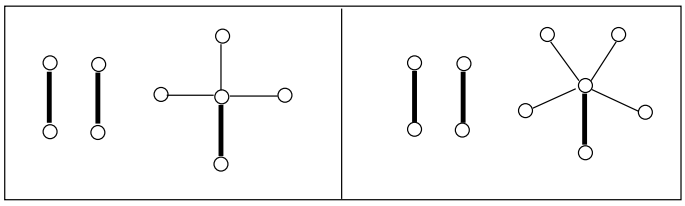


Figure 4.1: Red-blue colorings of the 3-galaxies $2S_1 + S_4$ and $2S_1 + S_5$

Let H be a 2-galaxy of size $m \geq 2$. If $m = 2$, then $AR(H) = 1$; while if $3 \leq m \leq 5$, then $AR(H) = 2$. Thus, we will only consider 2-galaxies of size 6 or more. First, we show that the Ramsey indexes of 2-galaxies $S_x + S_y$ can be determined for some special pairs x, y of positive integers.

Proposition 4.1.4 *Let $H = S_x + S_y$ be a 2-galaxy such that $\binom{k+1}{2} \leq x+y < \binom{k+2}{2}$ where $k \geq 3$.*

(a) *If* $x + y = \binom{k+1}{2}$ *and* $\min\{x, y\} \leq k - 2$*, then* $AR(H) = k - 1$.

(b) *If* $\max\{x, y\} \geq \binom{k+1}{2}$*, then* $AR(H) = k$.

Proof. Assume, without loss of generality, that $x = \min\{x, y\}$ and $\max\{x, y\} = y$. To verify (a), suppose that $x + y = \binom{k+1}{2}$ and $x \leq k - 2$. Then $y = \binom{k+1}{2} - x \geq \binom{k+1}{2} - (k-2) = \binom{k}{2} + 2$ and so $S_{\binom{k}{2}} \subseteq S_y \subseteq H$. It then follows by Proposition 4.1.3 that $AR(H) \geq k-1$. Next, we show that $AR(H) \leq k-1$. Let c be a 2-edge coloring of H resulting in monochromatic subgraphs $H_1 = 2S_1$ and $H_2 = S_{x-1} + S_{y-1}$. We claim that there is no Ramsey chain of length k in H. Suppose that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ of length k in H. Since the size of H is $\binom{k+1}{2}$, it follows that $\{G_1, G_2, \ldots, G_k\}$ is a decomposition of H. Necessarily, $G_2 = H_1 = 2S_1$. This implies that $2S_1 \subseteq G_i \subseteq H_2$ for $3 \leq i \leq k$ and that G_i contains an edge of H_2 belonging to S_x. Since S_x contains at most $k - 3$ edges of H_2, this is impossible. Thus, $AR(H) = k - 1$.

To verify (b), suppose that $y \geq \binom{k+1}{2}$. Then $S_{\binom{k+1}{2}} \subseteq H$ and so $AR(H) \geq k$ by by Proposition 4.1.3. Furthermore, $AR(H) \leq k$ by Observation 4.1.1. Therefore, $AR(H) = k$. ∎

The following is a consequence of Proposition 4.1.4.

Corollary 4.1.5 *If* $H = S_1 + S_{m-1}$ *is a 2-galaxy of size* m *where* $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ *for some integer* $k \geq 3$*, then*

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \\ k & \text{otherwise.} \end{cases}$$

Proof. First, suppose that $m = \binom{k+1}{2}$. Since $k \geq 3$, it follows that

$$\min\{1, m-1\} = 1 \leq k - 2$$

and so $AR(H) = k-1$ by Proposition 4.1.4. Next, suppose that $\binom{k+1}{2} < m < \binom{k+2}{2}$. Then $\max\{1, m-1\} = m - 1 \geq \binom{k+1}{2}$ and so $AR(H) = k$ by Proposition 4.1.4. ∎

Since $S_1 + S_{m-1}$ is the union of the matching S_1 and the star S_{m-1}, it follows that Corollary 4.1.5 is also a consequence of the result on $AR(aS_1 + S_b)$ for integers $a \geq 1$ and $b \geq 2$ stated in Chapter 1.

4.2 Ramsey Indexes of 2-galaxies $S_2 + S_{m-2}$

Let H be a 2-galaxy of size m where $\binom{3+1}{2} = 6 \leq m < 10 = \binom{3+2}{2}$ for $k = 3$. If $m = 6$, then $AR(S_1+S_5) = 2$ and $AR(S_2+S_4) = AR(2S_3) = 3$. If $7 \leq m \leq 9$, then H contains a subgraph isomorphic to one of S_6, S_2+S_4, or $2S_3$. Thus, $AR(H) \geq 3$ and so $AR(H) = 3$. Hence, we may assume that $k \geq 4$. In this section, we determine $AR(S_2 + S_{m-2})$ where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. To do this, we recall the following lemma stated in Chapter 3.

Lemma 4.2.1 *Let $m = \binom{k+1}{2}$ for some integer $k \geq 4$. For every two positive integers m_1 and m_2 with $m = m_1 + m_2$ and $m_1, m_2 \notin \{2, 4\}$, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into the subsets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $1 = a_1 < a_2 < \cdots < a_{k_1} \leq \lceil \frac{m_1}{2} \rceil$, and*

$$b_1 < b_2 < \cdots < b_{k_2} = k \leq \left\lceil \tfrac{m_2}{2} \right\rceil \text{ such that } \sum_{i=1}^{k_1} a_i = m_1 \text{ and } \sum_{i=1}^{k_2} b_i = m_2.$$

With the aid of Lemma 4.2.1, we now determine the Ramsey index of every 2-galaxy of sufficiently large size where one component is $S_2 = P_3 = K_{1,2}$.

Theorem 4.2.2 *If $H = S_2+S_{m-2}$ is a 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$, then*

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \\ k & \text{otherwise.} \end{cases}$$

Proof. By Proposition 4.1.4, if $m = \binom{k+1}{2}$, then $AR(H) = k-1$ (since $2 \leq k-2$ when $k \geq 4$); while if $\binom{k+1}{2} + 2 \leq m < \binom{k+2}{2}$, then $AR(H) = k$. Thus, we may assume that $m = \binom{k+1}{2} + 1$. By Observation 4.1.1, it suffices to show that $AR(H) \geq k$. Let c be a 2-edge coloring of H using the colors 1 and 2. We show that there is a Ramsey chain of length k in H. For $i = 1, 2$, let H_i be the subgraph of size m_i in H induced by the edges of H colored i, where $1 \leq m_1 \leq m_2$ and $m_1 + m_2 = m = \binom{k+1}{2} + 1$. We now consider six cases, according to whether $m_1 \in \{1, 2, 3, 4, 5\}$ or $m_1 \geq 6$.

Case 1. $m_1 = 1$. Then $H_1 = S_1$ and $H_2 \in \{S_1 + S_{m-2}, S_2 + S_{m-3}\}$.

⋆ First, suppose $H_2 = S_1 + S_{m-2}$. Since $m-2 = \binom{k+1}{2} - 1$ and $S_{m-2} \subseteq H_2$, it follows that S_{m-2} can be decomposed into $S_2, S_3, \ldots, S_k$. Thus, there is a Ramsey chain $S_1, S_2, S_3, \ldots, S_k$ of length k in H.

⋆ Next, suppose that $H_2 = S_2 + S_{m-3}$. Since $m-4 = \binom{k+1}{2} - 3 = 3+4+\cdots+k$, there is an ascending subgraph sequence $S_3, S_4, \ldots, S_k$ in $S_{m-3} \subseteq H_2$. Let $G_1 = H_1 = S_1$ and $G_2 = S_2 \subseteq H_2$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

Case 2. $m_1 = 2$. Thus, $H_1 \in \{2S_1, S_2\}$, $m_2 = m-2 = \binom{k+1}{2} - 1$, and

$$H_2 \in \{S_1 + S_{m-3}, S_{m-2}, S_2 + S_{m-4}\}.$$

⋆ First, suppose that $H_2 = S_1 + S_{m-3}$. Since

$$m-3 = \binom{k+1}{2} - 2 = 2 + 3 + \cdots + (k-1) + (k-1),$$

it follows that S_{m-3} can be decomposed into $S_2, S_3, \ldots, S_{k-1}, S_{k-1}$. Let $G_1 = S_1 \subseteq H_1$ and $G_k = S_1 + S_{k-1}$. Thus, $S_1, S_2, S_3, \ldots, S_{k-1}, S_1 + S_{k-1}$ is a Ramsey chain of length k in H.

⋆ Next, suppose that $H_2 = S_{m-2}$. Since $1 + 3 + 4 + \cdots + k = \binom{k+1}{2} - 2$, there is an ascending subgraph sequence $S_1, S_3, S_4, \ldots, S_k$ in $S_{m-2} \subseteq H_2$. Let $G_2 = H_1 = S_2$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

⋆ Finally, suppose that $H_2 = S_2 + S_{m-4}$. Since $m-4 = \binom{k+1}{2} - 3 = 3+4+\cdots+k$, it follows that S_{m-4} can be decomposed into $S_3, S_4, \ldots, S_k$. Let $G_1 = S_1 \subseteq S_2 \subseteq H_2$, $G_2 = H_1 = S_2$, Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

Case 3. $m_1 = 3$. Thus, $H_1 \in \{S_3, S_1 + S_2\}$, $m_2 = \binom{k+1}{2} - 2$, and

$$H_2 \in \{S_2 + S_{m-5}, S_1 + S_{m-4}, S_{m-3}\}.$$

⋆ First, suppose that $H_1 = S_3$ and $H_2 = S_2 + S_{m-5}$ where $m - 5 = \binom{k+1}{2} - 4$. Since $1 + 4 + 5 + \cdots + k = \binom{k+1}{2} - 5$, there is an ascending subgraph sequence $S_1, S_4, S_5, \ldots, S_k$ in $S_{m-5} \subseteq H_2$. Let $G_2 = S_2 \subseteq H_2$ and $G_3 = H_1 = S_3$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

⋆ Next, suppose that $H_1 = S_1 + S_2$ and $H_2 \in \{S_1 + S_{m-4}, S_{m-3}\}$. Since $m - 4 = \binom{k+1}{2} - 3 = 3 + 4 + \cdots + k$, there is an ascending subgraph sequence $S_3, S_4, \ldots, S_k$ in H_2. Let $G_1 = S_1 \subseteq H_1$ and $G_2 = S_2 \subseteq H_1$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

Case 4. $m_1 = 4$. Thus, $H_1 \in \{S_1 + S_3, 2S_2, S_4\}$ and $m_2 = \binom{k+1}{2} - 3$.

⋆ First, suppose that $H_1 = S_1 + S_3$. Then $H_2 = S_1 + S_{m-5}$, where $m - 5 = \binom{k+1}{2} - 4$. Since $2 + 4 + \cdots + k = \binom{k+1}{2} - 4$, there is an ascending subgraph sequence $S_2, S_4, S_5, \ldots, S_k$ in $S_{m-5} \subseteq H_2$. Let $G_1 = S_1 \subseteq H_1$ and $G_3 = S_3 \subseteq H_1$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

⋆ Next, suppose that $H_1 = S_2 + S_2$. Then $H_2 = S_{m-4}$ where $m-4 = \binom{k+1}{2} - 3 = 3 + 4 + \cdots + k$. Then S_{m-4} can be decomposed into $S_3, S_4, \ldots, S_k$. Let $G_1 = S_1 \subseteq H_1$ and $G_2 = S_2 \subseteq H_1$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

⋆ Finally, suppose that $H_1 = S_4$. Then $H_2 = S_2 + S_{m-6}$, where $m - 6 = \binom{k+1}{2} - 5 = 2 + 3 + 5 + 6 \cdots + k$. Then S_{m-5} can be decomposed into $S_2, S_3, S_5, S_6, \ldots, S_k$. Let $G_1 = S_1 \subseteq S_2 \subseteq H_2$ and $G_4 = H_1 = S_4$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

Case 5. $m_1 = 5$. Thus, $H_1 \in \{S_1 + S_4, S_2 + S_3, S_5\}$ and $m_2 = \binom{k+1}{2} - 4$.

⋆ First, suppose that $H_1 = S_1 + S_4$. Then $H_2 = S_1 + S_{m-6}$ where $m - 6 = \binom{k+1}{2} - 5$. Since $2 + 3 + 5 + 6 + \cdots + k = \binom{k+1}{2} - 5$, there is an ascending subgraph sequence $S_2, S_3, S_5, S_6, \ldots, S_k$ in $S_{m-6} \subseteq H_2$. Let $G_1 = S_1 \subseteq H_1$ and $G_4 = S_4 \subseteq H_1$. Thus, $S_1, S_2, S_3, S_4, S_5, \ldots, S_k$ is a Ramsey chain of length k in H.

⋆ Next, suppose that $H_1 = S_2 + S_3$. Then $H_2 = S_{m-5}$ where $m - 5 = \binom{k+1}{2} - 4$. Since $1 + 4 + 5 + \cdots + k = \binom{k+1}{2} - 5$, the subgraph S_{m-5} in H_2 can be decomposed into $S_1, S_4, S_5, \ldots, S_k$. Let $G_2 = S_2 \subseteq H_1$ and $G_3 = S_3 \subseteq H_1$. Thus, $S_1, S_2, S_3, S_4, \ldots, S_k$ is a Ramsey chain of length k in H.

⋆ Finally, suppose that $H_1 = S_5$. Then $H_2 = S_2 + S_{m-7}$ where $m - 7 = \binom{k+1}{2} - 6 = 2 + 3 + 4 + 6 + 7 + \cdots + k$. Then S_{m-7} can be decomposed into $S_2, S_3, S_4, S_6, S_7, \ldots, S_k$. Let $G_1 = S_1 \subseteq S_2 \subseteq H_2$ and $G_5 = H_1 = S_5$. Thus, $S_1, S_2, S_3, S_4, S_5, \ldots, S_k$ is a Ramsey chain of length k in H.

Case 6. $m_1 \geq 6$. Thus, $H_1 \in \{S_1 + S_{m_1-1}, S_2 + S_{m_1-2}, S_{m_1}\}$ and

$$m_2 = \binom{k+1}{2} - m_1 + 1.$$

$\star$ First, suppose that $H_1 = S_1 + S_{m_1-1}$. Then $H_2 = S_1 + S_{\binom{k+1}{2}-m_1}$. Since $m_1 \geq 6$, it follows that $m_2 - 1 \geq 5$. Since $m = m_1 + m_2 = \binom{k+1}{2} + 1$, we have $m_1 + (m_2 - 1) = \binom{k+1}{2}$. By Lemma 4.2.1, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into the subsets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \lceil \frac{m_1}{2} \rceil$, and $b_1 < b_2 < \cdots < b_{k_2} \leq \lceil \frac{m_2-1}{2} \rceil$. Furthermore, $a_1 = 1$, $b_{k_2} = k$, $\sum_{i=1}^{k_1} a_i = m_1$, and $\sum_{i=1}^{k_2} b_i = m_2 - 1$. Then H_1 can be decomposed into $S_1 = S_{a_1}, S_{a_2}, \ldots, S_{a_{k_1}}$ and the subgraph $S_{\binom{k+1}{2}-m_1}$ can be decomposed into $S_{b_1}, S_{b_2}, \ldots, S_{b_{k_2}}$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H.

$\star$ Next, suppose that $H_1 = S_2 + S_{m_1-2}$. Then $H_2 = S_{\binom{k+1}{2}-m_1+1}$. Then $m_1 - 1 \geq 5$ and $(m_1 - 1) + m_2 = \binom{k+1}{2}$. By Lemma 4.2.1, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into the subsets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \lceil \frac{m_1-1}{2} \rceil$, and $b_1 < b_2 < \cdots < b_{k_2} \leq \lceil \frac{m_2}{2} \rceil$. Furthermore, $a_1 = 1$, $b_{k_2} = k$, $\sum_{i=1}^{k_1} a_i = m_1$, and $\sum_{i=1}^{k_2} b_i = m_2$. The subgraph $S_1 + S_{m_1-2}$ of H_1 can be decomposed into $S_1 = S_{a_1}, S_{a_2}, \ldots, S_{a_{k_1}}$ and H_2 can be decomposed into $S_{b_1}, S_{b_2}, \ldots, S_{b_{k_2}}$. Thus, $S_1, S_2, S_3, \ldots, S_k$ is a Ramsey chain of length k in H. ∎

$\star$ Finally, suppose that $H_1 = S_{m_1}$. Then $H_2 = S_2 + S_{\binom{k+1}{2}-m_1-1}$. Since $m_2 \geq m_1 \geq 6$ and $m = m_1 + m_2 = \binom{k+1}{2} + 1$, it follows that $\binom{k+1}{2} - m_1 \geq 5$. By Lemma 4.2.1, there exists a partition of $[k] = \{1, 2, \ldots, k\}$ into the subsets $A = \{a_1, a_2, \ldots, a_{k_1}\}$ and $B = \{b_1, b_2, \ldots, b_{k_2}\}$, where $k_1 + k_2 = k$, $a_1 < a_2 < \cdots < a_{k_1} \leq \lceil \frac{m_1}{2} \rceil$, and $b_1 < b_2 < \cdots < b_{k_2} \leq \left\lceil \frac{\binom{k+1}{2}-m_1}{2} \right\rceil$. Furthermore, $a_1 = 1$, $b_{k_2} = k$, $\sum_{i=1}^{k_1} a_i = m_1$, and $\sum_{i=1}^{k_2} b_i = \binom{k+1}{2} - m_1$. Then $H_1 = S_{m_1}$ can be decomposed into $\{S_{a_1}, S_{a_2}, \ldots, S_{a_{k_1}}\}$ and $S_{\binom{k+1}{2}-m_1}$ can be decomposed into $\{S_{b_1}, S_{b_2}, \ldots, S_{b_{k_2}}\}$ where $S_{b_{k_2}} = S_k$. Since either $S_{a_{k_1}} = S_{k-1}$ or $S_{b_{k_2-1}} = S_{k-1}$, it follows that $S_{\binom{k+1}{2}-m_1-1}$ can be decomposed

into $\{S_{b_1}, S_{b_2}, \ldots, S_{b_{k_2-1}}, S_{b_{k_2}-1}\}$, where the last two stars in the decomposition may be the same (that is, it is possible that $b_{k_2-1} = b_{k_2} - 1$). Thus, $S_1 + S_{\binom{k+1}{2}-m_1-1}$ can be decomposed into $\{S_{b_1}, S_{b_2}, \ldots, S_{b_{k_2-1}}, S_1 + S_{b_{k_2}-1}\}$. Therefore, $S_1, S_2, S_3, \ldots, S_{k-1}, S_1+S_{k-1}$ is a Ramsey chain of length k in H. ■

4.3 The Ramsey Index of the 2-Galaxy $S_3 + S_{m-3}$

In this section, we determine the Ramsey indexes of all 2-galaxies $S_3 + S_{m-3}$ of size $m \geq 6$ where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. We saw that $AR(S_3 + S_{m-3}) = 3$ for $\binom{3+1}{2} = 6 \leq m < 10 = \binom{3+2}{2}$. First, we show that $AR(S_3 + S_{m-3}) = 4$ for $\binom{4+1}{2} = 10 \leq m < 15 = \binom{4+2}{2}$.

Proposition 4.3.1 *For each integer m with $10 \leq m \leq 14$, $AR(S_3 + S_{m-3}) = 4$.*

Proof. Let m be an integer with $10 \leq m \leq 14$. Since the size of $S_3 + S_{m-3}$ is less than $\binom{5+1}{2} = 15$, it follows by Observation 4.1.1 that $AR(S_3 + S_{m-3}) \leq 4$. Thus, it remains to show that $AR(S_3 + S_{m-3}) \geq 4$. By Observation 4.1.2, it suffices to show that $AR(S_3 + S_7) \geq 4$.

Let $H = S_3 + S_7$. We show that for every red-blue coloring of H, there is a Ramsey chain in H of length 4. Let c be an arbitrary red-blue coloring of H with red subgraph H_1 of size m_1 and blue subgraph H_2 of size m_2. We may assume that $1 \leq m_1 \leq m_2$ and so $1 \leq m_1 \leq 5$. We consider five cases, according to the value of m_1. In each case, we show that there is a Ramsey chain G_1, G_2, G_3, G_4 of length 4 in H. Hence, $|E(G_i)| = i$ for $1 \leq i \leq 4$ and so $\{G_1, G_2, G_3, G_4\}$ is a decomposition of H.

Case 1. $(m_1, m_2) = (1, 9)$. Then

$$(H_1, H_2) \in \{(S_1, S_2 + S_7), (S_1, S_3 + S_6)\}.$$

Since S_7 can be decomposed into S_3, S_4 and S_6 can be decomposed into S_2, S_4, it follows that S_1, S_2, S_3, S_4 is a Ramsey chain of length 4 in H.

Case 2. $(m_1, m_2) = (2, 8)$. Thus,

$$(H_1, H_2) \in \{(S_2, S_1 + S_7), (S_2, S_3 + S_5), (2S_1, S_2 + S_6)\}.$$

Since $\{G_1, G_2, G_3, G_4\}$ is a decomposition of H, it follows that $G_2 = H_1$. First, suppose that $(H_1, H_2) \in \{(S_2, S_1+S_7), (S_2, S_3+S_5)\}$. Since S_7 can be decomposed into S_3, S_4 and S_5 can be decomposed into S_1, S_4, it follows that S_1, S_2, S_3, S_4 is a Ramsey chain of length 4 in H. Next, suppose that $(H_1, H_2) = (2S_1, S_2 + S_6)$. Then S_2 can be decomposed into S_1, S_1 and S_6 can be decomposed into S_1, S_2, S_3. Let $G_1 = S_1 \subseteq H_2$, $G_2 = 2S_1 = H_1$, $G_3 = S_1 + S_2 \subseteq H_2$, and $G_4 = S_1 + S_3 \subseteq H_2$. Then G_1, G_2, G_3, G_4 is a Ramsey chain of length 4 in H.

Case 3. $(m_1, m_2) = (3, 7)$. Thus,

$$(H_1, H_2) \in \{(S_3, S_7), (S_3, S_3 + S_4), (S_1 + S_2, S_2 + S_5), (S_1 + S_2, S_1 + S_6)\}.$$

First, suppose that $(H_1, H_2) \in \{(S_3, S_7), (S_3, S_3 + S_4)\}$. Since S_3 can be decomposed into S_1, S_2 and S_7 can be decomposed into S_3, S_4, it follows that S_1, S_2, S_3, S_4 is a Ramsey chain of length 4. Next, suppose that $(H_1, H_2) = (S_1 + S_2, S_2 + S_5)$. Since S_5 can be decomposed into S_1, S_2, S_2, we let $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 = H_1$ and $G_4 = 2S_2 \subseteq H_2$, resulting in a Ramsey chain of length 4 in H. Finally, suppose that $(H_1, H_2) = (S_1 + S_2, S_1 + S_6)$. Since S_6 can be decomposed into S_1, S_2, S_3, we let $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 = H_1$, and $G_4 = S_1 + S_3 \subseteq H_2$, resulting in a Ramsey chain of length 4 in H.

Case 4. $(m_1, m_2) = (4, 6)$. In this case,

$$(H_1, H_2) \in \{(S_4, 2S_3), (S_1 + S_3, S_2 + S_4), (S_1 + S_3, S_6), (2S_2, S_1 + S_5)\}.$$

First, suppose that $(H_1, H_2) \in \{(S_4, 2S_3), (S_1+S_3, S_2+S_4), (S_1+S_3, S_6)\}$. Since S_3 can be decomposed into S_1, S_2 and S_6 can be decomposed into S_2, S_4, it follows that S_1, S_2, S_3, S_4 is a Ramsey chain of length 4 in H. Next suppose that $(H_1, H_2) = (2S_2, S_1 + S_5)$. Since S_5 can be decomposed into S_1, S_2, S_2, we let $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 \subseteq H_2$, and $G_4 = 2S_2 = H_1$, resulting a Ramsey chain of length 4 in H.

Case 5. $(m_1, m_2) = (5, 5)$. Thus,

$$(H_1, H_2) \in \{(S_5, S_3 + S_2), (S_1 + S_4, S_2 + S_3), (S_2 + S_3, S_1 + S_4), (S_2 + S_3, S_5)\}.$$

Since S_5 can be decomposed into S_1, S_4, it follows that S_1, S_2, S_3, S_4 is a Ramsey chain of length 4 in H.

Therefore, $AR(H) = 4$ and so $AR(S_3 + S_{m-3}) = 4$ for $10 \leq m \leq 14$. ∎

To determine $AR(S_3 + S_{m-3})$ for all integers $m \geq 10$, we first introduce some additional notation. For each positive integer k, the set of k pairwise edge-disjoint stars $S_1, S_2, \ldots, S_k$ is denoted by

$$\mathcal{S}_k = \{S_1, S_2, \ldots, S_k\}. \tag{4.1}$$

Thus, $\mathcal{S}_k$ is a decomposition of the star $S_{\binom{k+1}{2}}$ of size $\binom{k+1}{2}$. If x is an integer with $1 \leq x \leq k$, then $\mathcal{S}_k - \{S_x\}$ is a decomposition of the star $S_{\binom{k+1}{2}-x}$ of size $\binom{k+1}{2} - x$. In the following, it is also convenient to denote the star S_m of size m by $S_0 + S_m$.

Theorem 4.3.2 *Let $H = S_3 + S_{m-3}$ be a 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. Then*

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \text{ and } k \geq 5 \\ k & \text{if } (i)\ m = \binom{k+1}{2} \text{ and } k = 4 \text{ or } (ii)\ \binom{k+1}{2} + 1 \leq m < \binom{k+2}{2}. \end{cases}$$

Proof. Let $H = S_3 + S_{m-3}$ be a 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ where $k \geq 4$. If $k = 4$, then $\binom{4+1}{2} = 10 \leq m < 15 = \binom{4+2}{2}$. Since $AR(H) = 4$ for $10 \leq m \leq 14$ by Proposition 4.3.1, the result follows when $k = 4$. Thus, we now assume that $k \geq 5$. If $m = \binom{k+1}{2}$ and $k \geq 5$, then $AR(H) = k-1$ by Theorem 4.1.4. If $\binom{k+1}{2}+3 \leq m < \binom{k+2}{2}$, then $AR(H) = k$, again by Theorem 4.1.4. If $m = \binom{k+1}{2}+2$, then $H' = S_2 + S_{m-3} \subseteq H$. Since the size of H' is $m-1 = \binom{k+1}{2}+1$, it follows by Theorem 4.2.2 that $AR(H') = k$. By Observations 4.1.2, $AR(H) \geq AR(H') = k$ and so $AR(H) = k$ by Observation 4.1.1. Hence, we have the following.

$$\text{If } k \geq 5, \text{ then } AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \\ k & \text{if } \binom{k+1}{2} + 2 \leq m < \binom{k+2}{2}. \end{cases} \tag{4.2}$$

It remains to verify the truth of the following statement:

$$\text{If } H = S_3 + S_{m-3} \text{ where } m = \binom{k+1}{2} + 1 \text{ for some integer } k \geq 5, \text{ then } AR(H) = k. \tag{4.3}$$

Since $AR(H) \leq k$ by Observation 4.1.1, we show that $AR(H) \geq k$. Let c be an arbitrary red-blue coloring of H with red subgraph H_1 of size m_1 and blue

subgraph H_2 of size m_2 where $m_1+m_2 = m = \binom{k+1}{2}+1$ and so $m_2 = \binom{k+1}{2}+1-m_1$. We may assume that $1 \leq m_1 \leq m_2$. Let $H = S_3 + S_{m-3} = S_3 + S_{\binom{k+1}{2}-2}$. We consider two cases, according to whether $1 \leq m_1 \leq 5$ or $m_1 \geq 6$.

Case 1. $1 \leq m_1 \leq 5$. We show that S_1, S_2, ..., S_k is a Ramsey chain in H.

- ⋆ If $m_1 = 1$, then $H_1 = S_1$ and $H_2 = S_x + S_{\binom{k+1}{2}-x}$ where $x = 2, 3$. Since $S_{\binom{k+1}{2}-x}$ (in H_2) can be be decomposed into $\mathcal{S}_k - \{S_x\}$, it follows that H_2 can be decomposed into $\mathcal{S}_k$.

- ⋆ If $m_1 = 2$, then $H_1 \in \{2S_1, S_2\}$ and $H_2 = S_x + S_{\binom{k+1}{2}-(x+1)}$ where $x = 1, 2, 3$. Since $S_{\binom{k+1}{2}-(x+1)}$ (in H_2) can be be decomposed into $\mathcal{S}_k - \{S_1, S_x\}$, it follows that H_2 can be decomposed into $\mathcal{S}_k - \{S_1\}$. The subgraph H_1 contains a copy of S_1.

- ⋆ If $m_1 = 3$, then $H_1 \in \{S_3, S_1 + S_2\}$ and $H_2 = S_x + S_{\binom{k+1}{2}-(x+2)}$ where $x = 0, 1, 2, 3$. Since $\binom{k+1}{2} - (x+2) = (1+3+4+\cdots+k) - x$, it follows that $S_{\binom{k+1}{2}-(x+2)}$ (in H_2) can be be decomposed into $\mathcal{S}_k - \{S_2, S_x\}$ and so H_2 can be decomposed into $\mathcal{S}_k - \{S_2\}$. The subgraph H_1 contains a copy of S_2.

- ⋆ If $m_1 = 4$, then $H_1 \in \{S_1 + S_3, 2S_2, S_4\}$ and $H_2 = S_x + S_{\binom{k+1}{2}-(x+3)}$ where $x = 0, 1, 2, 3$. Since $\binom{k+1}{2} - (x+3) = (3+4+\cdots+k) - x$, it follows that $S_{\binom{k+1}{2}-(x+3)}$ (in H_2) can be decomposed into $\mathcal{S}_k - \{S_1, S_2, S_x\}$ and so H_2 can be decomposed into $\mathcal{S}_k - \{S_1, S_2\}$. The subgraph H_1 contains edge-disjoint copies of S_1 and S_2.

- ⋆ If $m_1 = 5$, then $H_1 \in \{S_1 + S_4, S_2 + S_3, S_5\}$ and $H_2 = S_x + S_{\binom{k+1}{2}-(x+4)}$ where $x = 0, 1, 2, 3$. Since $\binom{k+1}{2} - (x+4) = (2+4+5+\cdots+k) - x$, it follows that $S_{\binom{k+1}{2}-(x+4)}$ (in H_2) can be decomposed into $\mathcal{S}_k - \{S_1, S_3, S_x\}$and so H_2 can be decomposed into $\mathcal{S}_k - \{S_1, S_3\}$. The subgraph H_1 contains edge-disjoint copies of S_1 and S_3.

Thus, if $1 \leq m_1 \leq 5$, then S_1, S_2, ..., S_k is a Ramsey chain in H.

Case 2. $m_1 \geq 6$. Then (H_1, H_2) is one of the pairs in the set

$$\Big\{(S_a + S_{m_1-a}, S_{3-a} + S_{\binom{k+1}{2}-2-m_1+a}) : a = 0, 1, 2, 3\Big\},$$

where $S_0 + S_{m_1-0} = S_{m_1}$ and $S_0 + S_{\binom{k+1}{2}-m_1+1} = S_{\binom{k+1}{2}-m_1+1}$. We consider two subcases, according to whether $a = 0$ or $a \neq 0$.

Subcase 2.1. $a = 0$. In this case, $(H_1, H_2) = \left(S_{m_1}, S_3 + S_{\binom{k+1}{2}-2-m_1}\right)$. We show that S_1, S_2, ..., S_k is a Ramsey chain of length k in H. First, suppose that $m_1 \leq k-1$. Since

$$1 + 3 + 4 + \cdots + (m_1 - 1) + (m_1 + 1) + \cdots + k = \tbinom{k+1}{2} - 2 - m_1,$$

it follows that $S_{\binom{k+1}{2}-2-m_1}$ (in H_2) can be decomposed into S_1, S_3, S_4, ..., S_{m_1-1}, S_{m_1+1}, ..., S_k. We decompose S_3 (in H_2) into S_1 and S_2. Let $G_{m_1} = S_{m_1} = H_1$. Then $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. Next, suppose that $m_1 \geq k$. Then there exists an integer $t \geq 0$ such that

$$k + (k-1) + \cdots + (k-t) \leq m_1 < k + (k-1) + \cdots + (k-t) + (k-t-1).$$

Let $d = m_1 - [k + (k-1) + \cdots + (k-t)]$. Then $0 \leq d \leq k-t-2$ and

$$m_1 = k + (k-1) + \cdots + (k-t) + d. \tag{4.4}$$

We now consider four subcases, according to the value of d.

Subcase 2.1.1. $d \in \{0, 1\}$. Since

$$3 + 4 + \cdots + (k-t-1) = \tbinom{k+1}{2} - 3 - (m_1 - d) \leq \tbinom{k+1}{2} - 2 - m_1,$$

it follows that $S_{\binom{k+1}{2}-2-m_1}$ (in H_2) contains pairwise edge-disjoint subgraphs S_3, S_4, $\cdots$, S_{k-t-1}. We decompose S_3 (in H_2) into S_1 and S_2. By (4.4), the graph $H_1 = S_{m_1}$ contains pairwise edge-disjoint subgraphs S_{k-t}, S_{k-t+1}, $\cdots$, S_k. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Subcase 2.1.2. $d = 2$. Since

$$1 + 4 + 5 + \cdots + (k-t-1) \;=\; \binom{k+1}{2} - 3 - m_1 < \binom{k+1}{2} - 2 - m_1,$$

it follows that $S_{\binom{k+1}{2}-2-m_1}$ (in H_2) contains pairwise edge-disjoint subgraphs S_1, S_4, S_5, $\cdots$, S_{k-t-1}. By (4.4), the graph $H_1 = S_{m_1}$ can be decomposed into subgraphs $S_d = S_2$, S_{k-t}, S_{k-t+1}, $\cdots$, S_k. Hence, with $G_3 = S_3 \subseteq H_2$, it follows that $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Subcase 2.1.3. $d = 3$. Then S_3 (in H_2) can be decomposed into $G_1 = S_1$ and $G_2 = S_2$. Since

$$4 + 5 + \cdots + (k - t - 1) \;=\; \binom{k+1}{2} - 3 - m_1 < \binom{k+1}{2} - 2 - m_1,$$

it follows that $S_{\binom{k+1}{2}-2-m_1}$ (in H_2) contains pairwise edge-disjoint subgraphs S_4, S_5, $\cdots$, S_{k-t-1}. By (4.4), the graph $H_1 = S_{m_1}$ can be decomposed into subgraphs $S_d = S_3$, S_{k-t}, S_{k-t+1}, $\cdots$, S_k. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Subcase 2.1.4. $d \geq 4$. Then S_3 (in H_2) can be decomposed into $G_1 = S_1$ and $G_2 = S_2$. Since

$$[3 + 4 + \cdots + (k - t - 1)] - d \;=\; \binom{k+1}{2} - 3 - m_1 < \binom{k+1}{2} - 2 - m_1,$$

it follows that $S_{\binom{k+1}{2}-2-m_1}$ (in H_2) contains pairwise edge-disjoint subgraphs S_3, S_4, $\cdots$, S_{d-1}, S_{d+1}, $\ldots$, S_{k-t-1}. By (4.4), the graph $H_1 = S_{m_1}$ can be decomposed into subgraphs S_d, S_{k-t}, S_{k-t+1}, $\cdots$, S_k. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H when $a = 0$.

Subcase 2.2. $a \neq 0$. In this case, we show that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H. We saw in Subcase 2.1 where $a = 0$ that every subgraph in a Ramsey chain of length k is a star. However, this is not the case when $a \neq 0$. Suppose that $a = 1$ and so $(H_1, H_2) = \left(S_1 + S_{m_1-1}, S_2 + S_{\binom{k+1}{2}-m_1-1}\right)$. First, suppose that $m_1 - 1 \leq k - 1$. Since

$$3 + 4 + \cdots + (m_1 - 2) + m_1 + \cdots + k = \tbinom{k+1}{2} - 2 - m_1 < \tbinom{k+1}{2} - 1 - m_1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint subgraphs S_3, S_4, $\cdots$, S_{m_1-2}, S_{m_1}, $\ldots$, S_k. Let $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, and $G_{m_1-1} = S_{m_1-1} \subseteq H_1$. Thus, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. Next, suppose that $m_1 - 1 \geq k$. Then there exists an integer $t' \geq 0$ such that

$$k + (k-1) + \cdots + (k - t') \leq m_1 - 1 < k + (k-1) + \cdots + (k - t') + (k - t' - 1).$$

Let $d' = (m_1 - 1) - [k + (k-1) + \cdots + (k - t')]$. Then $0 \leq d' \leq k - t' - 2$ and

$$m_1 - 1 = k + (k-1) + \cdots + (k - t') + d'. \tag{4.5}$$

We consider four subcases, according to the value of d'.

Subcase 2.2.1. $d' \in \{0, 1\}$. By (4.5), the subgraph S_{m_1-1} (in H_1) contains pairwise edge-disjoint subgraphs $S_{k-t'}$, $S_{k-t'+1}$, $\cdots$, S_k. Since

$$3 + 4 + \cdots + (k - t' - 1) = \tbinom{k+1}{2} - (2 - d') - m_1 \leq \tbinom{k+1}{2} - m_1 - 1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of S_3, S_4, $\cdots$, $S_{k-t'-1}$. Let $G_1 = S_1 \subseteq H_1$ and $G_2 = S_2 \subseteq H_2$. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Subcase 2.2.2. $d' = 2$. Let $G_1 = S_1 \subseteq S_2 \subseteq H_2 = S_2 + S_{\binom{k+1}{2}-m_1-1}$.

- ⋆ First, suppose that $t' = 0$. Then $m_1 - 1 = k + 2 = (k - 1) + 3$ and so $H_1 = S_1 + S_{m_1-1} = S_1 + S_{(k-1)+3}$ can be decomposed into $G_3 = S_3$ and $G_k = S_1 + S_{k-1}$. Since

  $$2 + 4 + 5 + \cdots + (k - 1) = \tbinom{k+1}{2} - m_1 - 1,$$

  it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) can be decomposed into $G_2 = S_2$, $G_4 = S_4$, $G_5 = S_5$, $\cdots$, $G_{k-1} = S_{k-1}$. Hence, $G_1, G_2, \ldots, G_k$ is a Ramsey chain in H.

- ⋆ Next, suppose that $t' \geq 1$. Since

  $$m_1 - 1 = k + (k-1) + \cdots + (k-t') + 2 = (k-1) + (k-1) + \cdots + (k-t') + 3,$$

  it follows that $H_1 = S_1 + S_{m_1-1}$ can be decomposed into $G_k = S_1 + S_{k-1}$, $G_{k-1} = S_{k-1}$, $G_{k-2} = S_{k-2}$, …, $G_{k-t'} = S_{k-t'}$, $G_3 = S_3$. Since

  $$2 + 4 + 5 + \cdots + (k - t' - 1) = \tbinom{k+1}{2} - m_1 - 1,$$

  it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) can be decomposed into $G_2 = S_2$, $G_4 = S_4$, $G_5 = S_5$, $\cdots$, $G_{k-t'-1} = S_{k-t'-1}$. Let $G_1 = S_1 \subseteq S_2 \subseteq H_2$. Hence, $G_1, G_2, \ldots, G_k$ is a Ramsey chain in H.

Subcase 2.2.3. $d' = 3$. Let $G_1 = S_1 \subseteq H_1$ and $G_2 = S_2 \subseteq H_2$. Since

$$m_1 - 1 = k + (k-1) + \cdots + (k-t') + 3,$$

it follows that S_{m_1-1} (in H_1) can be decomposed into S_3, $S_{k-t'}$, $S_{k-t'+1}$, $\cdots$, S_k. Since

$$4+5+\cdots+(k-t'-1)=\tbinom{k+1}{2}-m_1-2<\tbinom{k+1}{2}-m_1-1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of S_4, S_5, $\cdots$, $S_{k-t'-1}$. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Subcase 2.2.4. $d' \geq 4$. Let $G_1 = S_1 \subseteq H_1$ and $G_2 = S_2 \subseteq H_2$. Since

$$[3+4+\cdots+(k-t'-1)]-d'=\tbinom{k+1}{2}-m_1-2<\tbinom{k+1}{2}-m_1-1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of S_3, S_4, $\cdots$, $S_{d'-1}$, $S_{d'+1}$, $\ldots$, $S_{k-t'-1}$. By (4.5), the subgraph S_{m_1-1} (in H_1) can be decomposed into $S_{d'}$, $S_{k-t'}$, $S_{k-t'+1}$, $\cdots$, S_k. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

A similar argument shows that there is a Ramsey chain of length k in H when $a = 2, 3$. This verifies the statement in (4.3). The result then follows by the statements in (4.2) and (4.3). ∎

4.4 The Ramsey Index of the 2-Galaxy $S_4 + S_{m-4}$

We now turn our attention to 2-galaxies $S_4 + S_{m-4}$ of size $m \geq 10$ where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. First, we determine $AR(S_4+S_{m-4})$ for $k = 4, 5$.

Proposition 4.4.1 *For each integer m with $\binom{4+1}{2} = 10 \leq m < 15 = \binom{4+2}{2}$,*

$$AR(S_4 + S_{m-4}) = 4.$$

Proof. Since the size of S_4+S_{m-4} is less than $\binom{4+2}{2} = 15$, it follows by Observation 4.1.1 that $AR(S_4+S_{m-4}) \leq 4$. Thus, it remains to show that $AR(S_4+S_{m-4}) \geq 4$. By Observation 4.1.2, it suffices to show that $AR(S_4 + S_6) \geq 4$.

Let $H = S_4 + S_6$. Since the size of H is $10 = \binom{4+1}{2} < \binom{4+2}{2} = 15$, it follows by Observation 4.1.1 that $AR(H) \leq 4$. Hence, it remains to show that $AR(H) \geq 4$; that is, we show that for every red-blue coloring c of H, there is a Ramsey chain in H of length 4 and so $AR_c(H) \geq 4$, which implies that $AR(H) \geq 4$. (In fact, the length of a longest Ramsey chain in H is 4. Hence, if there is a Ramsey chain of length 4 in H, then $AR_c(H) = 4$.)

Let c be an arbitrary red-blue coloring of H with red subgraph H_1 of size m_1 and blue subgraph H_2 of size m_2. We may assume that $1 \leq m_1 \leq m_2$ and so

$1 \leq m_1 \leq 5$. We consider five cases, according the value of m_1. In each case, we show that there is a Ramsey chain G_1, G_2, G_3, G_4 of length 4 in H. Hence, $|E(G_i)| = i$ for $1 \leq i \leq 4$ and $\{G_1, G_2, G_3, G_4\}$ is a decomposition of H.

Case 1. $m_1 = 1$. Then $H_1 = S_1$ and $H_2 \in \{S_3 + S_6, S_4 + S_5\}$.

- ⋆ For $(H_1, H_2) = (S_1, S_3 + S_6)$, we decompose the subgraph S_6 of H_2 into S_2, S_4. Then $G_1 = S_1 = H_1$, $G_2 = S_2$, $G_3 = S_3$ and $G_4 = S_4$.

- ⋆ For $(H_1, H_2) = (S_1, S_4 + S_5)$, we decompose the subgraph S_5 of H_2 into S_2, S_3. Then $G_1 = S_1 = H_1$, $G_2 = S_2$, $G_3 = S_3$ and $G_4 = S_4$.

Case 2. $m_1 = 2$. Then $H_1 \in \{S_2, 2S_1\}$ and $H_2 \in \{S_2 + S_6, 2S_4, S_3 + S_5\}$. Since $\{G_1, G_2, G_3, G_4\}$ is a decomposition of H, it follows that $G_2 = H_1$ and $G_i \in H_2$ for $i = 1, 3, 4$.

- ⋆ For $(H_1, H_2) = (S_2, S_2 + S_6)$, we decompose the subgraph S_2 of H_2 into $2S_1$ and decompose the subgraph S_6 of H_2 into S_1, S_2, S_3. Then $G_1 = S_1$, $G_2 = S_2 = H_1$, $G_3 = S_1 + S_2$, and $G_4 = S_1 + S_3$.

- ⋆ For $(H_1, H_2) = (S_2, 2S_4)$, we decompose one subgraph S_4 of H_2 into $2S_1, S_2$ and decompose the other subgraph S_4 into $2S_2$. Then $G_1 = S_1$, $G_2 = S_2 = H_1$, $G_3 = S_1 + S_2$, and $G_4 = 2S_2$.

- ⋆ For $(H_1, H_2) = (2S_1, S_3 + S_5)$, we decompose the subgraph S_3 of H_2 into $3S_1$ and decompose the subgraph S_5 of H_2 into S_2, S_3. Then $G_1 = S_1$, $G_2 = 2S_1 = H_1$, $G_3 = S_1 + S_2$, and $G_4 = S_1 + S_3$.

Case 3. $m_1 = 3$. Then $H_1 \in \{S_3, S_1 + S_2\}$ and $H_2 \in \{S_1 + S_6, S_4 + S_3, S_3 + S_4, S_2 + S_5\}$.

- ⋆ For $(H_1, H_2) = (S_3, S_1 + S_6)$, we decompose the subgraph S_6 of H_2 into S_2, S_4. Then $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 = H_1$, and $G_4 = S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_3, S_4 + S_3)$, we decompose H_1 into S_1, S_2. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_3 \subseteq H_2$, and $G_4 = S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_1 + S_2, S_3 + S_4)$, let $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_3 \subseteq H_2$, and $G_4 = S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_1 + S_2, S_2 + S_5)$, we decompose the subgraph S_2 of H_2 into $2S_1$ and decompose the subgraph S_5 of H_2 into S_2, S_3. Then $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 = H_1$, and $G_4 = S_1 + S_3 \subseteq H_2$.

Case 4. $m_1 = 4$. Then $H_1 \in \{S_4, S_1 + S_3, 2S_2\}$ and $H_2 \in \{S_6, S_4 + S_2, 2S_3, S_1 + S_5, S_2 + S_4\}$.

- ⋆ For $(H_1, H_2) = (S_4, S_6)$, we decompose the subgraph $S_4 = H_1$ into S_1, S_3 and decompose the subgraph $S_6 = H_2$ into S_2, S_4. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 = H_1$, and $G_4 = S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_4, S_4 + S_2)$, we decompose the subgraph $S_4 = H_1$ into S_1, S_3. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_1$, and $G_4 = S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_1 + S_3, 2S_3)$, we decompose one of the subgraphs $S_3 \subseteq H_2$ into S_1, S_2. Then $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_2$, and $G_4 = S_1 + S_3 = H_1$.

- ⋆ For $(H_1, H_2) = (S_1 + S_3, S_1 + S_5)$, we decompose one of the subgraphs S_5 into S_2, S_3. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_1$, and $G_4 = S_1 + S_3 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (2S_2, S_2 + S_4)$, we decompose the subgraph $S_2 \subseteq H_2$ into $2S_1$ and decompose the subgraph $S_4 \subseteq H_2$ into $2S_2$. Then $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 \subseteq H_2$, and $G_4 = 2S_2 = H_1$.

Case 5. $m_1 = 5$. Then $H_1 \in \{S_5, S_1 + S_4, S_2 + S_3\}$ and $H_2 \in \{S_4 + S_1, S_3 + S_2, S_5, S_2 + S_3, S_1 + S_4\}$.

- ⋆ For $(H_1, H_2) = (S_5, S_4 + S_1)$, we decompose the subgraph $S_5 = H_1$ into S_2, S_3 Then $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_3 \subseteq H_1$, and $G_4 = S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_1 + S_4, S_3 + S_2)$, let $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_2$, and $G_4 = S_4 \subseteq H_1$.

- ⋆ For $(H_1, H_2) = (S_1 + S_4, S_5)$, we decompose the subgraph $S_5 = H_2$ into S_2, S_3 Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_2$, and $G_4 = S_4 \subseteq H_1$.

- ⋆ For $(H_1, H_2) = (S_2 + S_3, S_2 + S_3)$, we decompose the subgraph $S_2 \subseteq H_1$ into $2S_1$. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_2$, and $G_4 = S_1 + S_3 \subseteq H_1$.

- ⋆ For $(H_1, H_2) = (S_2 + S_3, S_1 + S_4)$, let $G_1 = S_1 \subseteq H_2$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_3 \subseteq H_1$, and $G_4 = S_4 \subseteq H_2$.

Therefore, $AR_c(H) = 4$ for every red-blue coloring of H and so $AR(H) = 4$. ∎

Proposition 4.4.2 *For each integer m with $\binom{5+1}{2} = 15 \leq m < 21 = \binom{5+2}{2}$,*

$$AR(S_4 + S_{m-4}) = 5.$$

Proof. Since the size of $S_4 + S_{m-4}$ is less than $\binom{5+2}{2} = 21$, it follows by Observation 4.1.1 that $AR(S_4+S_{m-4}) \leq 5$. Thus, it remains to show that $AR(S_4+S_{m-4}) \geq 5$. By Observation 4.1.2, it suffices to show that $AR(S_4 + S_{11}) \geq 5$.

Let $H = S_4 + S_{11}$. Since the size of H is $15 = \binom{4+1}{2} < \binom{5+2}{2} = 21$, it follows by Observation 4.1.1 that $AR(H) \leq 5$. Hence, it remains to show that $AR(H) \geq 5$; that is, we show that for every red-blue coloring c of H, there is a Ramsey chain in H of length 5 and so $AR_c(H) \geq 5$, which implies that $AR(H) \geq 5$.

Let c be an arbitrary red-blue coloring of H with red subgraph H_1 of size m_1 and blue subgraph H_2 of size m_2. We may assume that $1 \leq m_1 \leq m_2$ and so $1 \leq m_1 \leq 7$. We consider seven cases, according the value of m_1. In each case, we show that there is a Ramsey chain G_1, G_2, G_3, G_4, G_5 of length 5 in H. Hence, $|E(G_i)| = i$ for $1 \leq i \leq 5$ and $\{G_1, G_2, G_3, G_4, G_5\}$ is a decomposition of H.

Case 1. $m_1 = 1$. Then $H_1 = S_1$ and $H_2 \in \{S_3 + S_{11}, S_4 + S_{10}\}$.

- ⋆ For $(H_1, H_2) = (S_1, S_3+S_{11})$, we decompose the subgraph S_{11} of H_2 into S_2, S_4, S_5.

- ⋆ For $(H_1, H_2) = (S_1, S_4+S_{10})$, we decompose the subgraph S_{10} of H_2 into S_2, S_3, S_5.

Then $G_i = S_i$ for $1 \leq i \leq 5$ In each situation.

Case 2. $m_1 = 2$. Then $H_1 \in \{S_2, 2S_1\}$ and $H_2 \in \{S_2 + S_{11}, S_4 + S_9, S_3 + S_{10}\}$. Since $\{G_1, G_2, \ldots, G_5\}$ is a decomposition of H, it follows that $G_2 = H_1$ and $G_i \in H_2$ for $i = 1, 3, 4, 5$.

- ⋆ For $(H_1, H_2) = (S_2, S_2 + S_{11})$, we decompose the subgraph S_2 of H_2 into $2S_1$ and decompose the subgraph S_{11} of H_2 into S_1, S_3, S_3, S_4. Then $G_1 = S_1$, $G_2 = S_2 = H_1$, $G_3 = S_3$, $G_4 = S_1 + S_3$, $G_5 = S_1 + S_4$.

- ⋆ For $(H_1, H_2) = (S_2, S_4+S_9)$, we decompose the subgraph S_9 of H_2 into S_1, S_3, S_5. Thus, $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (2S_1, S_3+S_{10})$, we decompose the subgraph S_3 of H_2 into $3S_1$ and decompose the subgraph S_{10} of H_2 into S_1, S_2, S_3, S_4. Then $G_1 = S_1 \subseteq S_{10}$, $G_2 = 2S_1 = H_1$, $G_3 = S_1 + S_2$, $G_4 = S_1 + S_3$, and $G_4 = S_1 + S_4$.

Case 3. $m_1 = 3$. Then $H_1 \in \{S_3, S_1 + S_2\}$ and $H_2 \in \{S_1 + S_{11}, S_4 + S_8, S_3 + S_9, S_2 + S_{10}\}$.

- ⋆ For $(H_1, H_2) = (S_3, S_1 + S_{11})$, we decompose the subgraph $S_{11} \subseteq H_2$ into S_2, S_4, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_3, S_4 + S_8)$, we decompose the subgraph $S_8 \subseteq H_2$ into S_1, S_2, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_2, S_3 + S_9)$, we decompose the subgraph $S_9 \subseteq H_2$ into S_4, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_2, S_2 + S_{10})$, we decompose the subgraph $S_2 \subseteq H_2$ into $2S_1$ and decompose the subgraph $S_{10} \subseteq H_2$ into $2S_3, S_4$. Then let $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_3 \subseteq H_2$, $G_4 = S_1 + S_3 \subseteq H_2$, and $G_5 = S_1 + S_4 \subseteq H_2$.

Case 4. $m_1 = 4$. Then $H_1 \in \{S_4, S_1 + S_3, 2S_2\}$ and $H_2 \in \{S_{11}, S_4 + S_7, S_3 + S_8, S_1 + S_{10}, S_2 + S_9\}$.

- ⋆ For $(H_1, H_2) = (S_4, S_{11})$, we decompose the subgraph $S_{11} \subseteq H_2$ into S_1, S_2, S_3, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_4, S_4 + S_7)$, we decompose the subgraph $S_4 \subseteq H_2$ into S_1, S_3 and decompose the subgraph $S_7 \subseteq H_2$ into S_2, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_3, S_3 + S_8)$, we decompose the subgraph $S_3 \subseteq H_2$ into S_1, S_2 and decompose the subgraph $S_8 \subseteq H_2$ into $2S_4$. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_1$, $G_4 = S_4 \subseteq H_2$, and $G_5 = S_1 + S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_1+S_3, S_1+S_{10})$, we decompose the subgraph $S_{10} \subseteq H_2$ into $S_2, 2S_4$. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_1$, $G_4 = S_4 \subseteq H_2$, and $G_5 = S_1 + S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (2S_2, S_2 + S_9)$, we decompose one subgraph $S_2 \subseteq H_1$ into $2S_1$, decompose the other subgraph $S_2 \subseteq H_2$ into $2S_1$, and decompose the subgraph $S_9 \subseteq H_2$ into S_2, S_3, S_4. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 \subseteq H_1$, $G_4 = S_1 + S_3 \subseteq H_2$, and $G_5 = S_1 + S_4 \subseteq H_2$.

Case 5. $m_1 = 5$. Then $H_1 \in \{S_5, S_1 + S_4, S_2 + S_3\}$ and $H_2 \in \{S_4 + S_6, S_3 + S_7, S_{10}, S_2 + S_8, S_1 + S_9\}$.

- ⋆ For $(H_1, H_2) = (S_5, S_4 + S_6)$, we decompose the subgraph $S_6 \subseteq H_2$ into S_1, S_2, S_3. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_4, S_3 + S_7)$, we decompose the subgraph $S_7 \subseteq H_2$ into S_2, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_4, S_{10})$, we decompose the subgraph $S_{10} \subseteq H_2$ into S_2, S_3, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_2 + S_3, S_2 + S_8)$, we decompose the subgraph $S_2 \subseteq H_1$ into $2S_1$ and decompose the subgraph $S_8 \subseteq H_2$ into $S_1, 2S_2, S_3$. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_1 + S_2 \subseteq H_2$, $G_4 = S_1 + S_3 \subseteq H_1$, and $G_5 = S_2 + S_3 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_2 + S_3, S_1 + S_9)$, we decompose the subgraph $S_9 \subseteq H_2$ into S_4, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

Case 6. $m_1 = 6$. Then $H_1 \in \{S_6, S_1 + S_5, S_2 + S_4, 2S_3\}$ and $H_2 \in \{S_4 + S_5, S_3 + S_6, S_2 + S_7, S_9, S_1 + S_8\}$.

- ⋆ For $(H_1, H_2) = (S_6, S_4 + S_5)$, we decompose the subgraph $S_6 \subseteq H_1$ into S_1, S_2, S_3. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_5, S_3 + S_6)$, we decompose the subgraph $S_6 \subseteq H_2$ into S_2, S_4. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_2 + S_4, S_2 + S_7)$, we decompose the subgraph $S_4 \subseteq H_1$ into $2S_1, S_2$, decompose the subgraph $S_2 \subseteq H_2$ into $2S_1$, and decompose the subgraph $S_7 \subseteq H_2$ into S_3, S_4. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_1 + S_2 \subseteq H_1$, $G_4 = S_1 + S_3 \subseteq H_2$, and $G_5 = S_1 + S_4 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_2 + S_4, S_9)$, we decompose the subgraph $S_9 \subseteq H_2$ into S_1, S_3, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (2S_3, S_1+S_8)$, we decompose one subgraph $S_3 \subseteq H_1$ into S_1, S_2 and decompose the subgraph $S_8 \subseteq H_2$ into $S_2, 2S_3$. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_2$, $G_3 = S_3 \subseteq H_2$, $G_4 = S_1 + S_3 \subseteq H_2$, and $G_5 = S_2 + S_3 \subseteq H_1$.

Case 7. $m_1 = 7$. Then $H_1 \in \{S_7, S_1+S_6, S_2+S_5, S_3+S_4\}$ and $H_2 \in \{2S_4, S_3+S_5, S_2+S_6, S_1+S_7, S_8\}$.

- ⋆ For $(H_1, H_2) = (S_7, 2S_4)$, we decompose $S_7 = H_1$ into S_2, S_5 and decompose one subgraph $S_4 \subseteq H_2$ into S_1, S_3. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_1 + S_6, S_3 + S_5)$, we decompose $S_6 \subseteq H_1$ into S_2, S_4. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_2+S_5, S_2+S_6)$, we decompose $S_2 \subseteq H_1$ into $2S_1$, decompose $S_5 \subseteq H_1$ into S_2, S_3, and decompose $S_6 \subseteq H_2$ into $2S_3$. Then $G_1 = S_1 \subseteq H_1$, $G_2 = S_2 \subseteq H_1$, $G_3 = S_3 \subseteq H_2$, $G_4 = S_1 + S_3 \subseteq H_1$, and $G_5 = S_2 + S_3 \subseteq H_2$.

- ⋆ For $(H_1, H_2) = (S_3 + S_4, S_1 + S_7)$, we decompose $S_7 \subseteq H_2$ into S_2, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

- ⋆ For $(H_1, H_2) = (S_3 + S_4, S_8)$, we decompose $S_8 = H_2$ into S_1, S_2, S_5. Then $G_i = S_i$ for $1 \leq i \leq 5$.

Therefore, $AR(H) \geq 5$ and so $AR(H) = AR(S_4 + S_{11}) = 5$. ∎

We now determine $AR(S_4 + S_{m-4})$ for all integers $m \geq 10$.

Theorem 4.4.3 *Let $H = S_4 + S_{m-4}$ be a 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. Then*

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \text{ and } k \geq 6 \\ k & \text{if (i) } m = \binom{k+1}{2} \text{ and } k \in \{4,5\} \text{ or (ii) } \binom{k+1}{2} + 1 \leq m < \binom{k+2}{2}. \end{cases}$$

Proof. Let $H = S_4 + S_{m-4}$ be a 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. By Propositions 4.4.1 and 4.4.2, it follows that $AR(H) = 4$ when $10 \leq m < 15$ and $AR(H) = 5$ when $15 \leq m < 21$. Hence, the statement is true if $k = 4$ or $k = 5$. Therefore, we may assume that $k \geq 6$. By an argument similar to that in the proof of Theorem 4.3.2, it follows that

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \\ k & \text{if } \binom{k+1}{2} + 4 \leq m < \binom{k+2}{2}. \end{cases} \tag{4.6}$$

Thus, it remains to verify the following statement:

$$\text{If } \binom{k+1}{2} + 1 \leq m \leq \binom{k+1}{2} + 3, \text{ then } AR(H) = k. \tag{4.7}$$

Since $m \leq \binom{k+1}{2} + 3 < \binom{k+2}{2}$, it follows by Observation 4.1.1 that $AR(H) \leq k$. Hence, it remains to show that $AR(H) \geq k$. By Observation 4.1.2, it suffices to show that $AR(H) \geq k$ when $m = \binom{k+1}{2} + 1$ where $k \geq 6$. Let $H = S_4 + S_{m-4} = S_4 + S_{\binom{k+1}{2}-3}$ and let there be given a red-blue coloring of H with red subgraph H_1 of size m_1 and blue subgraph H_2 of size m_2. We show that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ of length k in H. Assume, without loss of generality, that $1 \leq m_1 \leq m_2$.

- ⋆ If $1 \leq m_1 \leq 5$, then an argument similar to that in Case 1 of the proof of Theorem 4.3.2 shows that S_1, S_2, …, S_k is a Ramsey chain in H.

- ⋆ If $m_1 = 6$, then (H_1, H_2) is one of the five pairs

$$(S_6, S_4 + S_{\binom{k+1}{2}-9}),\ (S_1 + S_5, S_3 + S_{\binom{k+1}{2}-8}),$$
$$(S_2 + S_4, S_2 + S_{\binom{k+1}{2}-7}),\ (S_2 + S_4, S_{\binom{k+1}{2}-5}),\ (2S_3, S_1 + S_{\binom{k+1}{2}-6}).$$

  Thus, $H_2 = S_x + S_{\binom{k+1}{2}-(x+5)}$ for $x = 0, 1, 2, 3, 4$. Then $\mathcal{S}_k - \{S_2, S_3\}$ is a decomposition of H_2. Since $H_1 \in \{S_6, S_1 + S_5, S_2 + S_4, 2S_3\}$, it follows that H_1 contains edge-disjoint copies of S_2 and S_3. Hence, S_1, S_2, …, S_k is a Ramsey claim in H.

- ⋆ If $m_1 = 7$, then (H_1, H_2) is one of the five pairs

$$(S_7, S_4 + S_{\binom{k+1}{2}-10}),\ (S_1 + S_6, S_3 + S_{\binom{k+1}{2}-9}),$$
$$(S_2 + S_5, S_2 + S_{\binom{k+1}{2}-8}),\ (S_3 + S_4, S_1 + S_{\binom{k+1}{2}-7}),\ (S_3 + S_4, S_{\binom{k+1}{2}-6}).$$

Thus, $H_2 = S_x + S_{\binom{k+1}{2}-(x+6)}$ for $x = 0, 1, 2, 3, 4$. Then $\mathcal{S}_k - \{S_2, S_4\}$ is a decomposition of H_2. Since $H_1 \in \{S_7, S_1 + S_6, S_2 + S_5, S_3 + S_4\}$, it follows that H_1 contains edge-disjoint copies of S_2 and S_4. Hence, S_1, S_2, ..., S_k is a Ramsey claim in H.

Thus, we may assume that $m_1 \geq 8$. Then (H_1, H_2) is one of the pairs in the following set:

$$\{(S_a + S_{m_1-a}, S_{4-a} + S_{\binom{k+1}{2}-3-m_1+a}) : a = 0, 1, 2, 3, 4\}.$$

We consider two cases, according to whether $a \neq 2$ or $a = 2$.

Case 1. $a \neq 2$. In this case, we show that $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. Suppose that $a = 0$. Then $(H_1, H_2) = \left(S_{m_1}, S_4 + S_{\binom{k+1}{2}-m_1-3}\right)$. First, suppose that $m_1 \leq k - 1$. Since

$$[1 + 2 \cdots + k] - (1 + 3 + m_1) = \tbinom{k+1}{2} - 4 - m_1 < \tbinom{k+1}{2} - m_1 - 3,$$

it follows that $S_{\binom{k+1}{2}-m_1-3}$ (in H_2) contains pairwise edge-disjoint copies of S_2, S_4, ..., S_{m_1-1}, S_{m_1+1}, ..., S_k. Since S_4 (in H_2) can be decomposed into S_1, S_3 and $H_1 = S_{m_1}$, it follows that $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. Next, suppose that $m_1 \geq k$. Then there exists an integer $t \geq 0$ such that

$$k + (k-1) + \cdots + (k-t) \leq m_1 < k + (k-1) + \cdots + (k-t) + (k-t-1).$$

Thus, $m_1 = k + (k-1) + \cdots + (k-t) + d$ where $0 \leq d \leq k - t - 2$.

- If $d \in \{0, 1\}$, then $m_1 \geq k + (k-1) + \cdots + (k-t)$ and so $H_1 = S_{m_1}$ contains pairwise edge-disjoint copies of $S_{k-t}, S_{k-t+1}, \ldots, S_k$. Since

  $$2 + 4 + \cdots + (k-t-1) = \tbinom{k+1}{2} - (4-d) - m_1 \leq \tbinom{k+1}{2} - 3 - m_1,$$

  it follows that $S_{\binom{k+1}{2}-m_1-3}$ (in H_2) contains pairwise edge-disjoint copies of S_2, S_4, S_5, ..., S_{k-t-1}. Furthermore, S_4 (in H_2) can be decomposed into S_1, S_3. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

- If $d = 2$, then $m_1 = k + (k-1) + \cdots + (k-t) + 2$ and so $H_1 = S_{m_1}$ can be decomposed into $S_2, S_{k-t}, S_{k-t+1}, \ldots, S_k$. Since

$$4 + 5 + \cdots + (k-t-1) = \tbinom{k+1}{2} - 4 - m_1 < \tbinom{k+1}{2} - 3 - m_1,$$

it follows that $S_{\binom{k+1}{2}-m_1-3}$ (in H_2) contains pairwise edge-disjoint copies of S_4, S_5, …, S_{k-t-1}. Furthermore, S_4 (in H_2) can be decomposed into S_1, S_3. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

- If $d = 3$, then $m_1 = k + (k-1) + \cdots + (k-t) + 3$ and so $H_1 = S_{m_1}$ can be decomposed into $S_1, S_2, S_{k-t}, S_{k-t+1}, \ldots, S_k$. Since

$$4 + 5 + \cdots + (k-t-1) = \tbinom{k+1}{2} - 3 - m_1,$$

it follows that $S_{\binom{k+1}{2}-m_1-3}$ (in H_2) can be decomposed into $S_4, S_5, \ldots, S_{k-t-1}$. Furthermore, S_4 (in H_2) can be decomposed into S_1, S_3. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

- If $d = 4$, then $m_1 = k + (k-1) + \cdots + (k-t) + 4$ and so $H_1 = S_{m_1}$ can be decomposed into S_1, S_3, S_{k-t}, S_{k-t+1}, …, S_k. Since

$$2 + 5 + \cdots + (k-t-1) = \tbinom{k+1}{2} - 4 - m_1 < \tbinom{k+1}{2} - 3 - m_1,$$

it follows that $S_{\binom{k+1}{2}-m_1-3}$ (in H_2) contains pairwise edge-disjoint copies of $S_2, S_5, \ldots, S_{k-t-1}$. Furthermore, S_4 (in H_2) can be decomposed into S_1, S_3. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

- If $d \geq 5$, then $m_1 = k + (k-1) + \cdots + (k-t) + d$ and so $H_1 = S_{m_1}$ can be decomposed into $S_d, S_{k-t}, S_{k-t+1}, \ldots, S_k$. Since

$$[2 + 4 + \cdots + (k-t-1)] - d = \tbinom{k+1}{2} - 4 - m_1 < \tbinom{k+1}{2} - 3 - m_1,$$

it follows that $S_{\binom{k+1}{2}-m_1-3}$ (in H_2) contains pairwise edge-disjoint copies of S_2, S_4, …, S_{d-1}, S_{d+1}, …, S_{k-t-1}, Furthermore, S_4 (in H_2) can be decomposed into S_1, S_3. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Therefore, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H when $a = 0$. A similar argument shows that $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H when $a = 1, 3, 4$.

Case 2. $a = 2$. Then $(H_1, H_2) = (S_2 + S_{m_1-2}, S_2 + S_{\binom{k+1}{2}-m_1-1})$. First, suppose that $m_1 - 2 \leq k-1$. Since $[3+4+\cdots+k]-(m_1-2) = \binom{k+1}{2}-m_1-1$, it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) can be decomposed into $S_3, S_4, \ldots, S_{m_1-3}, S_{m_1-1}, \ldots, S_k$. Since $H_1 = S_2 + S_{m_1-2}$ can be decomposed into S_2, S_{m_1-2}, it follows that $S_1, S_2, \ldots, S_k$ is also a Ramsey chain in H. Next, suppose that $m_1 - 2 \geq k$. Then there exists an integer $t' \geq 0$ such that

$$k + (k-1) + \cdots + (k-t') \leq m_1 - 2 < k + (k-1) + \cdots + (k-t') + (k-t'-1).$$

Thus, $m_1 - 2 = k + (k-1) + \cdots + (k-t') + d'$ where $0 \leq d' \leq k - t' - 2$.

- If $d' = 0$, then $m_1 - 2 = k + (k-1) + \cdots + (k-t')$ and so S_{m_1-2} (in H_1) can be decomposed into $S_{k-t'}, S_{k-t'+1}, \ldots, S_k$. Since

  $$3 + 4 + \cdots + (k-t'-1) = \tbinom{k+1}{2} - (1-d') - m_1 = \tbinom{k+1}{2} - 1 - m_1,$$

  it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) can be decomposed into $S_3, S_4, \ldots, S_{k-t'-1}$. Furthermore, let $G_1 = S_1 \subseteq S_2 \subseteq H_1$ and let $G_2 = S_2 \subseteq H_2$. Thus, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

- If $d' = 1$, then $m_1 - 2 = k + (k-1) + \cdots + (k-t') + 1$. Thus, $t' = 0$, $t' = 1$, or $2 \leq t' \leq k - d' - 2$.

  - If $t' = 0$, then $m_1 - 2 = k + 1 = (k-1) + 2$ and so S_{m_1-2} (in H_1) can be decomposed into $G_2 = S_2$ and $G_{k-1} = S_{k-1}$ (where $k - 1 \neq 2$ since $k \geq 5$). Since

    $$3+4+\cdots+(k-3)+(k-3)+(k-2) = \tbinom{k+1}{2}-m_1-2 < \tbinom{k+1}{2}-m_1-1,$$

    it follows that $S_{\binom{k+1}{2}-1-m_1}$ (in H_2) contains pairwise edge-disjoint copies of $G_i = S_i$ ($i = 3, 4, \ldots, k-3$), S_{k-3}, and S_{k-2}. We decompose both S_2 (in H_1) and S_2 (in H_2) into S_1, S_1. Let $G_1 = S_1 \subseteq S_2 \subseteq H_1$, $G_{k-2} = S_1 + S_{k-3} \subseteq H_2$, $G_{k-1} = S_1 + S_{k-2} \subseteq H_2$, and $G_k = S_1 + S_{k-1} \subseteq H_1$. Thus, $G_1, G_2, \ldots, G_k$ is also a Ramsey chain in H.

- If $t' = 1$, then $m_1 - 2 = k + (k-1) + 1 = (k-1) + (k-2) + 3$ and so S_{m_1-2} (in H_1) can be decomposed into $S_{k-1}, G_{k-2} = S_{k-2}, G_3 = S_3$. [Observe that $k - 2 \neq 3$, for otherwise, $k = 5$ and $m = 16$. Thus, $m_1 \leq 8$. However, $m_1 - 2 = 5 + (5-1) + 1 = 10$, which is impossible.] Since

$$4 + 5 + \cdots + (k-3) + (k-1) = \tbinom{k+1}{2} - m_1 - 2 < \tbinom{k+1}{2} - m_1 - 1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of $G_i = S_i$ for $i \in \{4, 5, \ldots, k-3, k-1\}$. We decompose S_2 (in H_1) into S_1, S_1. Let $G_1 = S_1 \subseteq S_2 \subseteq H_1$, let $G_2 = S_2 \subseteq H_2$, and let $G_k = S_1 + S_{k-1} \subseteq H_1$. Thus, $G_1, G_2, \ldots, G_k$ is a Ramsey chain in H.

- If $2 \leq t' \leq k - d' - 2$, then $4 \leq t' + 2 \leq k - d'$. Since

$$\begin{aligned} m_1 - 2 &= k + (k-1) + \cdots + (k-t') + 1 \\ &= (k-1) + (k-2) + \cdots + (k-t') + (k-t'-1) + (t'+2), \end{aligned}$$

it follows that S_{m_1-2} (in H_1) can be decomposed into G_i for $i \in \{k-2, k-3, \cdots, k-t', k-t'-1, t'+2\}$ and S_{k-1}. Since

$$\begin{aligned} &3 + 4 + \cdots + (t'+1) + (t'+3) + \cdots + (k-t'-2) + (k-1) \\ = \ &\binom{k+1}{2} - m_1 - 2 < \binom{k+1}{2} - m_1 - 1, \end{aligned}$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of $G_i = S_i$ for $i \in \{3, 4, \ldots, t'+1, t'+3, \ldots, k-t'-2, k-1\}$. We decompose $S_2 \subseteq H_1$ into S_1, S_1. Let $G_1 = S_1 \subseteq S_2 \subseteq H_1$, let $G_2 = S_2 \subseteq H_2$, and let $G_k = S_1 + S_{k-1} \subseteq H_1$. Thus, $G_1, G_2, \ldots, G_k$ is a Ramsey chain in H.

- If $d' = 2$, then $m_1 - 2 = k + (k-1) + \cdots + (k-t') + 2$.

  - If $t' = 0$, then $m_1 - 2 = k + 2 = (k-1) + 3$ and so $H_1 = S_2 + S_{m_1-2}$ can be decomposed into $G_1 = S_1$, $G_3 = S_3$, and $G_k = S_1 + S_{k-1}$. Let $G_2 = S_2 \subseteq H_2$. Since

$$4 + 5 + \cdots + (k-2) + (k-1) = \tbinom{k+1}{2} - m_1 - 2 < \tbinom{k+1}{2} - m_1 - 1.$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of $G_i = S_i$ for $i \in \{4, 5, \ldots, k-2, k-1\}$. Thus, $G_1, G_2, \ldots, G_k$ is a Ramsey chain in H.

- If $t' \geq 1$, then since

$$4 + 5 + \cdots + (k - t' - 1) < \tbinom{k+1}{2} - m_1 - 1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) contains pairwise edge-disjoint copies of $G_i = S_i$ for $i \in \{4, 5, \ldots, k-t'-1\}$. Since

$$m_1 - 2 = k + (k-1) + \cdots + (k-t') + 2 = (k-1) + (k-1) + \cdots + (k-t') + 3,$$

it follows that S_{m_1-2} (in H_1) can be decomposed into S_{k-1}, $G_{k-1} = S_{k-1}$, $G_{k-2} = S_{k-2}$, …, $G_{k-t'} = S_{k-t'}$, $G_3 = S_3$. We decompose S_2 (in H_1) into S_1, S_1. Let $G_1 = S_1 \subseteq S_2 \subseteq H_1$, let $G_2 = S_2 \subseteq H_2$, and let $G_k = S_1 + S_{k-1} \subseteq H_1$. Thus, $G_1, G_2, \ldots, G_k$ is a Ramsey chain in H.

- If $d' = 3$, then $m_1 - 2 = k + (k-1) + \cdots + (k-t') + 3$ and so $H_1 = S_2 + S_{m_1-2}$ can be decomposed into $S_k, S_{k-1}, \ldots, S_{k-t'}, S_3, S_2$. Since

$$4 + 5 + \cdots + (k - t' - 1) = \tbinom{k+1}{2} - 3 - (m_1 - 2) = \tbinom{k+1}{2} - m_1 - 1,$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) can be decomposed into $S_4, S_5, \ldots, S_{k-t'-1}$. Let $S_1 \subseteq S_2 \subseteq H_2$. Then $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

- We now assume that $d' \geq 4$. Since

$$m_1 - 2 = k + (k-1) + \cdots + (k-t') + d',$$

it follows that $H_1 = S_2 + S_{m_1-2}$ (in H_1) can be decomposed into S_k, S_{k-1}, …, $S_{k-t'}$, $S_{d'}$, S_2. Since

$$3 + 4 + \cdots + (k - t' - 1) - d' = \tbinom{k+1}{2} - 3 - (m_1 - 2) = \tbinom{k+1}{2} - m_1 - 1.$$

it follows that $S_{\binom{k+1}{2}-m_1-1}$ (in H_2) can be decomposed into S_3, S_4, $\cdots$, $S_{d'-1}$, $S_{d'+1}$, …, $S_{k-t'-1}$. Let $S_1 \subseteq S_2 \subseteq H_2$. Then $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. ∎

As a consequence of Corollary 4.1.5 , Theorems 4.2.2, 4.3.2, and 4.4.3, we have the following.

Theorem 4.4.4 *Let $H = S_x + S_{m-x}$ be a 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$. If $1 \leq x \leq 4$, then*

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \text{ and } x \leq k-2 \\ k & \text{otherwise.} \end{cases}$$

4.5 Conjectures and Problems

For integers x and m where $5 \leq x \leq \lfloor \frac{m}{2} \rfloor$ and $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$, the values of $AR(S_x + S_{m-x})$ are not known in general. However, all such 2-galaxies $S_x + S_{m-x}$ whose Ramsey indexes have been determined lead us to the following conjecture.

Conjecture 4.5.1 *If $H = S_x + S_{m-x}$ is a 2-galaxy of size m where $5 \leq x \leq \lfloor \frac{m}{2} \rfloor$ and $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$, then*

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \text{ and } x \leq k-2 \\ k & \text{otherwise.} \end{cases}$$

It is straightforward to show that Conjecture 4.5.1 is true for 2-galaxies $S_5 + S_{m-x}$ of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for small values of k. For example, if $H = S_5 + S_{m-5}$ where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ and $k = 4, 5, 6$, then $AR(H) = k$. By Theorem 4.1.4, if $H = S_x + S_{m-x}$ is a 2-galaxy of size m where $5 \leq x \leq \frac{m}{2}$ and $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 4$, then $AR(H) = k-1$ when $m = \binom{k+1}{2}$ and $x \leq k-2$. Also, $AR(H) = k$ when $\binom{k+1}{2} + x \leq m < \binom{k+2}{2}$. Thus, it remains to investigate the situation when

(i) $m = \binom{k+1}{2}$ and $x \geq k-1$ and

(ii) $\binom{k+1}{2} + 1 \leq m \leq \binom{k+1}{2} + (x-1)$.

Furthermore, if H' is a 2-galaxy of size m' such that $H' \subseteq H$, then $AR(H') \leq AR(H) \leq k$ for every 2-galaxy H of size m where $m' \leq m < \binom{k+2}{2}$ by Observation 4.1.2. Therefore, to verify Conjecture 4.5.1, it suffices to verify the following conjecture.

Conjecture 4.5.2 *Let $H = S_x + S_{m-x}$ be a 2-galaxy of size m where $5 \leq x \leq \left\lfloor \frac{m}{2} \right\rfloor$ and $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 5$. If (i) $m = \binom{k+1}{2}$ and $x \geq k-1$ or (ii) $m = \binom{k+1}{2} + 1$ and $x \leq k-2$, then $AR(H) = k$.*

For 2-galaxies H of size $\binom{k+1}{2}$ where $k \geq 5$, we have obtained some sufficient conditions for red-blue colorings of H to produce a Ramsey chain of length k. These sufficient conditions involve the structures of the red or blue subgraphs of H. For example, the next two results provide such conditions.

Proposition 4.5.3 *For integers x and k where $\max\{5, k-1\} \leq x \leq \frac{1}{2}\binom{k+1}{2}$ and $k \geq 5$, let $H = S_x + S_{\binom{k+1}{2}-x}$ be a 2-galaxy of size $\binom{k+1}{2}$. Suppose that H has a red-blue coloring with red subgraph H_1 of size m_1 and blue subgraph H_2 of size m_2 with $1 \leq m_1 \leq m_2$ such that $H_1 = S_a + S_{m_1-a}$ where $0 \leq a < m_1 - a \leq k$ and $x - a \leq k$. If either (i) $x = m_1 - a$ and $H_2 = S_{\binom{k+1}{2}-m_1}$ or (ii) $x \notin \{2a, m_1, 2(m_1-a)\}$, then $S_1, S_2, \ldots, S_k$ is a Ramsey chain of H.*

Proof. First, suppose that (i) occurs. If $a = 0$, then $H_1 = S_{m_1}$ and $H_2 = S_{\binom{k+1}{2}-m_1}$. Since $m_1 \leq k$, it follows that $H_2 = S_{\binom{k+1}{2}-m_1}$ can be decomposed into $\mathcal{S}_k - \{S_{m_1}\}$ and so $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. Thus, we may assume that $a > 0$. Since $a < m_1 - a \leq k$ and $(1 + 2 + \cdots + k) - [a + (m_1 - a)] = \binom{k+1}{2} - m_1$, it follows that $H_2 = S_{\binom{k+1}{2}-m_1}$ can be decomposed into $\mathcal{S}_k - \{S_a, S_{m_1-a}\}$. Furthermore, H_1 can be decomposed into S_a, S_{m_1-a} where $S_a \neq S_{m_1-a}$. Thus, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H.

Next, suppose that (ii) occurs. If $a = 0$, then $H_1 = S_{m_1}$ and $H_2 = S_x + S_{\binom{k+1}{2}-x-m_1}$. Since $x \neq m_1$, $x \leq k$, and $m_1 \leq k$, it follows that $S_{\binom{k+1}{2}-x-m_1}$ (in H_2) can be decomposed into $\mathcal{S}_k - \{S_x, S_{m_1}\}$ and so H_2 can be decomposed into $\mathcal{S}_k - \{S_{m_1}\}$. Hence, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. Thus, we may assume that $a > 0$. Then

$$H_2 = S_{x-a} + S_{\binom{k+1}{2}-x-(m_1-a)} \text{ or } H_2 = S_{x-m_1+a} + S_{\binom{k+1}{2}-x-a}.$$

Since $x \notin \{2a, m_1, 2(m_1-a)\}$, it follows that

$$x - a \notin \{a, m_1 - a\} \text{ and } x - (m_1 - a) \neq m_1 - a.$$

$\star$ First, suppose that $H_2 = S_{x-a} + S_{\binom{k+1}{2}-x-(m_1-a)}$. Since

$$[1+2+\cdots+k]-[a+(m_1-a)+(x-a)]=\tbinom{k+1}{2}-x-(m_1-a),$$

it follows that $S_{\binom{k+1}{2}-x-(m_1-a)} \subseteq H_2$ can be decomposed into

$$\mathcal{S}_k-\{S_a, S_{m_1-a}, S_{x-a}\}$$

and so H_2 can be decomposed into $\mathcal{S}_k-\{S_a, S_{m_1-a}\}$.

$\star$ Next, suppose that $H_2 = S_{x-m_1+a}+S_{\binom{k+1}{2}-x-a}$. Since

$$[1+2+\cdots+k]-[a+(m_1-a)+(x-m_1+a)]=\tbinom{k+1}{2}-x-a,$$

it follows that $S_{\binom{k+1}{2}-x-a} \subseteq H_2$ can be decomposed into

$$\mathcal{S}_k-\{S_a, S_{m_1-a}, S_{x-m_1+a}\}$$

and so H_2 can be decomposed into $\mathcal{S}_k-\{S_a, S_{m_1-a}\}$.

In either case, H_1 can be decomposed into S_a, S_{m_1-a} where $S_a \neq S_{m_1-a}$. Thus, $S_1, S_2, \ldots, S_k$ is a Ramsey chain in H. ∎

In Proposition 4.5.3, the red subgraph in a red-blue coloring of a 2-galaxy consists of two non-isomorphic stars. The next result involves a red-blue coloring of a 2-galaxy whose red subgraph consists of two isomorphic stars. The proof of this result employs an argument similar to that employed in the proof of Proposition 4.5.3.

Proposition 4.5.4 *For integers x and k with $\max\{5, k-1\} \leq x \leq 2(k-2)$ and $k \geq 5$, let $H = S_x + S_{\binom{k+1}{2}-x}$ be a 2-galaxy of size $\binom{k+1}{2}$ with a red-blue coloring. If the red subgraph H_1 belongs to $\{2S_1, 2S_2, 2S_3\}$, then H has a Ramsey chain $G_1, G_2, \ldots, G_k$ of length k such that*

(1) *if $H_1 = 2S_1$, then $G_1 = S_1$ and $G_i \in \{S_1+S_{i-1}, S_2+S_{i-2}\}$ for $2 \leq i \leq k$;*

(2) *if $H_1 = 2S_2$, then $G_1 = S_i$ for $i = 1, 2$, and $G_i \in \{S_1+S_{i-1}, S_2+S_{i-2}\}$ for $3 \leq i \leq k$;*

(3) *if $H_1 = 2S_3$, then $G_1 = S_i$ for $i = 1, 2, 3$, $G_4 \in \{S_4, S_1+S_3\}$, and $G_i \in \{S_1+S_{i-1}, S_2+S_{i-2}\}$ for $5 \leq i \leq k$.*

Proof. Let $x = k + t$ for some integer t. Since $\max\{5, k-1\} \leq x \leq 2(k-2)$, it follows that $-1 \leq t \leq \frac{1}{2}\left[\binom{k}{2} - k\right]$ and $k \geq \max\{5-t, t+4\}$. We consider three cases, depending on whether $H_1 = 2S_1$, $H_1 = 2S_2$, or $H_1 = 2S_3$.

Case 1. $H_1 = 2S_1$. Since $H_1 = 2S_1$, it follows that $H_2 = S_{k+t-1} + S_{\binom{k+1}{2}-k-t-1}$. We show that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H. Necessarily, $G_2 = H_1 = 2S_1$.Thus, it remains to show that H_2 can be decomposed into $G_1, G_3, G_4, \ldots, G_k$ where each G_i is a 2-galaxy for $3 \leq i \leq k$. There are three possibilities, according to whether $t = -1$, $t = 0$, or $t \geq 1$, which we now consider.

Subcase 1.1. $t = -1$. In this case, $k \geq 6$. Then $H = S_{k-1} + S_{\binom{k+1}{2}-k+1}$ and $H_2 = S_{k-2} + S_{\binom{k+1}{2}-k}$ Since $S_{k-2} \subseteq H_2$ can be decomposed into $k-2$ copies of S_1 and $S_{\binom{k+1}{2}-k} \subseteq H_2$ can be decomposed into $S_1, S_2, \ldots, S_{k-1}$, it follows that H_2 can be decomposed into $G_1 = S_1$, $G_i = S_1 + S_{i-1}$ for $3 \leq i \leq k$.

Subcase 1.2. $t = 0$. In this case, $k \geq 5$. Then $H = S_k + S_{\binom{k+1}{2}-k-1}$ and $H_2 = S_{k-1} + S_{\binom{k+1}{2}-k-1}$ Since $S_{k-1} \subseteq H_2$ can be decomposed into $k-1$ copies of S_1 and $S_{\binom{k+1}{2}-k} \subseteq H_2$ can be decomposed into $S_2, S_3, \ldots, S_{k-1}$, it follows that H_2 can be decomposed into $G_1 = S_1$, $G_i = S_1 + S_{i-1}$ for $3 \leq i \leq k$.

Subcase 1.3. $t \geq 1$. Then $H = S_{k+t} + S_{\binom{k+1}{2}-k-t}$ and $H_2 = S_{k+t-1} + S_{\binom{k+1}{2}-k-t-1}$ Since

$$k + t - 1 = 2(t+1) + (k-t-3) \text{ where } k - t - 3 \geq 1,$$

it follows that $S_{k+t-1} \subseteq H_2$ can be decomposed into $t+1$ copies of S_2 and $k-t-3$ copies of S_1. Furthermore, since

$$\binom{k+1}{2} - k - t - 1 = 1 + 2 + \cdots + (k-1) - (t+1) = \sum_{i=1}^{k-(t+2)} i + \sum_{i=k-(t+2)}^{k-2} i,$$

it follows that $S_{\binom{k+1}{2}-k-t-1} \subseteq H_2$ can be decomposed into S_1, S_2, ..., $S_{k-(t+2)}$, $S_{k-(t+2)}$, $S_{k-(t+1)}$, ..., S_{k-2}. Thus, H_2 can be decomposed into $G_1 = S_1$, $G_i = S_1 + S_{i-1}$ for $3 \leq i \leq k-t-1$, and $G_i = S_2 + S_{i-2}$ for $k-t \leq i \leq k$.

Therefore, there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H and so $AR_c(H) = k$ if $H_1 = 2S_1$.

Case 2. $H_1 = 2S_2$. Since $H_1 = 2S_2$, it follows that $H_2 = S_{k+t-2} + S_{\binom{k+1}{2}-k-t-2}$. We show that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H. Necessarily,

$$H_1 \text{ is decomposed into } G_1 = S_1 \subseteq H_1 \text{ and } G_3 = S_1 + S_2.$$

Thus, it remains to show that H_2 can be decomposed into $G_2, G_4, G_5, \ldots, G_k$ where each graph G_i is a 2-galaxy for $i = 2$ and $4 \leq i \leq k$. There are two possibilities, according to whether $t = -1$ or $t \geq 1$.

Subcase 2.1. $t = -1$. Then $H = S_{k-1} + S_{\binom{k+1}{2}-k+1}$ and $H_2 = S_{k-3} + S_{\binom{k+1}{2}-k-1}$ Then S_{k-3} can be decomposed into $k-3$ copies of S_1. Since

$$\tbinom{k+1}{2} - k - 1 = 2 + 3 + \cdots + (k-1),$$

it follows that $S_{\binom{k+1}{2}-k-1} \subseteq H_2$ can be decomposed into $S_2, S_3, \ldots, S_{k-1}$. Therefore, H_2 can be decomposed into $G_2 = S_2$, $G_i = S_1 + S_{i-1}$ for $4 \leq i \leq k$.

Subcase 2.2. $t \geq 0$. Then $H = S_{k+t} + S_{\binom{k+1}{2}-k-t}$ and $H_2 = S_{k+t-2} + S_{\binom{k+1}{2}-k-t-2}$. We consider two subcases, according to whether $k = t+4$ or $k > t+4$.

Subcase 2.2.1. $k = t+4$. Then $H = S_{2t+4} + S_{\binom{t+5}{2}-(2t+4)}$ and $H_2 = S_{2t+2} + S_{\binom{t+5}{2}-(2t+4)-2}$. Then S_{2t+2} can be decomposed into $t+1$ copies of S_2. Since

$$\begin{aligned}
\binom{t+5}{2} - (2t+4) - 2 &= 1 + 2 + \cdots + (t+4) - 2t - 6 \\
&= 2 + 4 + 5 + \cdots + (t+4) - 2(t+1) \\
&= 2 + [2 + 3 + \cdots + (t+2)],
\end{aligned}$$

it follows that $S_{\binom{t+5}{2}-(2t+4)-2}$ can be decomposed into $G_2 = S_2$, S_2, S_3 …, S_{t+2}. Thus, H_2 can be decomposed into $G_2 = S_2$, $G_i = S_2 + S_{i-2}$ for $4 \leq i \leq t+4 = k$.

Subcase 2.2.2. $k > t+4$. Since

$$k + t - 2 = 2(t+1) + (k-t-4) \text{ where } k - t - 4 \geq 1,$$

it follows that $S_{k+t-1} \subseteq H_2$ can be decomposed into $t+1$ copies of S_2 and $k-t-4$ copies of S_1. Furthermore, since

$$\binom{k+1}{2} - k - t - 2 = 2 + 3 + \cdots + (k-1) - (t+1) = \sum_{i=2}^{k-t-2} i + \sum_{i=k-t-2}^{k-2} i,$$

it follows that $S_{\binom{k+1}{2}-k-t-2} \subseteq H_2$ can be decomposed into $G_2 = S_2$, S_3, ..., S_{k-t-2}, S_{k-t-2}, S_{k-t-1}, ..., S_{k-2}. Thus, H_2 can be decomposed into $G_2 = S_2$, $G_i = S_1 + S_{i-1}$ for $4 \leq i \leq k-t-2$, and $G_i = S_2 + S_{i-2}$ for $k-t-1 \leq i \leq k$.

Therefore, there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H and so $AR_c(H) = k$ if $H_1 = 2S_2$.

Case 3. $H_1 = 2S_3$. Since $H_1 = 2S_3$, it follows that $H_2 = S_{k+t-3} + S_{\binom{k+1}{2}-k-t-3}$. We show that there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H. First,

$$H_1 = 2S_3 \text{ can be decomposed into } G_i = S_i \text{ for } 1 \leq i \leq 3.$$

Thus, it remains to show that H_2 can be decomposed into $G_2, G_4, G_5, \ldots, G_k$ where each graph G_i is a 2-galaxy for $i = 2$ and $4 \leq i \leq k$. There are two possibilities, according to whether $t = -1$ or $t \geq 1$.

Subcase 3.1. $t = -1$. Then $H = S_{k-1} + S_{\binom{k+1}{2}-(k-1)}$ and $H_2 = S_{k-4} + S_{\binom{k+1}{2}-k-2}$ Furthermore, S_{k-4} can be decomposed into $k-4$ copies of S_1. Since

$$\tbinom{k+1}{2} - k - 2 = 4 + 4 + 5 + \cdots + (k-1),$$

it follows that $S_{\binom{k+1}{2}-k-2} \subseteq H_2$ can be decomposed into $S_4, S_4, S_5, \ldots, S_{k-1}$, it follows that H_2 can be decomposed into $G_4 = S_4$, $G_i = S_1 + S_{i-1}$ for $5 \leq i \leq k$.

Subcase 3.2. $t \geq 0$. Then $H = S_{k+t} + S_{\binom{k+1}{2}-k-t}$ and $H_2 = S_{k+t-3} + S_{\binom{k+1}{2}-k-t-3}$ Since

$$k + t - 3 = 2t + (k - t - 3) \text{ where } k - t - 3 \geq 1,$$

it follows that $S_{k+t-3} \subseteq H_2$ can be decomposed into t copies of S_2 and $k-t-3$ copies of S_1. Furthermore, since

$$\binom{k+1}{2} - k - t - 3 = 3 + 4 + \cdots + (k-1) - t = \sum_{i=3}^{k-t-1} i + \sum_{i=k-t-1}^{k-2} i,$$

it follows that $S_{\binom{k+1}{2}-k-t-3} \subseteq H_2$ can be decomposed into S_3, S_4, ..., S_{k-t-1}, S_{k-t-1}, S_{k-t}, ..., S_{k-2}. Thus, H_2 can be decomposed into $G_i = S_1 + S_{i-1}$ for $4 \leq i \leq k-t-1$ and $G_i = S_2 + S_{i-2}$ for $k-t \leq i \leq k$.

Therefore, there is a Ramsey chain $G_1, G_2, \ldots, G_k$ in H and so $AR_c(H) = k$ if $H = 2S_3$. ∎

A Ramsey chain is a *star Ramsey chain* if every subgraph in the chain is a star. In Proposition 4.5.3, each red-blue coloring produces a star Ramsey chain. However, this is not the case for Proposition 4.5.4. In fact, if $H_1 = 2S_1$ or $H_1 = 2S_2$, then no red-blue coloring of a 2-galaxy of size $\binom{k+1}{2}$ for some integer $k \geq 4$ can produce a star Ramsey chain of length k. These observations give rise to the following question.

Question 4.5.5 *Let H be 2-galaxy of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 5$. Is there a sufficient or a necessary condition for a red-blue coloring of H to produce a star Ramsey chain of length k?*

Question 4.5.5 can also be extended to galaxies of three or more components in general.

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Chapter 5

Maximal and Alternating Ramsey Chains

For a graph G and a red-blue edge coloring c of G, a Ramsey chain R in G is *maximal* with respect to c if R cannot be extended to a Ramsey chain of greater length in G. Recall that a Ramsey chain having maximum length $AR_c(G)$ is a maximum Ramsey chain of G with respect to c. Thus, every maximum Ramsey chain in a graph is maximal but the converse is not true. For example, consider the red-blue coloring of $G = C_{15} = (e_1, e_2, \ldots, e_{15})$ shown in Figure 5.1, where again a bold edge indicates a red edge and a thin edge indicates a blue edge. Again, let Q_n be a path of length n. Thus, the red subgraph of G is $G_r = Q_7 + Q_2 + 2K_2$ and the blue subgraph of G is $G_b = 4K_2$. For a Ramsey chain R in a graph, we write $E(R)$ for the union of the edge sets of the links in R.

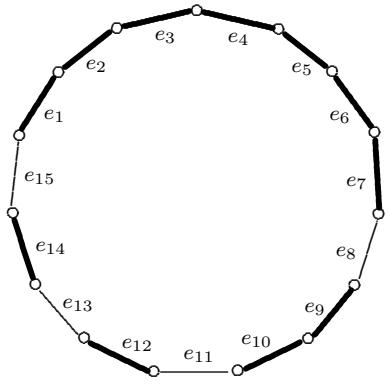


Figure 5.1: A red-blue coloring of C_{15}

For $\ell = 3, 4, 5$, there is a maximal Ramsey chain R_ℓ of length ℓ in the graph

$G = C_{15}$ of Figure 5.1, as we now show.

- ⋆ Let $R_3 = (Q_1, Q_2, Q_3)$ where $E(Q_1) = \{e_1\}$, $E(Q_2) = \{e_2, e_3\}$, and $E(Q_3) = \{e_4, e_5, e_6\}$. Since $G - E(R_3)$ contains no monochromatic Q_4 or $Q_3 + K_2$, it follows that R_3 is a maximal Ramsey chain of length 3.

- ⋆ Let $R_4 = (K_2, Q_2, Q_2+K_2, Q_2+2K_2)$ where $E(Q_1) = \{e_1\}$, $E(Q_2) = \{e_3, e_4\}$, $E(Q_2 + K_2) = \{e_6, e_7, e_{12}\}$, and $E(Q_2 + 2K_2) = \{e_9, e_{10}, e_2, e_{14}\}$. Since $G - E(R_4)$ contains no monochromatic $Q_3 + 2K_2$ or $Q_2 + 3K_2$, it follows that R_4 is a maximal Ramsey chain of length 4.

- ⋆ Let $R_5 = (K_2, 2K_2, 3K_2, 4K_2, 5K_2)$ where $E(K_2) = \{e_{14}\}$, $E(2K_2) = \{e_{12}, e_9\}$, $E(3K_2) = \{e_6, e_4, e_2\}$, $E(4K_2) = \{e_{15}, e_{13}, e_{11}, e_8\}$, and $E(5K_2)$=$\{e_{10}, e_7, e_5, e_3, e_1\}$. Since $\{K_2, 2K_2, 3K_2, 4K_2, 5K_2\}$ is a decomposition of G, it follows that R_5 is a maximal Ramsey chain of length 5.

Since $AR(C_{15}) = 5$, it follows that R_3 and R_4 are not maximum Ramsey chains of G while R_5 is a maximum Ramsey chain of G.

The minimum length of a maximal Ramsey chain in G with respect to c is referred to as the *lower Ramsey index* $AR_c^-(G)$ of G *with respect to* c and the *lower Ramsey index* $AR^-(G)$ of G is

$$AR^-(G) = \min\{AR_c^-(G) : \ c \text{ is a red-blue coloring of } G\}.$$

Thus, for every graph G

$$AR^-(G) \leq AR(G). \tag{5.1}$$

For example, if $G = C_{15}$, then $AR(G) = 5$. It can be shown that there is no maximal Ramsey chain of length 2 in G and so $AR^-(G) \geq 3$. On the other hand, we saw that there is a maximal Ramsey chain of length 3 in G and so $AR^-(G) \leq 3$. Therefore, $AR^-(G) = 3$.

In this chapter, we investigate Ramsey chains and maximal Ramsey chains in stars, matchings, and cycles. It is shown that (1) for every two integers p and q with $2 \leq p < q$, there exists a graph with a red-blue coloring possessing a maximal Ramsey chain of length p and a maximum Ramsey chain of length q and (2) for every positive integer k, there exists a graph with a red-blue coloring possessing at least k maximal Ramsey chains of distinct lengths with prescribed conditions. A conjecture and additional results on Ramsey chains with prescribed conditions are presented.

5.1 Ramsey Chains in Stars and Matchings

We saw that if G is the star $K_{1,m}$ or the matching mK_2 of size $m \geq 3$, then $AR(G) = k$ for an integer $k \geq 2$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. In this section, we determine the lower Ramsey indices of stars and matchings. In order to do this, we first study a class of graphs G for which $AR(G) = k$ in terms of the size of G. Recall that if G is a graph of size m where $2 \leq m < \binom{k+2}{2}$ for a positive integer k, then $AR(G) \leq k$. On the other hand, if G is a graph of size m such that $AR(G) \geq k$, then $m \geq \binom{k+1}{2}$. The following result presents a class of graphs G for which $AR(G) = k$ in terms of the size of G.

Theorem 5.1.1 *Let G be a graph of size $m \geq 2$ without isolated vertices such that for every two subgraphs F and H of G without isolated vertices, $|E(F)| < |E(H)|$ implies $F \subseteq H$. Then $AR(G) = k$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$.*

Proof. First, we verify the following claim.

> **Claim.** Let G be a graph of size $m \geq 2$ without isolated vertices such that for every two subgraphs F and H of G without isolated vertices, $|E(F)| < |E(H)|$ implies $F \subseteq H$. If $m \geq \binom{k+1}{2}$ for some positive integer k, then $AR(G) \geq k$.

We proceed by induction on k. The truth of the claim is immediate if $k = 1$ or $k = 2$. Assume for an integer $k \geq 2$ that a graph G' without isolated vertices has $AR(G') \geq k$ if G' has size $m' \geq \binom{k+1}{2}$ such that for every two subgraphs F' and H' of G' without isolated vertices, $|E(F')| < |E(H')|$ implies $F' \subseteq H'$. Let G be a graph without isolated vertices having size $m \geq \binom{k+2}{2}$ such that for every two subgraphs F and H of G without isolated vertices, $|E(F)| < |E(H)|$ implies $F \subseteq H$. We show that $AR(G) \geq k + 1$. Let there be given a red-blue coloring c of G. Since $k \geq 2$, it follows that $\frac{1}{2}\binom{k+2}{2} \geq k + 1$. Thus, G has a monochromatic subgraph G_{k+1} of size $k + 1$. Let $G' = G - E(G_{k+1})$, where G' has size $m' = m - (k + 1)$. Then the restriction c' of c to G' is a red-blue coloring of G'. Since $m \geq \binom{k+2}{2}$, it follows that $m' = m - (k + 1) \geq \binom{k+1}{2}$. By the induction hypothesis, G' has a Ramsey chain $(G_1, G_2, \ldots, G_k)$ of length k with respect to c'. Then $(G_1, G_2, \ldots, G_k, G_{k+1})$ is a Ramsey chain of length $k + 1$ in G. Thus, the claim holds.

Now, let G be a graph of size $m \geq 2$ without isolated vertices such that for every two subgraphs F and H of G without isolated vertices, $|E(F)| < |E(H)|$ implies $F \subseteq H$. First, assume that $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. Since $m < \binom{k+2}{2}$, it follows that $AR(G) \leq k$. Since $m \geq \binom{k+1}{2}$, it follows by the claim that $AR(G) \geq k$. Therefore, $AR(G) = k$.

For the converse, assume that $AR(G) = k$. Since $AR(G) = k$, there is a Ramsey chain of length k for every red-blue coloring of G. Thus, $m \geq \binom{k+1}{2}$. By the claim, if $m \geq \binom{k+2}{2}$, then $AR(G) \geq k+1$. Since $AR(G) = k$, it follows that $m < \binom{k+2}{2}$. Consequently, $\binom{k+1}{2} \leq m < \binom{k+2}{2}$. ■

For example, the 5-cycle C_5 satisfies the hypothesis of Theorem 5.1.1 and so $AR(C_5) = 2$. The 6-cycle C_6 does not satisfy the hypothesis of Theorem 5.1.1, however, since both P_3 and $3K_2$ are subgraphs of C_6 but $P_3 \not\subseteq 3K_2$. In fact, there are only two classes of graphs of size 6 or more that satisfy the hypothesis of Theorem 5.1.1.

Proposition 5.1.2 *Let G be a graph of size at least 6 without isolated vertices such that for every two subgraphs F and H of G without isolated vertices, $|E(F)| < |E(H)|$ implies $F \subseteq H$. Then G is either a star or a matching.*

Proof. Assume, to the contrary, that G is neither a star nor a matching. Thus, G contains two adjacent edges and two nonadjacent edges. If G contains a vertex of degree at least 3 or a matching of size at least 3, then G contains subgraphs F and H where F has a smaller size than H and F is not isomorphic to a subgraph of H. Therefore, we may assume that $\Delta(G) = 2$ and $2K_2$ is a maximum matching in G. The graph G contains no triangle since $2K_2 \not\subseteq K_3$. If G contains a k-cycle C_k where $k \geq 4$, then C_k is a component of G and so G contains a matching of size 3, a contradiction. Hence, G is a linear forest with two components. Since the size of G is at least 6, it follows that G contains a matching of size 3, a contradiction. ■

As a consequence of Theorem 5.1.1 and Proposition 5.1.2, we have the following result which was stated in Chapter 1.

Corollary 5.1.3 [6] *Let $k \geq 2$ be an integer and let G be the star $K_{1,m}$ or the matching mK_2. Then $AR(G) = k$ if and only if $\binom{k+1}{2} \leq m < \binom{k+2}{2}$.*

To determine the lower Ramsey indices of stars and matchings, we return to a class of graphs encountered in Theorem 5.1.1. Recall for a Ramsey chain R in a graph, that we write $E(R)$ for the union of the edge sets of the links in R.

Theorem 5.1.4 *Let G be a graph of size $m \geq 6$ without isolated vertices such that for every two subgraphs F and H of G without isolated vertices, $|E(F)| < |E(H)|$ implies $F \subseteq H$.*

(1) *If $\binom{k+1}{2} \leq m \leq \binom{k+2}{2} - 3$, then $AR^-(G) = k-1$.*

(2) *If $\binom{k+2}{2} - 2 \leq m \leq \binom{k+2}{2} - 1$, then $AR^-(G) = k$.*

Proof. To verify (1), we first show that if $m = \binom{k+1}{2}$, then $AR^-G) = k-1$. Let c be the red-blue coloring of G that assigns red to all edges of G except one and let s_c be a Ramsey chain of length $k-1$ consisting of $k-1$ red subgraphs of G. Then s_c is maximal and so $AR^-(G) \leq k-1$. Next, we show that $AR^-(G) \geq k-1$. We claim that every Ramsey chain $R_j = (G_1, G_2, \ldots, G_j)$ of length j, where $j \leq k-2$, can be extended to a Ramsey chain $(G_1, G_2, \ldots, G_j, G_{j+1})$ of length $j+1$. Observe that

$$|E(G) - E(R_j)| \;=\; \binom{k+1}{2} - \binom{j+1}{2} = (j+1) + (j+2) + \cdots + k \geq 2k-1$$

and so $\left\lceil \frac{\binom{k+1}{2} - \binom{j+1}{2}}{2} \right\rceil \geq k-1$. Hence, $G - E(R_j)$ contains a monochromatic subgraph of size at least $k-1$. Since the required size of G_{j+1} is $j+1$ and $j+1 \leq k-1$, the chain R_j can be extended to $(G_1, G_2, \ldots, G_j, G_{j+1})$. Thus, $AR^-(G) \geq k-1$ and so $AR^-(G) = k-1$.

We now show that if $m = \binom{k+1}{2} - 3$, then $AR^-(G) = k-1$. Since $k \geq 2$, it follows that $\binom{k+2}{2} - 3 \geq \binom{k+1}{2}$ and so

$$AR^-(G) \geq AR^-\left(\tbinom{k+1}{2} K_2\right) = k-1.$$

Thus, it remains to show that $AR^-(G) \leq k-1$. Let c be the red-blue coloring of G that assigns blue to $k-1$ edges of G and assigns red to the remaining $\binom{k+2}{2} - 3 - (k-1)$ edges of G. Since $\binom{k+2}{2} - 3 - (k-1) = \binom{k+1}{2} - 1$, there is a Ramsey chain $R_c = (G_1, G_2, \ldots, G_{k-1})$ of length $k-1$ where each subgraph G_i

$(1 \leq i \leq k-1)$ is a red subgraph of G but no such sequence of length k where all subgraphs are red exists. Since there are only $k-1$ blue edges, there is no blue subgraph of size k. Thus, s_c cannot be extended and so s_c is maximal. Therefore, $AR^-(G) \leq k-1$ and so $AR^-(G) = k-1$.

If $\binom{k+1}{2} \leq m \leq \binom{k+2}{2} - 3$, then

$$AR^-\left(\binom{k+1}{2}K_2\right) \leq AR^-(G) \leq AR^-\left(\left[\binom{k+2}{2} - 3\right]K_2\right)$$

and so $AR^-(G) = k-1$.

Next, we verify (2). Let $m = \binom{k+2}{2} - 2$. Since $AR^-(G) \leq AR(G) \leq k$ by Corollary 5.1.3, it remains to show that $AR^-(G) \geq k$. We claim that every Ramsey chain $R_j = (G_1, G_2, \ldots, G_j)$ of length j, where $j \leq k-1$, can be extended to a Ramsey chain $(G_1, G_2, \ldots, G_j, G_{j+1})$ of length $j+1$. Observe that

$$\begin{aligned} |E(G) - E(R_j)| &= \left[\binom{k+2}{2} - 2\right] - \binom{j+1}{2} \\ &= [(j+1) + (j+2) + \cdots + k + (k+1)] - 2 \\ &\geq 2k-1 \end{aligned}$$

and so $\left\lceil \frac{\left[\binom{k+2}{2}-2\right]-\binom{j+1}{2}}{2} \right\rceil \geq k$. Hence, $G - E(R_j)$ contains a monochromatic subgraph of size at least k. Since the required size of G_{j+1} is $j+1$ and $j+1 \leq k$, the chain R_j can be extended to a Ramsey chain $(G_1, G_2, \ldots, G_j, G_{j+1})$. Thus, $AR^-(G) \geq k$. Therefore, $AR^-(G) = k$. ∎

The following is a consequence of Proposition 5.1.2 and Theorem 5.1.4.

Corollary 5.1.5 *Let $k \geq 2$ be an integer and let G be the star $K_{1,m}$ or the matching mK_2.*

(1) *If $\binom{k+1}{2} \leq m \leq \binom{k+2}{2} - 3$, then $AR^-(G) = k-1$.*

(2) *If $\binom{k+2}{2} - 2 \leq m \leq \binom{k+2}{2} - 1$, then $AR^-(G) = k$.*

5.2 Comparing Two Ramsey Indices

We have seen that $AR^-(G) \leq AR(G)$ for every graph G. By Theorems 5.1.3 and 5.1.5, if $G \in \{K_{1,m}, mK_2\}$ where $\binom{k+2}{2} - 2 \leq m \leq \binom{k+2}{2} - 1$, then $AR^{-1}(G) = AR(G) = k$; while if $\binom{k+1}{2} \leq m < \binom{k+2}{2} - 3$, then $AR^{-1}(G) = k-1$ and $AR(G) = k$. Therefore, there are graphs G for which $AR^{-1}(G) = AR(G)$ and graphs G for which $AR(G) = AR^{-1}(G) + 1$. This brings up the question as to how large the number $AR(G) - AR^-(G)$ may be for some graph G. In order to answer this question, we first present a lemma.

Lemma 5.2.1 *Let $q \geq 3$ be an integer. For each integer m with*

$$\frac{1}{2}\binom{q+1}{2} < m < \binom{q+1}{2},$$

there exist integers $k_1, k_2, \ldots, k_t$ with $1 \leq k_1 < k_2 < \cdots < k_t = q$ such that

$$\sum_{i=1}^{t} k_i = m.$$

Proof. We proceed by induction on q. If $q = 3$, then $\binom{q+1}{2} = \binom{4}{2} = 6$. If m is an integer such that $3 < m < 6$, then $m = 4$ or $m = 5$. If $m = 4$, then $1+3 = 4$; while if $m = 5$, then $2 + 3 = 5$. Thus, the statement is true for $m = 3$. Assume that the statement is true for an integer q where $q \geq 3$. We show that the statement is true for $q + 1$. Let m be an integer such that $\frac{1}{2}\binom{q+2}{2} < m < \binom{q+2}{2}$. Since $q + 1 \geq 4$, it follows that $q + 1 < \frac{1}{2}\binom{q+2}{2}$. Let $m' = m - (q + 1)$. Then

$$\frac{1}{2}\binom{q+2}{2} - (q+1) < m' < \binom{q+2}{2} - (q+1).$$

Hence, $q \leq \frac{1}{2}\binom{q+1}{2} < m' < \binom{q+1}{2}$ for each integer $q \geq 3$. By the induction hypothesis, there exist integers $k_1, k_2, \ldots, k_t$ with $1 \leq k_1 < k_2 < \cdots < k_t = q$ such that $\sum_{i=1}^{t} k_i = m'$. Letting $k_{t+1} = q + 1$, we obtain $\sum_{i=1}^{t+1} k_i = m$. ∎

With the aid of Lemma 5.2.1 and Ramsey chains of cycles, we now show that $AR(G) - AR^-(G)$ can be arbitrarily large.

Theorem 5.2.2 *For every two integers p and q with $2 \leq p < q$, there exists a cycle with a red-blue coloring possessing a maximal Ramsey chain of length p and a maximum Ramsey chain of length q.*

Proof. If $\binom{p+1}{2}$ and $\binom{q+1}{2}$ are of opposite parity, let $n = \binom{q+1}{2}$; while if $\binom{p+1}{2}$ and $\binom{q+1}{2}$ are of the same parity, let $n = \binom{q+1}{2} + 1$. Let $G = C_n$ where the n consecutive edges of G are denoted by $e_1, e_2, \ldots, e_n$. We now define a red-blue coloring of G where e_i is colored red if $1 \leq i \leq \binom{p+1}{2}$, e_i is colored blue if $i = \binom{p+1}{2} + 1, \binom{p+1}{2} + 3, \cdots, n$, and all remaining edges of G are colored red. Therefore, the red subgraph of G is $G_r = Q_{\binom{p+1}{2}} + \left\lfloor \frac{n-\binom{p+1}{2}}{2} \right\rfloor K_2$, where $Q_{\binom{p+1}{2}}$ is a path of size $\binom{p+1}{2}$ in G_r and the blue subgraph of G is $G_b = \left\lceil \frac{n-\binom{p+1}{2}}{2} \right\rceil K_2$. Let $m_r = \binom{p+1}{2} + \left\lfloor \frac{n-\binom{p+1}{2}}{2} \right\rfloor$ be the number of red edges in G and let $m_b = \left\lceil \frac{n-\binom{p+1}{2}}{2} \right\rceil$ be the number of blue edges in G. Then $m_r > m_b$ and $m_r + m_b = n$.

For example, let $p = 3$ and $q = 6$. Since $\binom{p+1}{2} = \binom{4+1}{2} = 4$ and $\binom{q+1}{2} = \binom{6+1}{2} = 21$ are of opposite parity, the integer n in this example is $\binom{q+1}{2} = 21$ and so $G = C_{21}$. In the red-blue coloring of G in Figure 5.2, a bold edge indicates a red edge and a thin edge indicates a blue edge. The red subgraph of G is $G_r = Q_6 + 7K_2$ and the blue subgraph of G is $G_b = 8K_2$. Thus, $m_r = 13$, $m_b = 8$, and $m_r + m_b = n = 21$.

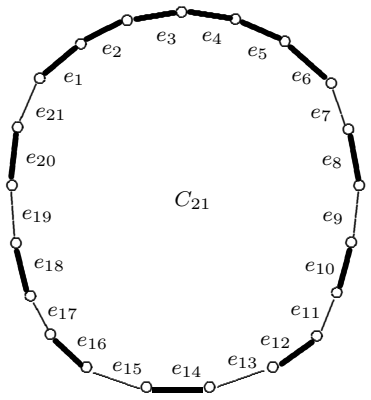


Figure 5.2: A red-blue coloring of C_{21}

First, we show that there is a maximal Ramsey chain of length p in G. The

subgraph $Q_{\binom{p+1}{2}}$ of G_r can be decomposed into $\{Q_1, Q_2, \ldots, Q_p\}$ where Q_i is a path of size i for $1 \leq i \leq p$. Thus, $R = (Q_1, Q_2, \ldots, Q_p)$ is a Ramsey chain in G_r and in G. Since $G - E(R)$ contains no monochromatic subgraph isomorphic to either Q_{p+1} or $Q_p + K_2$, the chain R is a maximal Ramsey chain of length p in G.

For example, the red subgraph $Q_6 \subseteq G_r$ in $G = C_{21}$ of Figure 5.2 can be decomposed into Q_1, Q_2, Q_3, where $E(Q_1) = \{e_1\}$, $E(Q_2) = \{e_2, e_3\}$, and $E(Q_3) = \{e_4, e_5, e_6\}$. Since $G - \cup_{i=1}^{3} E(Q_i)$ contains no monochromatic subgraph isomorphic to Q_4 or $Q_3 + K_2$, it follows that (Q_1, Q_2, Q_3) is a maximal Ramsey chain of length 3 in $G = C_{21}$.

Next, we show that there is a maximum Ramsey chain of length q in G. Define a sequence $\mathcal{S}$ of the m_r red edges of $G_r = Q_{\binom{p+1}{2}} + \left\lfloor \frac{n-\binom{p+1}{2}}{2} \right\rfloor K_2$ as follows:

$\star$ If $\binom{p+1}{2}$ is even, then let

$$\mathcal{S} = \left(\underline{e_1, e_3, \ldots, e_{\binom{p+1}{2}-1}}, \quad \underline{e_{\binom{p+1}{2}+2}, e_{\binom{p+1}{2}+4}, \ldots, e_{n-1}}, \quad \underline{e_2, e_4, \ldots, e_{\binom{p+1}{2}}} \right).$$

$\star$ If $\binom{p+1}{2}$ is odd, the let

$$\mathcal{S} = \left(\underline{e_1, e_3, \ldots, e_{\binom{p+1}{2}}}, \quad \underline{e_{\binom{p+1}{2}+2}, e_{\binom{p+1}{2}+4}, \ldots, e_{n-1}}, \quad \underline{e_2, e_4, \ldots, e_{\binom{p+1}{2}-1}} \right).$$

Then no two consecutive edges in $\mathcal{S}$ are adjacent. Denote the sequence $\mathcal{S}$ by $(f_1, f_2, \ldots, f_{m_r})$, where then $f_i f_{i+1} \notin E(G)$ for $1 \leq i \leq m_r - 1$. To construct a maximum Ramsey chain of length q in G, we consider two cases, according to whether $n = \binom{q+1}{2}$ or $n = \binom{q+1}{2} + 1$.

For example, if $p = 3$ and $q = 6$, then the sequence

$$\mathcal{S} = (f_1, f_2, \ldots, f_{13}) = (e_1, e_3, e_5, e_8, e_{10}, e_{12}, e_{14}, e_{16}, e_{18}, e_{20}, e_2, e_4, e_6)$$

is illustrated in Figure 5.3 for the red-blue coloring of $G = C_{21}$ of Figure 5.2.

Case 1. $n = \binom{q+1}{2}$. Since $\frac{1}{2}\binom{q+1}{2} < m_r < \binom{q+1}{2}$, it follows by Lemma 5.2.1 that there exist integers $a_1, a_2, \ldots, a_t$ with $1 \leq a_1 < a_2 < \cdots < a_t = q$ such that

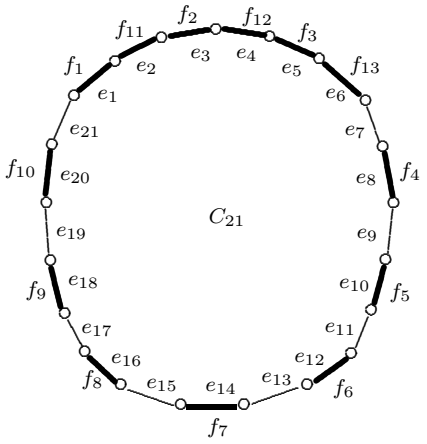


Figure 5.3: Revisiting the red-blue coloring of C_{21}

$\sum_{i=1}^{t} a_i = m_r$. Define a labeling ℓ of $\mathcal{S}$ by

$$\ell(f_i) = \begin{cases} t & \text{if } 1 \le i \le a_t = q \\ t-1 & \text{if } a_t + 1 \le i \le a_t + a_{t-1} \\ \vdots & \vdots \\ 1 & \text{if } a_t + a_{t-1} + \cdots + a_2 + 1 \le i \le m_r. \end{cases}$$

Since $q \le \frac{1}{2}\binom{q+1}{2} < \frac{1}{2}m_r$, it follows that for every pair i, j of distinct integers with $1 \le i, j \le t$, if $\ell(f_i) = \ell(f_j)$, then f_i and f_j are not adjacent. Thus, for $1 \le i \le t$, the a_i edges labeled i form the matching a_iK_2 and so G_r can be decomposed into the matchings a_1K_2, a_2K_2, ..., $a_tK_2 = qK_2$. Since

$$\left(\sum_{i=1}^{t} a_i\right) + m_b = \binom{q+1}{2} = \sum_{i=1}^{q} i,$$

it follows that there exist t' distinct integers $b_1, b_2, \ldots, b_{t'}$, where $t' = q - t$ and $1 \le b_1 < b_2 < \cdots < b_{t'} \le q - 1$ such that (i) $\sum_{i=1}^{t'} b_i = m_b$ and (ii) $a_i \ne b_j$ for every pair i, j of integers with $1 \le i \le t$ and $1 \le j \le t'$. That is,

$$\{a_1, a_2, \ldots, a_t\} \cup \{b_1, b_2, \ldots, b_{t'}\} = \{1, 2, \ldots, q\}.$$

The blue subgraph $G_b = m_bK_2$ can be decomposed into the matchings b_1K_2, b_2K_2, ..., $b_{t'}K_2$. Consequently, $(K_2, 2K_2, 3K_3, \ldots, qK_2)$ is a maximum Ramsey chain of length q in G.

For example, if $p = 3$ and $q = 6$, then $m_r = 13$, $m_b = 8$ and $n = 21$. Let $a_1 = 3$, $a_2 = 4$, and $a_3 = 6$ where $t = 3$. Thus, $a_1 + a_2 + a_3 = 3 + 4 + 6 = 13 = m_r$. Define a labeling ℓ of $\mathcal{S} = (f_1, f_2, \ldots, f_{13})$ by

$$\ell(f_i) = \begin{cases} 3 & \text{if } 1 \le i \le a_3 = 6 \\ 2 & \text{if } a_3 + 1 = 7 \le i \le 10 = a_3 + a_2 \\ 1 & \text{if } a_3 + a_2 + 1 = 11 \le i \le 13 = m_r. \end{cases}$$

The red subgraph $G_r = Q_6 + 7K_2$ can be decomposed into $3K_2, 4K_2, 6K_2$ where $E(3K_2) = \{f_{11}, f_{12}, f_{13}\}$, $E(4K_2) = \{f_7, f_8, f_9, f_{10}\}$, and $E(6K_2) = \{f_1, f_2, \ldots, f_6\}$. Let $b_1 = 1$, $b_2 = 2$, and $b_3 = 5$ where $t' = 3$. Then $b_1 + b_2 + b_3 = 1 + 2 + 5 = 8 = m_b$ and $\{a_1, a_2, a_3\} \cup \{b_1, b_2, b_3\} = \{3, 4, 6\} \cup \{1, 2, 5\} = \{1, 2, 3, 4, 5, 6\}$. The blue subgraph $G_b = 8K_2$ is decomposed into $K_2, 2K_2, 5K_2$. Thus, $(K_2, 2K_2, \ldots, 6K_2)$ is a maximum Ramsey chain of length 6 in $G = C_{21}$.

Case 2. $n = \binom{q+1}{2} + 1$. Thus, $\binom{p+1}{2}$ and $\binom{q+1}{2}$ are of the same parity. Then

$$m_r = \binom{p+1}{2} + \left\lfloor \frac{\binom{q+1}{2} + 1 - \binom{p+1}{2}}{2} \right\rfloor = \frac{1}{2}\left[\binom{q+1}{2} + \binom{p+1}{2}\right].$$

Since $3 \le p < q$, it follows that

$$\frac{1}{2}\binom{q+1}{2} < \frac{1}{2}\left[\binom{q+1}{2} + \binom{p+1}{2}\right] < \binom{q+1}{2}$$

and so $\frac{1}{2}\binom{q+1}{2} < m_r < \binom{q+1}{2}$. By the argument in Case 1, there exist integers $a_1, a_2, \ldots, a_t$ with $1 \le a_1 < a_2 < \cdots < a_t = q$ such that $\sum_{i=1}^{t} a_i = m_r$ and the red subgraph G_r can be decomposed into the matchings a_1K_2, a_2K_2, …, a_tK_2. In this case, $m_r + (m_b - 1) = \binom{q+1}{2}$. By the argument in Case 1, there exist t' distinct integers $b_1, b_2, \ldots, b_{t'}$, where $t' = q - t$ and $1 \le b_1 < b_2 < \cdots < b_{t'} \le q - 1$ such that (i) $\sum_{i=1}^{t'} b_i = m_b - 1$ and (ii) $a_i \ne b_j$ for every pair i, j of integers with $1 \le i \le t$ and $1 \le j \le t'$. The blue subgraph $(m_b - 1)K_2 \subseteq G_b$ can be decomposed into the matchings b_1K_2, b_2K_2, …, $b_{t'}K_2$. Since the size of G is $n = \binom{q+1}{2} + 1 < \binom{q+2}{2}$, there is no Ramsey chain of length $q + 1$ and so $(K_2, 2K_2, 3K_3, \ldots, qK_2)$ is a maximum Ramsey chain of G. ∎

The following is therefore a consequence of Theorem 5.2.2.

Corollary 5.2.3 *For each positive integer N, there is a graph G such that*

$$AR(G) - AR^{-}(G) > N.$$

5.3 Alternating Ramsey Chains

In the proof of Theorem 5.2.2, every link of both the maximal Ramsey chain and the maximum Ramsey chain has the same color, namely red. We now show that Corollary 5.2.3 can be obtained without all the links having the same color.

An *alternating Ramsey chain* in a graph with a red-blue coloring is a Ramsey chain in which the colors of every two consecutive links are distinct. For example, consider the red-blue coloring of $G = C_{10}$ in Figure 5.4 where $G_r = 2Q_2 + 2K_2$ and $G_b = 4K_2$. Since $AR(G) = 4$, the length of a Ramsey chain of G is at most 4.

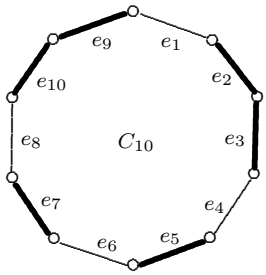


Figure 5.4: A red-blue coloring of C_{10}

(1) Let $R_1 = (K_2, 2K_2, 3K_2, 4K_2)$ where

$$E(K_2) = \{e_1\},\ E(2K_2) = \{e_2, e_{10}\},$$
$$E(3K_2) = \{e_4, e_6, e_8\},\ E(4K_2) = \{e_3, e_5, e_7, e_9\}.$$

Then R_1 is an alternating maximum Ramsey chain of length 4 in G.

(2) Let $R_2 = (K_2, 2K_2, Q_2 + K_2)$, where

$$E(K_2) = \{e_{10}\},\ E(2K_2) = \{e_1, e_4\},\ E(Q_2 + K_2) = \{e_2, e_3, e_5\}.$$

Since $G - E(R_2)$ contains no monochromatic subgraph isomorphic to Q_2, it follows that R_2 is an alternating maximal Ramsey chain of length 3 in G.

Consequently, $G = C_{10}$ has two alternating maximal Ramsey chains of distinct lengths. This example illustrates a more general result on alternating maximal Ramsey chains of cycles.

For integers p and q with $1 \leq p < q$, we write $Q_q(p)$ to denote the subpath of length p obtained by selecting the first p edges (in clockwise direction) from a path Q_q of length q in a cycle. We now show that there is a red-blue coloring of a cycle that produces arbitrarily many maximal alternating Ramsey chains of distinct lengths.

Theorem 5.3.1 *For every positive integer k, there exists a cycle with a red-blue coloring possessing at least k maximal alternating Ramsey chains of distinct lengths.*

Proof. The statement is true trivially for $k = 1$ and so we may assume that $k \geq 2$. Let $G = C_n$ where

$$n = \begin{cases} \frac{11k^2-5k}{2} & \text{if } k \text{ is odd} \\ \frac{11k^2-5k+2}{2} & \text{if } k \text{ is even.} \end{cases}$$

We now describe a red-blue coloring of G as follows.

- ⋆ Select an arbitrary edge of G and color it red. This is a red Q_1, which we denote by F_1. As we proceed clockwise about G, the next two edges are colored blue. This results in a blue Q_2, which we denote by F_2. The next three edges are colored red, resulting in a red Q_3, which we denote by F_3. We continue this procedure until arriving at a blue Q_{2k}, denoted by F_{2k}. Thus, the sequence $(F_1, F_2, \ldots, F_{2k}) = (Q_1, Q_2, \ldots, Q_{2k})$ appears (in clockwise direction) on G where $F_i = Q_i$ for $1 \leq i \leq 2k$ and

$$\sum_{i=1}^{2k} |E(F_i)| = \sum_{i=1}^{2k} |E(Q_i)| = \sum_{i=1}^{2k} i = \binom{2k+1}{2}.$$

- ⋆ The next $2k-1$ edges following F_{2k} are colored red, resulting in a red Q_{2k-1}, denoted by H_1. The next $2k-2$ edges following H_1 are colored blue, resulting

in a red Q_{2k-2}, denoted by H_2. We continue this procedure until arriving at Q_{k+1}, denoted by H_{k-1}. If k is odd, then H_{k-1} is a blue Q_{k+1}; while if k is even, then H_{k-1} is a red Q_{k+1}. Thus, the sequence $(H_1, H_2, \ldots, H_{k-1}) = (Q_{2k-1}, Q_{2k-2}, \ldots, Q_{k+1})$ appears (after F_{2k} in clockwise direction) on G, where $H_i = Q_{2k-i}$ for $1 \leq i \leq k-1$ and

$$\sum_{i=1}^{k-1} |E(H_i)| = \sum_{i=1}^{k-1} |E(Q_{2k-i})| = \sum_{i=1}^{k-1} (2k-i) = \binom{2k}{2} - \binom{k+1}{2}.$$

$\star$ Let X be the set consisting of the remaining edges of G, namely

$$X = E(G) - [E(F_1) \cup E(F_2) \cup \cdots \cup E(F_{2k}) \cup E(H_1) \cup E(H_2) \cup \cdots \cup E(H_{k-1})].$$

Then

$$\begin{aligned} |X| &= n - \left[\binom{2k+1}{2} + \binom{2k}{2} - \binom{k+1}{2}\right] = n - \frac{7k^2-k}{2} \\ &= \begin{cases} \frac{11k^2-5k}{2} - \frac{7k^2-k}{2} = 2k(k-1) & \text{if } k \text{ is odd} \\ \frac{11k^2-5k+2}{2} - \frac{7k^2-k}{2} = 2k(k-1)+1 & \text{if } k \text{ is even.} \end{cases} \end{aligned}$$

The edges in X are alternately colored red and blue such that the edge following H_{k-1} is colored differently than the edges of H_{k-1} and the edge preceding F_1 is blue. Hence,

- if k is odd, then H_{k-1} is blue and so the edge following H_{k-1} is colored red. Since $|X| = 2k(k-1)$ is even, the edge preceding F_1 is blue as required;
- if k is even, then H_{k-1} is red and so the edge following H_{k-1} is colored blue. Since $|X| = 2k(k-1)+1$ is odd, the edge preceding F_1 is blue.

Consequently, if $|X| = t$, then $G[X] = (f_1, f_2, \ldots, f_t)$ is a subpath Q_t of size t where f_1 is the edge following H_{k-1} and f_t is the edge preceding F_1. The edges of $G[X]$ are alternately colored red and blue such that f_1 is colored differently than the edges of H_{k-1} and f_t is blue.

We now have the following.

Observation. *No red edge in X is adjacent to any red edge in G and no blue edge in X is adjacent to any blue edge in G. Thus, each edge in X is a monochromatic component Q_1 either in the red subgraph G_r of G or in the blue subgraph G_b of G.*

If k is odd, then the red subgraph G_r and the blue subgraph G_b of G are

$$\begin{aligned} G_r &= F_1 + F_3 + \cdots + F_{2k-1} + H_1 + H_3 + \cdots + H_{k-2} + k(k-1)K_2 \\ &= Q_1 + Q_3 + \cdots + Q_{2k-1} + Q_{2k-1} + Q_{2k-3} + \cdots + Q_{k+2} + k(k-1)K_2 \\ G_b &= F_2 + F_4 + \cdots + F_{2k} + H_2 + H_4 + \cdots + H_{k-1} + k(k-1)K_2 \\ &= Q_2 + Q_4 + \cdots + Q_{2k} + Q_{2k-2} + Q_{2k-4} + \cdots + Q_{k+1} + k(k-1)K_2. \end{aligned}$$

If k is even, then the red subgraph G_r and the blue subgraph G_b of G are

$$\begin{aligned} G_r &= F_1 + F_3 + \cdots + F_{2k-1} + H_1 + H_3 + \cdots + H_{k-1} + k(k-1)K_2 \\ &= Q_1 + Q_3 + \cdots + Q_{2k-1} + Q_{2k-1} + Q_{2k-3} + \cdots + Q_{k+1} + k(k-1)K_2 \\ G_b &= F_2 + F_4 + \cdots + F_{2k} + H_2 + H_4 + \cdots + H_{k-2} + [k(k-1)+1]K_2 \\ &= Q_2 + Q_4 + \cdots + Q_{2k} + Q_{2k-2} + Q_{2k-4} + \cdots + Q_{k+2} + [k(k-1)+1]K_2. \end{aligned}$$

We claim that G possesses k maximal alternating Ramsey chains $R_1, R_2, \ldots, R_k$ where R_i has length $2k-1+i$ for $1 \leq i \leq k$.

- First, $R_1 = (F_1, F_2, \ldots, F_{2k})$ is an alternating Ramsey chain of length $2k$ in G. Since $G - E(R_1)$ contains no monochromatic subgraph isomorphic to $F_{2k} = Q_{2k}$, it follows that R_1 is maximal.

- Next, let $R_2 = (F_1, F_2, \ldots, F_{2k-1}, F_{2k}(2k-1)+K_2, H_1+2K_2)$ where the edges of K_2 and $2K_2$ are taken from X such that each link in R_2 is monochromatic. Then R_2 is an alternating Ramsey chain of length $2k+1$ in G. Since $G-E(R_2)$ contains no monochromatic subgraph isomorphic to Q_{2k-1}, it follows that R_2 is maximal.

- Next, let

$$\begin{aligned} R_3 = (&F_1, F_2, \ldots, F_{2k-2}, F_{2k-1}(2k-2) + K_2, F_{2k}(2k-2) + 2K_2, \\ &H_1(2k-2) + 3K_2, H_2 + 4K_2) \end{aligned}$$

where the edges of $K_2, 2K_2, 3K_2, 4K_2$ are taken from X such that each link in R_3 is monochromatic. Then R_3 is an alternating Ramsey chain of length $2k+2$ in G. Since $G - E(R_3)$ contains no monochromatic subgraph isomorphic to Q_{2k-2}, it follows that R_3 is maximal.

- In general, for $2 \leq i \leq k$, let

$$\begin{aligned} R_i \;=\; (F_1, F_2, \ldots, F_{2k+1-i}, \quad & F_{2k+2-i}(2k+1-i) + K_2, \\ & F_{2k+3-i}(2k+1-i) + 2K_2, \ \ldots, \ F_{2k}(2k+1-i) + (i-1)K_2, \\ & H_1(2k+1-i) + iK_2, \quad H_2(2k+1-i) + (i+1)K_2, \ \ldots, \\ & H_{i-1} + 2(i-1)K_2) \end{aligned}$$

where the edges of $K_2, 2K_2, \ldots, 2(i-1)K_2$ are taken from X such that each link in R_i is monochromatic. Thus, R_i is an alternating Ramsey chain of length $2k-1+i$ in G. Since $G-E(R_i)$ contains no monochromatic subgraph isomorphic to Q_{2k+1-i}, it follows that R_i is maximal. In particular,

$$\begin{aligned} R_k \;=\; (F_1, F_2, \ldots, F_{k+1}, \quad & F_{k+2}(k+1) + K_2, \\ & F_{k+3}(k+1) + 2K_2, \ \ldots, \ F_{2k}(k+1) + (k-1)K_2, \\ & H_1(k+1) + kK_2, \quad H_2(k+1) + (k+1)K_2, \ \ldots, \\ & H_{k-1} + 2(k-1)K_2) \end{aligned}$$

where the edges of $K_2, 2K_2, \ldots, 2(k-1)K_2$ are taken from X such that each link in R_k is monochromatic. Thus, R_k is an alternating Ramsey chain of length $3k-1$ in G. Since $G - E(R_k)$ contains no monochromatic subgraph isomorphic to Q_{k+1}, it follows that R_k is maximal.

Finally, we show that each of the k maximal alternating Ramsey chains R_1, R_2, ..., R_k of distinct lengths in G can be constructed as described. Of the k alternating Ramsey chains R_1, R_2, ..., R_k in G, the longest chain R_k among them takes the maximum number of edges from X. This maximum number is

$$1 + 2 + \cdots + 2(k-1) = \tbinom{2k-1}{2} = (k-1)(2k-1).$$

- If k is odd, then the link $F_{k+2}(k+1) + K_2$ in R_k is red and so the number of red components Q_1 required in R_k from X is $1+3+\cdots+(2k-3) = (k-1)^2$ and the number of blue components Q_1 in R_k is

$$2+4+\cdots+2(k-1)=2[1+2+\cdots+(k-1)]=2\tbinom{k}{2}=k(k-1).$$

Since $|X|=2k(k-1)$, where $k(k-1)$ edges are red and $k(k-1)$ edges are blue, it follows by the observation that k such maximal alternating Ramsey chains $R_1, R_2, \ldots, R_k$ of distinct lengths in G can be constructed.

◦ If k is even, then the link $F_{k+2}(k+1)+K_2$ in R_k is blue and so the number of blue components Q_1 required in R_k is $1+3+\cdots+(2k-3)=(k-1)^2$ and the number of red components Q_1 from X is

$$2+4+\cdots+2(k-1)=2[1+2+\cdots+(k-1)]=2\tbinom{k}{2}=k(k-1).$$

Since $|X|=2k(k-1)+1$, where $k(k-1)$ edges are red and $k(k-1)+1$ edges are blue, it follows by the observation that k such maximal alternating Ramsey chains $R_1, R_2, \ldots, R_k$ of distinct lengths in G can be constructed. ■

We now illustrate the proof of Theorem 5.3.1 for $k \in \{2,3,4\}$.

⋆ For $k=2$, let $n=\frac{11k^2-5k+2}{2}=18$. The red-blue coloring of $G=C_{18}$ in the proof of Theorem 5.3.1 produces the red subgraph G_r and the blue subgraph G_b where

$$\begin{aligned} G_r &= F_1+F_3+H_1+2K_2=Q_1+Q_3+Q_3+2K_2 \\ G_b &= F_2+F_4+3K_2=Q_2+Q_4+3K_2. \end{aligned}$$

In this coloring, $H_{k-1}=H_1=Q_3$ is red, the edge following H_1 is blue, and the edge preceding $F_1=Q_1$ is blue. Since $k=2$, the set X consists of $k(k-1)=2$ red edges and $k(k-1)+1=3$ blue edges. No red edge in X is adjacent to any red edge of G and no blue edge in X is adjacent to any blue edge of G. The two maximal alternating Ramsey chains R_i of length $2k-1+i=3+i$ for $i=1,2$ are

$$\begin{aligned} R_1 &= (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4) \\ R_2 &= (\underline{Q_1}, Q_2, \underline{Q_3},\ Q_4(3)+K_2,\ \underline{H_1+2K_2}). \end{aligned}$$

Each red link in a chain is underlined. Since $k=2$, the chain $R_k=R_2$ uses the $k(k-1)=2$ red edges from X and uses only $(k-1)^2=1$ of the $k(k-1)+1=3$ blue edges from X.

⋆ For $k = 3$, let $n = \frac{11k^2-5k}{2} = 42$. The red-blue coloring of $G = C_{42}$ in the proof of Theorem 5.3.1 produces the red subgraph G_r and the blue subgraph G_b where

$$\begin{aligned} G_r &= F_1 + F_3 + F_5 + H_1 + 6K_2 = Q_1 + Q_3 + Q_5 + Q_5 + 6K_2 \\ G_b &= F_2 + F_4 + F_6 + H_2 + 6K_2 = Q_2 + Q_4 + Q_6 + Q_4 + 6K_2. \end{aligned}$$

In this coloring, $H_{k-1} = H_2 = Q_4$ is blue, the edge following H_2 is red, and the edge preceding $F_1 = Q_1$ is blue. Since $k = 3$, the set X consists of $k(k-1) = 6$ red edges and $k(k-1) = 6$ blue edges. No red edge in X is adjacent to any red edge of G and no blue edge in X is adjacent to any blue edge of G. The three maximal alternating Ramsey chains R_i of length $2k - 1 + i = 5 + i$ for $1 \le i \le 3$ are

$$\begin{aligned} R_1 &= (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5}, Q_6) \\ R_2 &= (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5}, Q_6(5) + K_2, \underline{H_1(5) + 2K_2}) \\ R_3 &= (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5(4) + K_2}, Q_6(4) + 2K_2, \underline{H_1(4) + 3K_2}, H_2(4) + 4K_2). \end{aligned}$$

Each red link in a chain is underlined. Since $k = 3$, the chain $R_k = R_3$ uses $(k-1)^2 = 4$ of the $k(k-1) = 6$ red edges from X and uses all $k(k-1) = 6$ blue edges from X.

⋆ For $k = 4$, let $n = \frac{11k^2-5k+2}{2} = 79$. The red-blue coloring of $G = C_{79}$ in the proof of Theorem 5.3.1 produces the red subgraph G_r and the blue subgraph G_b where

$$\begin{aligned} G_r &= F_1 + F_3 + F_5 + F_7 + H_1 + H_3 + 12K_2 \\ &= Q_1 + Q_3 + Q_5 + +Q_7 + Q_7 + Q_5 + 12K_2 \\ G_b &= F_2 + F_4 + F_6 + F_8 + H_2 + 13K_2 \\ &= Q_2 + Q_4 + Q_6 + Q_8 + Q_6 + 13K_2. \end{aligned}$$

In this coloring, $H_{k-1} = H_3 = Q_5$ is red, the edge following H_3 is blue, and the edge preceding $F_1 = Q_1$ is blue. Since $k = 4$, the set X consists of $k(k-1) = 12$ red edges and $k(k-1) + 1 = 13$ blue edges. No red edge in X is adjacent to any red edge of G and no blue edge in X is adjacent to any blue edge of G. The following are the four maximal alternating Ramsey

chains R_i of length $7+i$ for $1 \le i \le 4$:

$$R_1 = (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5}, Q_6, \underline{Q_7}, Q_8)$$

$$R_2 = (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5}, Q_6, \underline{Q_7}, Q_8(7) + K_2, \underline{Q_7 + 2K_2})$$

$$R_3 = (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5}, Q_6, \underline{Q_7(6) + K_2}, Q_8(6) + 2K_2, \underline{H_1(6) + 3K_2}, H_2(6) + 4K_2)$$

$$R_4 = (\underline{Q_1}, Q_2, \underline{Q_3}, Q_4, \underline{Q_5}, Q_6(5) + K_2, \underline{Q_7(5) + 2K_2}, Q_8(5) + 3K_2, \underline{H_1(5) + 4K_2},$$
$$H_2(5) + 5K_2, \underline{H_3(5) + 6K_2}).$$

Each red link in a chain is underlined. Since $k = 4$, the chain $R_k = R_4$ uses all $k(k-1) = 12$ red edges from X and uses only $(k-1)^2 = 9$ of the $k(k-1) + 1 = 13$ blue edges from X.

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Chapter 6

In Closing

We have seen that the concept of Ramsey chains in graphs combines the features of red-blue colorings in Ramsey theory and the well-known Ascending Subgraph Decomposition Conjecture and that this provides a new perspective of Ramsey concepts and ascending subgraph decompositions. We now summarize the main results that have been obtained.

- If G is a path or a cycle of size $m \geq 3$, then

$$AR(C_m) = k \text{ if and only if } \binom{k+1}{2} \leq m < \binom{k+2}{2}.$$

  Consequently, $AR(G) = \left\lfloor \frac{-1+\sqrt{1+8m}}{2} \right\rfloor$.

- If H is a linear forest of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ where $k \geq 3$, then

$$AR(H) = \begin{cases} k & \text{if } \binom{k+1}{2} \leq m < \binom{k+2}{2} \\ k \text{ or } k-1 & \text{if } m = \binom{k+1}{2} \text{ or } H \text{ is } k\text{-binomial.} \end{cases}$$

  We have also obtained the following results on linear forests.

  (a) If H is a k-binomial linear forest where $k \geq 4$ having at most $\binom{k-1}{2}$ components, then $AR(H) = k$.

  (b) Let $H = Q_{q_1} + Q_{q_2} + \cdots + Q_{q_t}$ be a k-binomial linear forest of size

$$m = \sum_{i=1}^{t} q_i = \binom{k+1}{2}$$

for some integer $k \geq 4$ with t components where

$$\binom{k-1}{2} < t < \binom{k+1}{2} \text{ and } s = \sum_{i=1}^{t} \left\lfloor \frac{q_i}{2} \right\rfloor$$

is the maximum number of pairwise edge-disjoint copies of Q_2 in H.

(1) If $s > k$, then $AR(H) = k$.

(2) If $s = k$ and H contains two components of even size 4 or more, then $AR(H) = k - 1$; otherwise, $AR(H) = k$.

(3) If $s = k - 1$ and H contains at least one component of even size 4 or more, then $AR(H) = k - 1$; otherwise, $AR(H) = k$.

(4) If $s \leq k - 2$, then $AR(H) = k - 1$.

(c) For every two integers t and k where $\binom{k-1}{2} < t < \binom{k+1}{2}$ and $k \geq 4$, there is a k-binomial linear forest H with t components such that

$$AR(H) = k - 1.$$

- Let $H = S_x + S_y$ be a 2-galaxy such that $\binom{k+1}{2} \leq x + y < \binom{k+2}{2}$ where $k \geq 3$.

(1) If $x + y = \binom{k+1}{2}$ and $\min\{x, y\} \leq k - 2$, then $AR(H) = k - 1$.

(2) If $\max\{x, y\} \geq \binom{k+1}{2}$, then $AR(H) = k$.

(3) If $1 \leq x \leq 4$ and $y = m - x$ where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ and $k \geq 4$, then

$$AR(H) = \begin{cases} k-1 & \text{if } m = \binom{k+1}{2} \text{ and } x \leq k-2 \\ k & \text{otherwise.} \end{cases}$$

Furthermore, when $5 \leq x \leq \left\lfloor \frac{m}{2} \right\rfloor$ and $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 5$, it is conjectured that if (i) $m = \binom{k+1}{2}$ and $x \geq k - 1$ or (ii) $m = \binom{k+1}{2} + 1$ and $x \leq k - 2$, then $AR(H) = k$.

(4) For $m = \binom{k+1}{2}$ where $k \geq 5$, several sufficient conditions have been established for red-blue colorings of H to produce a Ramsey chain of length k. These sufficient conditions involve the structures of the red or blue subgraphs of H.

- For an integer $k \geq 2$, if G is the star $K_{1,m}$ or the matching mK_2, then

$$AR^{-}(G) = \begin{cases} k-1 & \text{if } \binom{k+1}{2} \leq m \leq \binom{k+2}{2} - 3 \\ k & \text{if } \binom{k+2}{2} - 2 \leq m \leq \binom{k+2}{2} - 1. \end{cases}$$

Furthermore, for every two integers p and q with $2 \leq p < q$, there exists a cycle with a red-blue coloring possessing a maximal Ramsey chain of length p and a maximum Ramsey chain of length q. Consequently, for each positive integer N, there is a graph G such that $AR(G) - AR^-(G) > N$.

We have determined $AR(G)$ for many graphs G of size m with $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some positive integer k and have always shown that either $AR(G) = k$ or $AR(G) = k - 1$ in every example considered. This led us to the following conjecture.

The Ramsey Index Conjecture: If G is a graph of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ for some integer $k \geq 2$, then $AR(G) = k$ or $AR(G) = k - 1$.

The Ramsey Index Conjecture says that if G is a graph of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$, then not only does G contain ℓ pairwise edge-disjoint subgraphs $G_1, G_2, \ldots, G_\ell$ where $\ell \in \{k-1, k\}$ that satisfy the 1987 Ascending Subgraph Decomposition Conjecture, but in any red-blue coloring of G, there is always such a sequence where each subgraph G_i $(1 \leq i \leq \ell)$ is monochromatic as well. In particular, if G is a graph of size m where $\binom{k+1}{2} \leq m < \binom{k+2}{2}$ such that $AR(G) = k$, then the Ascending Subgraph Decomposition Conjecture is also true for the graph G. While many problems on this topic remain unresolved, there are some related and potentially intriguing classes of graphs that are worthy of further investigation, including the class of graphs each of whose component is a triangle and the class of graphs each of whose component is a cycle of fixed order.

The concept of Ramsey chains in red-blue colored graphs is a special case in a more general concept. For positive integers m and t with $m \geq t$, let G be a graph with m edges each of which is assigned one of t colors, where each color is assigned to at least one edge. For such a t-edge coloring c of G, an ascending Ramsey sequence in G with respect to c is a sequence $G_1, G_2, \ldots, G_k$ of pairwise edge-disjoint subgraphs of G such that each subgraph G_i $(1 \leq i \leq k)$ is monochromatic and G_i is isomorphic to a proper subgraph of G_{i+1} for $1 \leq i \leq k-1$. The ascending Ramsey index $AR_c(G)$ of G with respect to c is the maximum length of an ascending Ramsey sequence in G with respect to c. The ascending t-Ramsey index $AR_t(G)$ of G is the minimum value of $AR_c(G)$ among all t-edge colorings c of G. This concept was introduced by Gary Chartrand and first studied in [7]. With

the aid of results on partitions of sets of positive integers, the exact value of $AR_t(G)$ has been determined when G is a star or a matching for every integer $t \geq 2$.

Theorem 1 [6] *Let $t \geq 2$ and $m \geq t$ be integers and let $G \in \{K_{1,m}, mK_2\}$.*

(a) *If $m \leq t(t-1)$, then $AR_t(G) = k$ where k is the unique positive integer such that $(k-1)(t-1) + k \leq m \leq kt$. Thus, $AR_t(G) = \lceil \frac{m}{t} \rceil$.*

(b) *If $m > t(t-1)$, then $AR_t(G) = k$ where k is the unique positive integer such that $\binom{k+1}{2} + \binom{t-1}{2} \leq m \leq \binom{k+2}{2} + \binom{t-1}{2} - 1$. Thus,*

$$AR_t(G) = \left\lceil \frac{-3 + \sqrt{8m + 12t - 4t^2 + 1}}{2} \right\rceil.$$

The plan is to investigate $AR_t(G)$ where G is a path or a cycle of size m for all integers m and t with $m \geq t \geq 3$ and investigate $AR_3(G)$ for some well-known classes of graphs G.

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